

# On the representation and realization of the Ancient Greek acute

*Evidence from tone-tune mappings in Ancient Greek music*

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## Abstract

This study addresses the phonological representation and phonetic realization of pitch patterns found on or near prosodically prominent syllables in Ancient Greek, namely, the distinction between the so-called “acute” and “circumflex” accents. Empirically, we investigate in detail the correlation between tones and tunes in the Delphic hymns (DAGM 20 and 21) on syllables capable of bearing a circumflex accent (i.e., syllables containing a long vowel or diphthong = VV-syllables). This data supports two major findings. First, VV-syllables with circumflex accent are significantly more likely to be set to a melism than VV-syllables that are acute, grave, or unaccented, and, moreover, the proportion of melismatic settings among acute, unaccented, and grave VV-syllables does not significantly differ. Second, circumflex melisms consistently (always or nearly so) fall in pitch (on average, by three semitones), whereas acute and unaccented melisms may either rise or fall (on average, by 1.5–2.25 semitones in either direction). Taken together, this data conforms to the usual description of the circumflex as a falling pitch, [H L], but speaks against claims that the acute constitutes a rising pitch ([L H], or High alone aligned with the latter portion of a VV-syllable, [Ø H]). We instead conclude that the acute represents a single High pitch target phonologically mapped to the entirety of a VV-syllable, and discuss the implications for the phonological analysis of the prosody of Ancient Greek in light of the typology of contour tones.

## Keywords

Ancient Greek prosody – Ancient Greek music – pitch accent – tone-tune setting – prosodic phonology – melism – Delphic hymns

### 1 Introduction: acute and circumflex

Ancient Greek knows a distinction in its word-level prosody between two types of “accents”: the so-called “acute” and “circumflex”.<sup>1</sup> Genuine lexical minimal pairs are rare, but do exist (e.g., *κῆρ* ‘doom, fate’ vs. *κῆρ* ‘heart’), while minimal pairs internal to the inflection of a stem (e.g., *ἀγοράς* ‘meeting place:ACC.PL’ vs. *ἀγορᾶς* ‘id:GEN.SG’) are somewhat more common; the distinction is limited to final syllables.<sup>2</sup> That these “accents” occur on or near the prosodically most prominent syllable of a polysyllabic word is universally acknowledged, as is the fact that at least one such accent is obligatory in each prosodic word (see Goldstein 2015: Ch. 3 for a recent detailed treatment of the structure of prosodic words and cliticization in Ancient Greek).<sup>3</sup> The existence of mono-

1 The term “Ancient Greek” is used here to mean the dialects spoken in the 1st millennium BCE in which the distinction between “acute” and “circumflex” certainly existed, that is, mainly Attic-Ionic and Doric (see Probert 2003: 158–168 for a brief overview of accentuation in dialects other than Attic-Ionic), and the *κοινή* of the Hellenistic period. Between Attic-Ionic, Doric, and the *κοινή*, there existed both some minor, if systematic, differences in the rules of accentuation, as well as lexical differences in the accentuation of specific words or subclasses of words (see Probert 2006: 74–84 on differences in accentuation between the earlier dialects and the *κοινή*).

2 In the latter minimal pair, however, the accentuation of *ἀγοράς*, like very many instances of a circumflex in a final syllable on a polysyllabic stem, can well be derived synchronically via (post-lexical) vowel contraction following assignment of an acute to the penultimate syllable (see generally Noyer 1997, Probert 2010, and now also Ruan 2024 on such derivations). The historically older, uncontracted forms are sometimes attested in Homer (e.g., GEN.PL *ἀγοράων* *Il.* 2.275, *Od.* 4.818).

3 We do not concern ourselves here with the problem of an explicit phonological grammar for the assignment of accents in Ancient Greek. While restrictions on the distribution of accents within words (e.g., the Law of Limitation, the *συντήρησις*-Rule) are well known, the particular rules responsible for generating those distributions remain an object of ongoing investigation and dispute. Since the position of an accent within the rightmost three syllables of a word is not predictable based on syllable count and syllable weight alone, some purely lexical properties certainly play a role. The main points of contention in the phonological analysis of Ancient (Attic-Ionic) Greek accentuation concern whether it should be handled purely through the assignment of tones (e.g., Ito & Mester 2017, Revithiadou 2018) or whether an abstract prominence (“stress”) is assigned, which may then be realized as pitch or attract a prominence tone (“pitch accent”), either with (so, e.g., Steriade 1988, Sauzet 1989, Golston

syllabic minimal pairs (see discussion in Sandell 2023: 502–507) differing only in accentuation strongly suggests that the phonological underlying representations of lexical entries can bear distinct prosodic specifications of some kind; we return in section 3.3 to discussion of the analysis and representation of this contrast. The primary focus of this study lies, however, with discussion of typological, analytical, and most importantly, empirical evidence for the surface representation of the acute–circumflex contrast, the latter drawn from tone–tune mappings in Ancient Greek music.

Beginning with descriptions of accent provided by grammarians in antiquity (see, in brief, Probert 2003: 3–15, and in deeper detail, Probert 2006: 15–45), the difference between syllables bearing an acute or circumflex (“accented syllables”), on the one hand, and unaccented syllables, on the other, has been characterized primarily as one of **pitch**: accented syllables in general exhibit higher pitch, i.e., a rise in *F<sub>0</sub>* relative to unaccented syllables. The terms used by the ancient grammarians, ὀξύς ‘acute’ (lit. ‘sharp’) and ὀξύβαρεα ‘circumflex’ (lit. ‘sharp-deep’), point towards a pitch peak relative to the unaccented syllables, which are labelled βαρύς ‘grave’ (lit. ‘heavy, deep’).<sup>4</sup> That pitch on accented syllables was relatively higher is further confirmed by tone–tune mappings in settings of Ancient Greek music: in the Delphic hymns (DAGM 20 and 21), the single accented syllable of a word, whether acute or circumflex, is virtually never set to a note lower than any other syllable in that word (see Pöhlmann 1970: 140, West 1992: 197–200, Devine & Stephens 1994: Ch. 4, Probert 2003: 19–21, Probert 2006: 47–48, Cosgrove & Meyer 2006).<sup>5</sup>

For most types of syllable structures, an appropriate tonal representation in terms of high (H) and low (L) tones (or in some cases, perhaps absence of tonal specification, ∅) for accented syllables seems fairly clear. For the cir-

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1990, Kiparsky 2003, Sandell 2023) or without (so, e.g., Steriade 2014, Golston & Paulsen 2022) the construction of foot structure; the recent analysis of Ruan (2024) is agnostic regarding these questions. Furthermore, while most descriptions and analyses of Ancient Greek accentuation take the mora (μ) to be the tone-bearing unit (TBU), we consider the question of whether the TBU was the mora or syllable (σ) to remain open.

4 Some readers may be more familiar with the term *περισπόμενος* ‘bent, round’ with reference to the circumflex, but this term technically refers to the shape of the graphical mark employed for the circumflex, not a description of the pitch contour of the circumflex accent; see Allen 1968 [1987]: 125, fn. 25.

5 Each paeon apparently contains but one exception: *φερόπλοιο* (20.10) and *δικόρυφον* (21.2=2). In the latter, the poet may have set the phrase *δικόρυφον κλειτόν*, with an unsurprising peak at [kó] and a second, surprising one at [p<sup>h</sup>on], to musically express the notion ‘two-peaked slope’ (cf. Reinach 1909–1913: 164 and 1926: 18, followed by Pöhlmann 1960: 68 and West 1992: 201, 294, among others), resulting in an exceptional setting.

cumflex, which is restricted to syllables containing two vocalic morae (VV; i.e., syllables with a long vowel or diphthong as nucleus), an autosegmental-metrical representation as a high-low (HL) sequence straightforwardly captures the falling pitch universally assumed to characterize it; the HL tonal sequence may either have one tone each linked to one mora, as most descriptions, going back at least to Vendryes (1904: 44–49) and Jakobson (1937, applying to Ancient Greek notions developed in Jakobson 1931), seem to assume, or two tones linked to a single syllable;<sup>6</sup> compare the representation adapted from Jasanoff 2017: 11, shown in (1).<sup>7</sup> In syllables containing but a single vocalic mora, only an acute is permitted, which may be represented as a lone H tone linked to the single mora or to the syllable node. Uncertainty persists, however, concerning the representation of acute accents on VV-syllables. Most current descriptions of Ancient Greek accentuation (e.g., Allen 1968 [1987]: 125, Szemerényi 1996: 73, Probert 2003: 15–16, Probert 2006: 58, Olander 2009: 62, Jasanoff 2017: 11) imply or explicitly provide a representation for the acute and circumflex as mirror-images of one another: if the circumflex on a VV-syllable is [H L] (or alternatively, [H Ø]) then the acute is [L H] (or [Ø H]), as represented in (2), where the first mora is linked to a L tone and the second to a H.

Nevertheless, an alternative, shown in (3), is available: there, a single H spreads across two morae, or is linked to a syllable dominating two vocalic

6 In considering this question, the source of the surface H and L tones (i.e., whether both are underlying, whether they are lexical, or they are intonational) is not of primary relevance. The main question pursued here is rather: what do the surface outputs in terms of association of tones to TBUs concretely look like, after the application of all phonological processes? Thus, the level of analysis at which we are operating is the *OUTPUT* of the phonology, as opposed to the underlying or intermediate associations of tone, and as opposed to the exact phonetic implementation of the phonological output. We take the output of the phonology to be the level to which tone-tune setting is sensitive. See, however, the discussion in § 3.3 on the status of underlying tonal contrasts in Ancient Greek given the surface contrasts claimed to exist on the basis of the evidence treated here.

7 Jasanoff's autosegmental-metrical representations in fact omits any L tones; it is not clear whether he tacitly assumes that L tones are inserted by default rules in the intonation of prosodic words, or whether all TBU's except for the one linked to H remain toneless on the surface. Regardless, the representations that he offers make clear that acute and circumflex constitute mirror-images in terms of pitch movement. Jasanoff's explanation of his representations implies that he believes that, in cases where a H tone is (underlyingly) linked to the second of two vocalic morae, the (surface) output will be a rising tone ("On a long vowel or diphthong, there are two theoretical possibilities, depending on whether the tonal peak occurs relatively early or relatively late in the timing of the syllable. In the latter case, the resulting *rising contour* [emphasis ours] is phonologically and notationally identified with the acute accent on short vowels").

morae.<sup>8</sup> This view, according to which a syllable containing two vocalic mora would show a single (level) High tone, neither a bitonal (rising) Low–High tonal sequence nor an underspecified sequence  $\emptyset$ –H, with a toneless first mora a high-toned second mora, is forwarded in Bally 1945: 13–15, Sauzet 1989: 91–92, and Steriade 2014: 7. Perhaps surprisingly, Devine & Stephens (1994) seem nowhere to explicitly discuss the realization of syllables bearing an acute; however, in the context of melisms in the Delphic hymns (p. 193–194),<sup>9</sup> they do seem to expect a rise in pitch on VV-syllables with an acute: “When a rising melism is assigned to a syllable bearing an acute on a long vowel or diphthong, the melism is a direct musical reflection of the rise on the second nuclear mora, in contrast to the fall on the second mora of circumflex vowels and diphthongs” (p. 193). Later discussion (pp. 444–445) indicates that they assume a High–Low pitch movement for both circumflex and acute (within a single syllable versus across two syllables). Thus, while the circumflex uncontroversially signals a falling tone ([H L], or perhaps [H  $\emptyset$ ]), whether the acute on VV-syllables should be understood to represent a surface rising tone, as in (2), or a surface high level tone, as in (3), constitutes a heretofore unresolved issue in the description of Ancient Greek prosody.<sup>10</sup>

(1) Autosegmental–Metrical Representation of HL Circumflex (TBU =  $\mu$ )

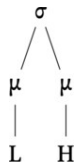


8 Steriade (2014: 7) offers the equivalence “HH = acute (acute)”, seemingly suggesting that *two* H tones might be present in the realization of an acute on a VV-syllable. While phonologically technically different from the single H used for the representations in (3), we see no way at present, empirically, to distinguish between single H vs. double HH for acutes on VV-syllables. Note, however, that two distinct H tones in such a configuration would constitute a violation of the Obligatory Contour Principle, which Steriade invokes with respect to patterns of accentuation in enclitic sequences (see section 3.2 below). For simplicity, we assume a single H throughout.

9 A *melism* is an instance in which a single syllable in the text is set to multiple pitches in the tune. In the Delphic hymns (DAGM 20 and 21), all melismatic syllables span exactly two pitches.

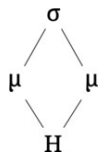
10 Since the second segment in each of the diagrams (1)–(3) can be either part of a vowel or a glide, we omit a segmental tier. The reader should keep in mind that the association of a tone is assumed to be licensed only when a  $\mu$  is projected from a [–consonantal] segment (i.e., vowel or glide).

- (2) Autosegmental–Metrical Representation of LH Acute on Bimoraic VV-Syllable (TBU =  $\mu$ )

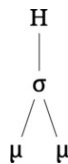


- (3) Autosegmental–Metrical Representation of H Acute on VV-Syllable

a. TBU =  $\mu$



b. TBU =  $\sigma$



While a mirror-image representation of acutes and circumflexes, as in (1) and (2), sets up a satisfying sort of structuralist symmetry, actual empirical or analytical reasons to posit the representation in (2) are thin on the ground. Jakobson (1937 [2002]: 264) draws a parallel with the word prosody of Čakavian Serbo-Croatian, but does not offer any motivation internal to Ancient Greek itself for characterizing the acute VV-syllables as “absence of accent on their first mora”.<sup>11</sup> Conversely, the phonological typology of tone (in canonical tone languages) and word-level intonation (in languages with consistent word-level pitch accents), one possible analysis of some aspects of enclitic accentuation, and, most importantly, empirical data from tone-tune mappings in musical settings of Ancient Greek texts provide convergent evidence in favor of a surface representation for the acute like that in (3): a level High tone linked to two morae or the syllable (depending on the TBU assumed).<sup>12</sup> The ultimate

11 See section 3.1.1 below on the supposed prosodic parallel between Serbo-Croatian dialects and Ancient Greek, with particular reference to the phonetics of “pitch accents” in Čakavian dialects.

12 Such a description would further accord with the terminology employed by the Alexandrian grammarians mentioned above; there is no parallel \*βαρυόξεια ‘low-high’ designation for acutes on VV-syllables, nor any other indication that the pitch of an acute on a VV-syllable was perceived as different than the pitch of an acute on a V-syllable. Jakob-

contention of this paper is thus that (3) constitutes the most plausible surface representation of acute accents on VV-syllables, and that such a representation ought to be assumed in future work on Ancient Greek prosody.

The core empirical argument of this paper is advanced in section 2. There, we examine the evidence from tone-tune mappings in Ancient Greek music and conduct statistical analyses thereof; we contend that the results of this analysis conclusively speak against the hypothesis of tonally parallel acute and circumflex accents on VV-syllables (i.e., [L H] and [H L] or [∅ H] and [H ∅]), and most readily lend themselves to the interpretation of acute as [H] (rather than [∅ H]) and circumflex as [H L]. We then consider how this result fits with evidence from phonological typology and the phonological analysis of Ancient Greek accentuation. In section 3.1, we consider the typology of surface tonal patterns in both canonical tone languages and “pitch-accent” languages, and find that languages that exclusively exhibit contour tones, but no level high tone, on syllables capable of hosting some contour tones at all, are comparatively less common. Section 3.2 then examines the facts of enclitic accentuation in Ancient Greek and shows that the assumption of a level [H] tone on VV-syllables with acute may provide a motivation for certain behaviors, and at least poses no additional obstacles. Next, section 3.3 discusses potential underlying representations of the surface accentual contrast in light of the foregoing arguments and analyses of similar binary contrasts in West and North Germanic languages. Finally, section 4 then summarizes the arguments of sections 2 and 3 and discusses implications for continuing research into the prosody of Ancient Greek.

## 2 The testimony of Ancient Greek music: melisms in DAGM 20 and 21

This section presents a systematic examination and statistical analysis of all certain and some relatively secure instances of VV-syllables (i.e., syllables containing a long vowel or diphthong) that are set to *melisms* (i.e., two musical notes) in the paean of Athenaios, and the paean and processional of Lime-nios (DAGM 20 and 21);<sup>13</sup> the two pieces are also known as the Delphic hymns.

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son (1937 [2002]: 266) found it “interesting that the complexity of a long syllable under the acute accent was not noticed [by the ancient grammarians]”; the failure to notice the tonal complexity expected by Jakobson could most straightforwardly be explained by the assumption that such syllables were not, in fact, tonally complex.

13 See footnote 9 above on the term *melism*.

DAGM 20 and 21 were selected for the present study because of their age, their length, the fact that they are attested as inscriptions, and because previous scholarship, both impressionistic (Winnington-Ingram 1955: 64–73, West 1992: 199) and statistical (Cosgrove & Meyer 2006), has found that they exhibit a particularly strong tendency towards tone-tune matching (e.g., the “pitch-height rule” that syllables with acute or circumflex are not set to notes lower than unaccented syllables within the same prosodic word is virtually always obeyed; cf. footnote 5 above). On other contributions of these two works to the reconstruction of Ancient Greek prosody, see Devine & Stephens 1994: 172–194. In section 2.1, we first summarize the conclusions drawn from detailed statistical analysis of the data and offer an introduction to tone-tune setting in the Delphic hymns generally and the setting of *melisms* in particular. Section 2.2 then explains how the data for the study was collected and processed and sets up the hypotheses to be tested in section 2.3.

### 2.1 *Summary of findings and introduction to melismatic setting in DAGM 20 and 21*

Analysis of the data on VV-syllable melisms in DAGM 20 and 21 admits of the following major conclusions, which are laid out here. First, syllables bearing a circumflex accent are significantly more likely to be set to a melism at all than acute, grave, or unaccented syllables, all of which are set to melisms at approximately the same (relatively low) *rate*. Second, while the circumflex melisms are consistently (perhaps exceptionlessly) set to falling melisms, acute and unaccented VV-syllables may be set to either rising or falling melisms.<sup>14</sup> Accent type is thus also a significant predictor of *directionality*: circumflex melisms fall, whereas acute or unaccented melisms may either rise or fall; a qualitative examination of unaccented and grave melisms in section 3.2.3 indicates that their tendency to be set to rising melisms—at a rate higher than the rate at which acute VV-syllables are set to rising melisms in the corpus—is plausibly attributable to the tendency of tunes to rise towards the next local pitch peak (as defined by an acute or circumflex). In addition the *size* of an interval between two pitches in a melism tends to be larger when the accent is circumflex; while there are not statistically significant differences in interval size between any of the four accent types (when examined separately from

14 The adjective *significant* and its adverb *significantly* are always used throughout in a statistical sense: the frequency distributions of the data are sufficiently extreme that they are unlikely to have arisen under the assumption that the predictor variables considered (here, usually accent type) have no effect on the independent variables examined (rate, directionality, and size of melisms).



the directionality of a melism), the difference in average size between a circumflex melism and the other three types is nevertheless larger than among the other three types (acute, unaccented, grave). Finally, whether a syllable bears a circumflex is a strong predictor of the second pitch in a melism given the first pitch; acute and grave syllables do not have an effect significantly different from the effect of unaccentedness. In sum, under the assumption that pitch movement in the tune of some Ancient Greek vocal music strove to match pitch targets of the phonological tones,<sup>15</sup> we conclude that the data examined here are most readily compatible with a [H L] pitch target on circumflex syllables, but speak against an interpretation of the acute as [L H] and offer no unambiguous support for a [Ø H] pitch movement on acute VV-syllables.

Before turning to the methodology employed, we now briefly preview how the poets typically set VV-syllables in DAGM 20 and 21, proceeding by accent type: circumflex, acute, unaccented, and grave. Readers may find it useful to have seen a number of settings before turning to the more detailed analysis and discussion below (section 2.2). The Delphic hymns are cited according to the document number in DAGM, the line number in our electronic corpus (see section 2.1), and the line number in the original inscription. For example, “20.10=12” may be read “DAGM 20 (i.e., the paeon of Athenaios), line 10 of the electronic corpus = line 12 of the original inscription.” Where settings are cited, *C–B* is the octave below middle *c* and *c–b* is the octave from middle *c* upward; postposed subscript *b* and # indicate flat and sharp.

Circumflex-accented VV-syllables are usually (~58–62 %) set to melisms (see Tables 5 and 9). Of the most textually secure melismatic settings, at least 95 % are set to falling melisms (see Table 16). Those melisms tend to fall relatively steeply, by about three semitones (i.e., three half-steps; see Table 19). In contrast, VV-syllables of other accent types are usually *not* set to melisms: acute VV-syllables are ~13–19 % melismatic, unaccented 14–16 %, and grave 15 % (cf. Tables 5 and 9 below). Instead, they are usually set to a single, long note, i.e., a quarter-note in the DAGM transcription. The setting of *χερσὶ γλαυκᾶς ἐλαίας* (21.6=6) [k<sup>h</sup>er.si<sup>?</sup>. glaυ.kâ:s e.lá.i.ja:s] is typical in that respect.<sup>16</sup> The circumflex-

15 For crosslinguistic overviews of tone-tune setting (also known as tone-melody matching), see Devine & Stephens 1994: 162–170, Schellenberg 2012, McPherson & Ryan 2018, and Ladd & Kirby 2020.

16 Where IPA transcriptions are given, we present the language as a conservative form of the Athenian dialect of the early Hellenistic period (cf. Horrocks 2010: 160–172). Note that in the IPA, the grave accent is transcribed with a superscript question mark, which

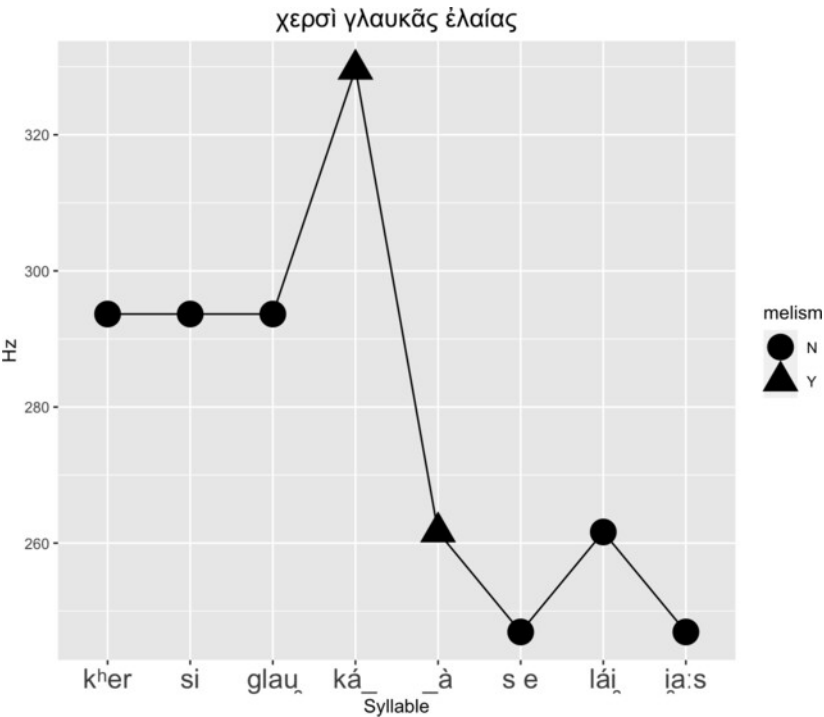


FIGURE 1 Pitch track for χερσὶ γλαυκᾶς ἐλαίας (21.6=6)  
Note: The term “pitch track” employed here is used in the general sense of a graph of fundamental frequency against time. The pitch tracks provided here are not, of course, based on any available audio recording of the musical fragments under discussion, but rather depend upon the careful philological and musicological work carried out to determine the correspondences between Ancient Greek musical notation and modern western musical notation (beginning with Beller-mann 1841 and 1847 and Westphal 1867; see the summaries in Hagel 2020 and West 1992: Ch. 9, especially pp. 254–259 and 273–276).

accented syllable [ká:] is set to a melism *g-e*, that falls four semitones (68 Hz in Figure 1 [see Table 2, note b]), whereas the VV-syllables of other accent types are set to a single note: acute [láí], unaccented [glau] and [iá:s]. Note that in Figure 1, melismatic [ká:] is represented as [ká\_à].

As noted above, acute-accented VV-syllables are usually *not* set to melisms; they are only ~13–19% melismatic. In the four relatively infrequent cases where acute-accented VV-syllables *are* unambiguously set to a melism, two rise and

reflects the fact that we are unsure whether it represents a deleted/delinked acute or a lowered/downstepped acute (cf. notes 29 and 30).

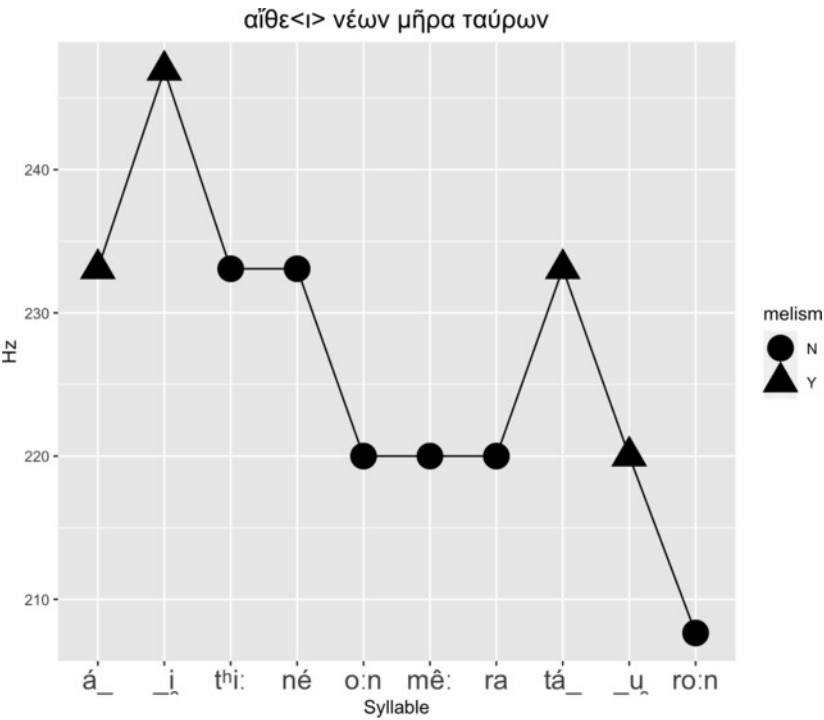


FIGURE 2 Pitch track for αἶθε<ι> νέων μῆρα ταύρων (20.10=12)

two fall. The median size of the interval for the falling two VV-acutes is 1.5 semitones, which is only half as steep as the median for the circumflex-accented comparanda. For instance, in the phrase αἶθε<ι> νέων μῆρα ταύρων (20.10=12) [áḱ.tʰi:. mē:ra. táḱ.ro:n] “(Hephaistos) burns the thigh-pieces of young bulls”, the first syllable of αἶθε<ι> is set to a melism  $d_b$ - $d$  that rises one semitone, and the first syllable of ταύρων is set to a melism  $d_b$ - $c$  that falls one semitone. Note that in Figure 2, melismatic [áḱ] and [táḱ] are represented as [á\_ \_í] and [tá\_ \_u].

Like acute-accented VV-syllables, unaccented VV-syllables are usually *not* set to melisms; they are only 14–16 % melismatic. In the seventeen comparatively infrequent cases where unaccented VV-syllables *are* set to a melism, eleven rise and six fall. Whether they rise or fall depends on their position relative to a local peak (i.e., acute or circumflex). For instance, most of the rising melisms (7/11) immediately precede a local peak in the same word, e.g., the rising melism  $c$ - $d_b$  on the immediately pre-accentual syllables of αἰόλοις (20.12=14) [aḱ.ió.loḱs]. Those rising melisms very likely reflect a rise in pitch over the syllable immediately preceding the acute/circumflex. Most of the falling melisms on unaccented VV-syllables (4/6) immediately follow a local

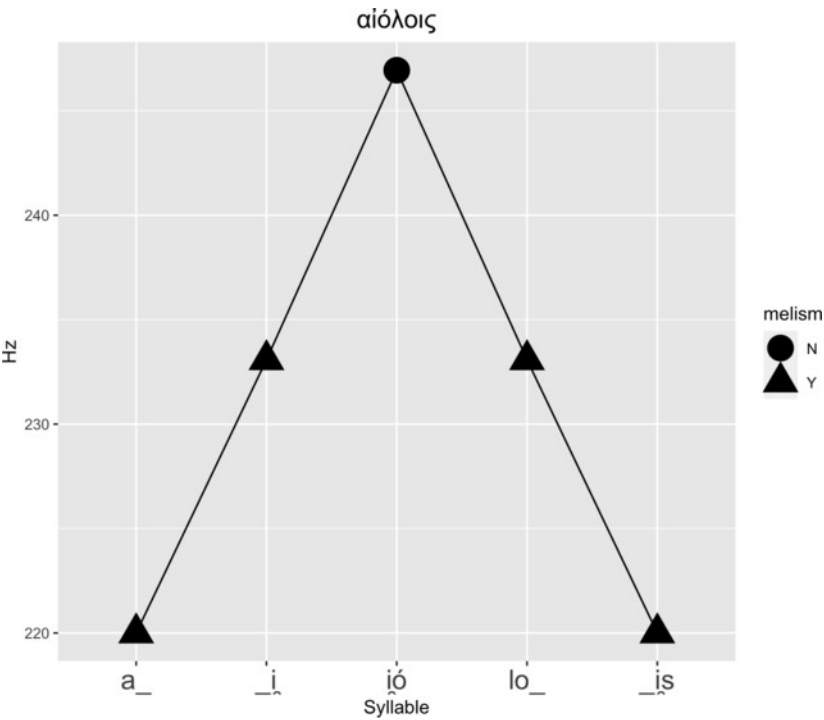


FIGURE 3 Pitch track for αἰόλοις (20.12=14)

peak in the same word, e.g., the falling melism  $d_b$ - $c$  on the post-accentual syllable of the same word αἰόλοις (20.12=14), whose setting  $c$ - $d_b$ ,  $d$   $d_b$ - $c$  thus exhibits both a pre-accentual rising melism and a post-accentual falling one (Figure 3).

Like acute-accented and unaccented VV-syllables, grave-accented VV-syllables are usually *not* set to melisms; only 15 % are melismatic. In the two relatively unusual cases where grave-accented VV-syllables *are* set to melisms, both rise. Despite the meager data, this is precisely what we would expect. Since grave-accented words form part of the pitch trajectory that rises to the next acute/circumflex (Devine & Stephens 1994: 354–355, 449), grave-accented syllables are comparable to unaccented syllables that are located along the rising pitch trajectory, e.g., the first syllable of αἰόλοις above. Compare the setting of the rising trajectory [aí.íó] of αἰόλοις (Figure 3) with the rising trajectory [daʔ.n.kré] of ὠιδᾶν κρέκει (20.13=15) in Figure 4. Like [aí], immediately pre-accentual [daʔ.n] is set to a melism  $B$ - $c$  that rises one semitone.

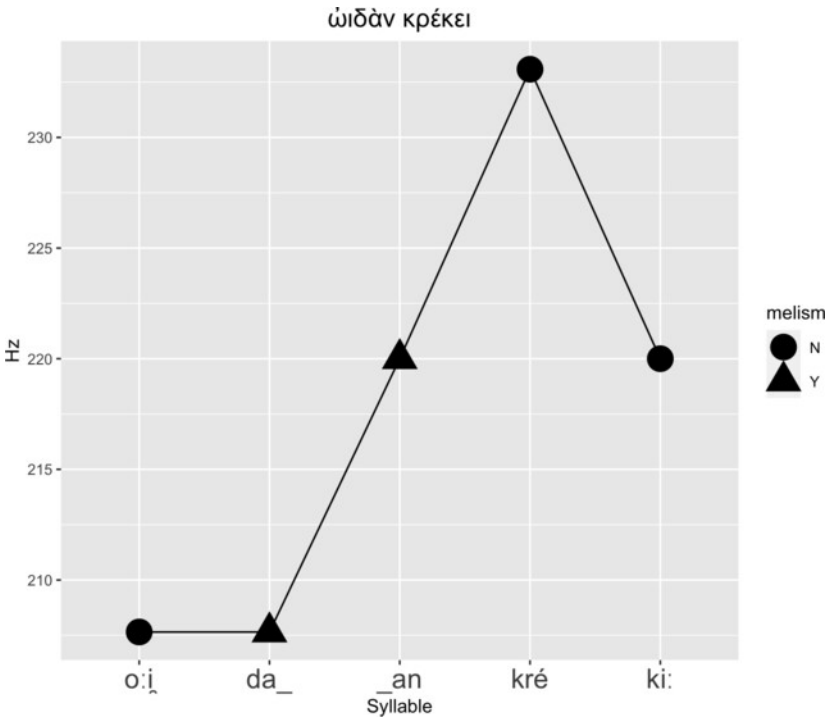


FIGURE 4 Pitch track for ὠιδὸν κρέκει (20.13=15)

## 2.2 Methodology: preparation of the corpus and its coding, hypotheses and statistical techniques

### 2.2.1 A database for *Documents of Ancient Greek Music*

To enable a systematic investigation of the melismatic settings of VV-syllables, an electronic corpus of seven documents in DAGM (DAGM 20, 21, 23–25, 27, 28) was manually prepared by the second author; from this larger corpus, a sub-corpus consisting of the paeans of Athenaios and Limenios (DAGM 20 and 21, pp. 62–85) was extracted.<sup>17</sup> The entire corpus is available as a plain-text

17 For readers departing from the modern musical notation in DAGM, three corrections were made in the electronic corpus. First, at 20.18=20, the third note of the setting of προφαίνεις (i.e., the first note of the melism on νεις), should be *F*, not *G*. It is correct in the ancient notation (Γ) on the preceding page (p. 66). Second, at 20.20=22f., the first preserved note of the setting of ἀπέστῃσας (i.e., the first note of the melism on τη) should be *B*, not *e<sub>♭</sub>*. See the discussion of the ancient note “O” on DAGM: 73 as well as Rocconi 2003: 71. Third, the second syllable of Πωμάων (21.44=39) should be set to an eighth note. The meter requires that μαί be light, and the shortening process is familiar from Attic αἰί vs. Ionic αἰέ; cf. the setting of Πατήονα at 21.21=19. The corpus also has one addition: the third and final note of μόλετε may be securely read as *c* (Hagel 2000: 128).

TABLE 1      Sample entries in database of DAGM 20 and 21

| Document | Line | Note | Note_Hz | Notevalue | Syllable | Accent | Juncture | Supplement |
|----------|------|------|---------|-----------|----------|--------|----------|------------|
| 20       | 2    | d    | 220     | 1         | mou      | u      | n        |            |
| 20       | 2    | c    | 246.94  | 1         | mou      | u      | l        |            |
| ...      | ...  | ...  | ...     | ...       | ...      | ...    | ...      | ...        |
| 21       | 18   | B    | 207.65  | 2         | toi      | a      |          |            |

comma-separated values file at [github.com/dcgunkel/agmusic](https://github.com/dcgunkel/agmusic). Each entry in the corpus (i.e., row in the database) corresponds to a single note, with its pitch and duration, or, where the note is missing entirely but the text is available, a syllable;<sup>18</sup> the total number of rows input based on DAGM 20 and 21 was 1107, which approximates the total number of notes. Table 1 presents three sample entries in the database, showing the relevant features (i.e., columns) for which an entry was coded, and Table 2 provides an explanation of these features.<sup>19</sup>

Once this electronic corpus was prepared and coded, the first author extracted the relevant data using R (Version 4.2.2; R Core Team 2022). Since the first portion of a melism was consistently coded in the **juncture** column with “n”, that row and the immediately following row could be extracted as the syllable comprising a melism made up of two notes. This subset of the data was then further restricted to only those syllables containing a long vocalic nucleus, i.e., a long vowel or diphthong, by applying a regular expression to the **syllable** column to identify only those syllables containing the appropriate character sequences.

From the resulting list of 65 melismatic syllables, eight were then excluded: in five in of these cases, the textual supplement provided by West 1992 is, for both portions of the melism, relatively insecure (coded with “i” in the electronic corpus); in three further cases, part of the textual restoration seems secure (coded with “s” in the corpus), but closer consideration of the relevant cases led to the conclusion that one cannot, in fact, say with certainty whether the

18

As is explained below, in cases where the note is entirely unattested, the data was generally excluded from analysis, but was included in the raw electronic corpus for completeness.

19

Entries were in fact coded for a number of further features, including larger prosodic structures (e.g., boundaries of minimal and maximal prosodic words, phonological and intonational phrases), but since these features play little role in the present analysis, they are not discussed further here.

TABLE 2      Features coded in database of DAGM 20 and 21

|            |                                                                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document   | Number of the document in DAGM                                                                                                                                                                                           |
| Line       | Line of transcription in DAGM <sup>a</sup>                                                                                                                                                                               |
| Note       | The pitch in modern musical notation in DAGM<br>(c = middle C; C = one octave below middle C; F = flat; S = sharp)                                                                                                       |
| Note_Hz    | Conversion of the DAGM pitch transcription into Hz. <sup>b</sup>                                                                                                                                                         |
| Notevalue  | Duration of the note in DAGM<br>(1 =eighth note; 2= quarter note; 3 = dotted quarter note)                                                                                                                               |
| Syllable   | The syllable corresponding to the note                                                                                                                                                                                   |
| Accent     | Type of accent found on the syllable<br>(a = acute, c = circumflex, g = grave, u = unaccented)                                                                                                                           |
| Juncture   | Type of morphoprosodic boundary<br>(e.g., n = no syllable boundary, i.e., syllable is mapped to multiple notes, s = syllable, l = boundary between two content words, lf = boundary between a lexical and function word) |
| Supplement | Note is supplemented by editors<br>(blank = note attested, s = relatively secure supplement, i = relatively insecure supplement)                                                                                         |

- a    Note that the printed line of transcription into modern musical notation in DAGM does not correspond to the line on which the note is found in the original inscriptions.
- b    Since there is general agreement that the standard transcription (as in DAGM) is about a minor third too high (cf. West 1992: 273–276), the Hz values were accordingly lowered by a minor third.

syllable was set to two notes or not.<sup>20</sup> Among the remaining 57 melisms, there are some 12 in which the pitch of one part of the melism cannot be discerned, as well as one case in which both pitches are missing but for which relatively good indications of a single syllable being set to two notes are available.<sup>21</sup> Below, analyses with both a more liberal criterion for categorization as a melism

20    The following eight syllables present in the original list of 65 were excluded on this basis; note that citations refer to the document, then the line number in the electronic corpus, which corresponds to the printed line in the musical transcription given in DAGM, followed by (=) the line number of the inscription: γᾱ (20.20=22); λῶν (21.9=9); γᾱ (21.12=11); γρᾱί (21.12=11); πρῶ (21.14=14); λῶ (21.16=15); χρῆ (21.25=21); τᾱν (21.26=22).

21    The thirteen less secure syllables excluded under the more conservation criterion are: κῶ (20.1=1); χᾱί (20.8=9); νεις (20.18=20); συ (20.21=24); δᾱ (20.27=29); ζ οἶδ (21.11=10 f.); δει (21.17=16); να (21.26=22); Γᾱς (21.31=27); σύ (21.33=30); νῶν (21.34=30); ρεις (21.35=31); Γᾱ (21.35=31).

(57 tokens) and a more conversative criterion (44 tokens) are presented. Complete code in R for extracting the relevant data and performing statistical analyses is available at [github.com/rpsandell/melisms](https://github.com/rpsandell/melisms).

### 2.2.2 Hypotheses and statistical techniques

On the basis of the available data and the exploratory observations outlined above, the relationship between the four distinct types of syllable-level accentuation uncontroversially found in Ancient Greek (unaccented, acute, grave, and circumflex) and melismatic settings in music can be compared with the predictions made by different phonological interpretations of the surface values of acute and circumflex on VV-syllables. As discussed in section 1, most introductory presentations of Ancient Greek accentuation implicitly or explicitly regard the acute and circumflex on VV-syllables as tonal mirror images of one another: the acute as rising ([L H] or [∅ H]), the circumflex as falling ([H L] or [H ∅]); many formal analyses of processes for the assignment of accent in Ancient Greek also explicitly adopt such a mirror-image representations (Steriade 1988, Golston 1990, Kiparsky 2003, Blumenfeld 2005, Ruan 2024).<sup>22</sup> Meanwhile, a smaller group of formal analyses—largely fairly recent—instead explicitly assumes a representation [∅ H] for the acute on VV-syllables, but HL for the circumflex (Ito & Mester 2017, Revithiadou 2018, Golston & Paulsen 2022). That is, under this latter view, the circumflex consists in two phonologically specified tonal targets, whereas the acute on a VV-syllable is defined by just a single high target, which results in a rising contour because it is aligned with the latter part (~ second mora) of a VV-syllable. The distinction between these two classes of analyses may appear minor, but may, in fact, be non-trivial as concerns the predictions made about the behavior of melismatic settings. Finally, that an acute VV-syllable represents a single high target aligned with the totality of the syllable is proposed in Sauzet 1989 and Steriade 2014. Thus, as concerns the surface phonological representation of the acute-circumflex contrast, three distinct arrangements of tonal patterns are found in the literature, as laid out in Table 3; for all three sets of representations, we allow that an unaccented VV-syllable may be associated with either no tone (∅) or a low tone (L).

Exploratory investigation of VV-melisms in DAGM 20 and 21 indicates that the relation between accentual type and melism is open to investigation along three parameters: 1) how frequently VV-syllables are set to melisms (*rate*); 2) whether VV-syllables are set to rising or falling melisms (*directionality*); 3) how

22 The ascription of this view to these analyses is made on the basis of the authors' explicit statements concerning the acute-circumflex contrast.



TABLE 3     Surface representations and representatives of three proposals on the acute–circumflex contrast

| Proposal                         | Acute VV-syllable | Circumflex VV-syllable | Unaccented VV-syllable | Representatives                                                        |
|----------------------------------|-------------------|------------------------|------------------------|------------------------------------------------------------------------|
| 1. Acute-circumflex mirror image | L-H or Ø-H        | H-L or H-Ø             | L or Ø                 | Steriade 1988, Golston 1990, Kiparsky 2003, Blumenfeld 2005, Ruan 2024 |
| 2. Three-way contrast            | Ø-H               | H-L                    | L or Ø                 | Ito & Mester 2017, Revithiadou 2018, Golston & Paulsen 2022            |
| 3. Acute-unaccented mirror image | H                 | H-L                    | L or Ø                 | Sauzet 1989, Steriade 2014                                             |

large the interval is (in Hz as a continuous value, or semitones as a discrete value) between the two pitches of a melism (*size*). The general null hypothesis to be evaluated throughout is that accent type has no (statistically demonstrable) impact on these parameters: regardless of the type of accent on a VV-syllable: 1) the rate of melismatic setting is not significantly different, 2) the proportion of rising versus falling melisms is not significantly different, and 3) the average interval within the melisms is not significantly different. If no evidence for the rejection of the respective null hypotheses along any of these parameters obtains, then we may conclude that melismatic settings in DAGM 20 and 21 simply offer no evidence that bears on the distinction between the acute and circumflex accents. Crucial in all regards is the role of *unaccented* syllables as a baseline of comparison: since all three proposals listed in Table 3 would be compatible with significant differences between circumflex and unaccented syllables, whether acute VV-syllables pattern with circumflex syllables, unaccented syllables, or with neither, provides a means of distinguishing between the three proposals.

More specifically, each of the three proposals above makes different predictions about the overall patterning of significant differences, in the event that the null hypothesis of no association between accent type and rate, directionality, and/or average size of melismatic setting can be rejected. Specifically, proposal 1 (acute-circumflex mirror image) would align most straightforwardly with similar effects in terms of rate and size for both acute and circumflex VV-syllables, and with opposite, parallel behavior in terms of directionality: acutes and circumflexes ought to condition significantly larger proportions of rising

and falling melisms, respectively. Proposal 2 (three-way distinction), meanwhile, would expect that circumflex, acute, and unaccented syllables all behave differently along all three parameters. Under the assumption that phonologically specified tonal targets might condition more “extreme” effects in order to obtain satisfactory tone-tune correspondences, a syllable with a [H L] contour should bring about stronger effects in terms of rate, directionality, and size of melisms than [Ø H] contours, which should in turn behave distinctly from a syllable with no contour ([Ø] or [L]). Proposal 2 would thus best align with the finding that a hierarchy circumflex > acute > unaccented emerges along all three parameters: 1) circumflex syllables are set to melisms more frequently than acutes, which are in turn set to melisms more frequently than unaccented syllables; 2) circumflex syllables are more regularly set to falling melisms than acute syllables are set to rising melisms, and acute melisms are in turn set to rising melisms more frequently than unaccented melisms are; 3) circumflex melisms are, on average, larger in size than acute melisms, while acute melisms are, on average, larger than unaccented melisms. Finally, proposal 3 (acute-unaccented mirror image) would find support in the event that circumflex VV-syllables behave significantly differently from both unaccented (as predicted by proposals 1 and 2) and acute (as predicted by proposal 2) syllables, but acute VV-syllables exhibit no significant differences in behavior from unaccented VV-syllables (in contrast to the predictions of both proposal 1 and proposal 2). The predictions that we ascribe to each of the three proposals concerning the surface phonological representation of acute, circumflex, and unaccented VV-syllables along the parameters of rate, directionality, and size in terms of melismatic setting are summarized in tabular form in Table 4.

The primary statistical tool that we employ in section 2.3 to assess hypotheses concerning association between accent type and melismatic setting and the predictions made by different proposals concerning the representation of acute VV-syllables is the **generalized linear model**. The purpose of constructing generalized linear models in this context is to obtain insights about and to compare the effects of estimates for the parameters (independent or predictor variables) in attempting to predict (the probability of) some outcome (the dependent or response variable). In models constructed below, the response variables are either real-valued continuous (positive) numbers (e.g., pitch in Hz) or are binary categories (e.g., melismatic or non-melismatic setting of a syllable); the latter is a special case, termed logistic regression, on account of the link function used in determining the probability of the response variable given the values of the predictor variables. The general form of a generalized linear model is stated in equation (1), and the mean distribution function used for cal-

TABLE 4      Summary of predictions for melismatic behavior of VV-syllables for the proposals in Table 3

| Proposal                                                                | Rate of melismatic setting                           | Directionality of melisms                                                                                                                                      | Average absolute size of melismatic interval         |
|-------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 1. Acute-circumflex mirror image<br>(Acute = LH/ØH, Circumflex = HL/HØ) | Circumflex ≈ acute<br>Circumflex, acute > unaccented | Falling on circumflex > falling on unaccented<br>Rising on acute > rising on unaccented                                                                        | Circumflex ≈ acute<br>Circumflex, acute > unaccented |
| 2. Three-way contrast<br>(Acute = ØH, Circumflex = HL)                  | Circumflex > acute > unaccented                      | Falling on circumflex > rising on acute > falling or rising on unaccented<br>Rising on acute > rising on unaccented<br>Falling on circumflex > rising on acute | Circumflex > acute > unaccented                      |
| 3. Acute-unaccented mirror image<br>(Acute = H, Circumflex = HL)        | Circumflex > acute, unaccented<br>Acute ≈ unaccented | Falling on circumflex > rising on acute<br>Falling on circumflex > falling on unaccented<br>Rising on acute ≈ rising on unaccented                             | Circumflex > acute, unaccented<br>Acute ≈ unaccented |

culating the probability of a success in logistic regression is shown in equation (2). That is, the estimated value of an outcome ( $\hat{y}$ ) is the sum of an intercept (“baseline”) coefficient  $\beta_0$  and the coefficients estimated for the respective predictor variables  $\beta_1$  to  $\beta_n$  (where  $n$  is the total number of predictor variables). In case the outcomes are binary, the probability of a “yes” outcome is calculated by exponentiating the inverse of the summed model coefficients multiplied by the values of the predictor variables. The model coefficients ( $\beta$ s) themselves are estimated through maximum likelihood estimation (i.e., by iteratively fitting coefficients until no other coefficient values that yield predicted values closer to the observed values can be found). For gentle introductions for linguists to generalized linear models and logistic regression in particular, see Baayen 2008: 195–214, Gries 2013: 247–311, or Levshina 2015: 253–276.

(1)  $\hat{y} = \beta_0 + \beta_1x_1 \dots \beta_nx_n$

(2)  $1 / (1 + \exp(-\beta_0 + \beta_1x_1 \dots \beta_nx_n))$

All models of this sort presented in this paper were generated through a call to the function `bayesglm()` made available through the package *arm* (version 1.13–1; Gelman et al. 2022) in R version 4.2.2 (R Core Team 2022). The func-

tion `bayesglm()` differs from R's built-in function `glm()` for generating generalized linear models primarily in that it imposes a weakly informative prior distribution on the coefficient estimates in the form of independent Cauchy *t*-distributions with centers at 0 and a scale of 2.5 (see Gelman et al. 2008 or Gelman et al. 2013: section 14.8 for further mathematical details). This weakly informative prior has the effect of making the coefficient estimates more conservative (i.e., more data is required to show that the estimate for the coefficient truly differs from 0, that is, that it has some real effect), which is desirable in observational studies (like the present one) in which the sample size for specific levels of factor variables may be rather small.<sup>23</sup> Lastly, important to note is that all models generated here include a predictor variable `ACCENT` with four levels corresponding to each of the surface accentual distinctions found in Ancient Greek (unaccented, acute, circumflex, grave) using a treatment contrast, with the level “unaccented” always set as the reference level. In this way, the model coefficients and corresponding tests of significance serve to explicitly compare the effect of the other three accent levels with the effect of unaccentedness.

### 2.3 *The melisms of DAGM 20 and 21: implications for the acute*

As outlined above, we aim in the following to test the hypothesis that VV-syllables bearing an acute accent (or its tonal allophone, the grave; cf. Devine & Stephens 1994: 180–183 for discussion of the nature of the grave based on DAGM 20 and 21, with conclusions that we do not necessarily endorse) carried multiple phonological pitch targets, more specifically [L H] or [Ø H] (rising tone). A grounding assumption adopted here is that the circumflex certainly exhibited a falling tonal contour, either [H L] or [H Ø] (though more likely the former; cf. section 3.3), whereas an unaccented syllable certainly bore either a single pitch target, [L], or no pitch target, [Ø].<sup>24</sup> On this basis, one can assess whether VV-syllables that certainly exhibited a phonologically relevant pitch contour (i.e., circumflex VV-syllables) behave differently from VV-syllables that certainly carry only a single pitch target. In other words: does accent type impact choices relating to tone-tune setting in DAGM 20 and 21 (given that

23 For further examples of the specific use of `bayesglm()` in linguistic research, see, for instance, Zuraw et al. 2019 or Gouskova & Becker 2013.

24 Under the view that specified tone in Ancient Greek occurred only towards the right edge of the word (cf. discussion in section 1), some unaccented syllables (specifically, those immediately following an acute syllable) might be specified for Low tone, while syllables preceding an acute would be unspecified for tone.

“opposing” settings are avoided; cf. under section 2 and note 5).<sup>25</sup> Possible distinctions conditioned by tonal targets are evaluated in terms of the likelihood that a syllable is set to a melism whatsoever (section 2.3.1) as well as the direction and size of melisms (section 2.3.2). Concretely, if syllables bearing a circumflex behave differently from unaccented syllables in these respects, evidence is available that tonal targets indeed impacted choices in terms of musical setting. From there, cases of melisms on acute (and potentially grave) VV-syllables may be compared with circumflex and unaccented VV-syllables to determine whether their profile more closely resembles one or the other, or is clearly distinct from either. The three basic null hypotheses to be tested in following are thus:

1. There is no association between accent type and *rate* of melismatic setting.
2. There is no association between accent type and *directionality* of melismatic setting.
3. There is no association between accent type and *size* of interval in a melism.

If these null hypotheses can be confidently rejected, then, depending upon the exact differences observable in terms of rate, directionality, and size for circumflex, acute, and unaccented syllables, the inferences stated just below may be drawn (cf. Table 4). Note here that comparison between acute/grave and unaccented syllables along the three dimensions studied serves to establish the presence of a pitch contour, while comparison with the circumflex, if the circumflex is assumed to bear a fully specified [H L] sequence, can indicate whether acute VV-syllables have a full or partial specification for tone.

- 1a. If acute and/or grave VV-syllables are set to melisms at a significantly higher rate than that of unaccented syllables, then some pitch contour on acute VV-syllables may be inferred.
- 1b. If acute and/or grave VV-syllables are set to melisms at a rate not significantly lower than that of circumflex syllables, the presence of multiple phonologically specified pitch targets may be inferred; if they are set to melisms at a rate significantly lower than that of circumflex syllables, an underspecified tonal contour may be inferred.
- 2a. If acute and/or grave VV-syllables set to melisms exhibit a rising pitch contour at significantly greater rates than the rate of rising melisms on

25 On “opposing” tone-tune settings, see Schellenberg 2012, McPherson & Ryan 2018: 119–122, and Ladd & Kirby 2020.

unaccented VV-syllables, then some rising pitch contour on acute VV-syllables may be inferred.

- 2b. If acute and/or grave VV-syllables set to melisms exhibit a rising pitch contour at a rate not significantly lower than the rate of falling melisms on circumflex VV-syllables, the presence of multiple phonologically specified pitch targets may be inferred; if the rate of rising melisms is lower than the rate of falling melisms on circumflex syllables, an underspecified tonal contour may be inferred.
- 3a. If acute and/or grave VV-syllables set to melisms exhibit an average interval size significantly larger than the average interval size on unaccented VV-syllables, then some pitch contour on acute VV-syllables may be inferred.
- 3b. If acute and/or grave VV-syllables exhibit an average interval size not significantly smaller than the average interval size of melisms on circumflex VV-syllables, the presence of multiple phonologically specified pitch targets may be inferred; if the average size of melisms is smaller than the average size of melisms on circumflex syllables, an underspecified tonal contour may be inferred.

### 2.3.1 Accent type and likelihood of melismatic setting

In order to establish the rate at which VV-syllables in DAGM 20 and 21 are set to melisms, to assemble a list of all VV-syllables *not* set to melisms was also necessary, in addition to the 57 plausible tokens of VV-syllable melisms discussed under section 2.2.1 above. To obtain such a list of cases where a VV-syllable was certainly or likely set to just a single note, the list of all VV-syllables in the electronic subcorpus was taken, from which the cases of melisms were excluded, and then further all cases in which the column **notevalue** contained the value “NA” (and thus, by extension, the column **note** contained no value), since the syllable in such cases might, in fact, have been set as a melism. We also exclude one instance of a reconstructed grave on a proclitic  $\acute{\omega}\varsigma$  at 20.19=21.<sup>26</sup> The resulting table of secure non-melismatic VV-syllables contained 165 tokens; together

26 If the 74 tokens of likely non-melismatic VV-syllables whose note is unknown were maintained in the analysis here, rates of melismatic setting by accent would be lower across the board (28/60 = ~46% for circumflexes, 6/39 = ~15.3% for acutes, 21/174 = ~12% for unaccented syllables, and 2/21 = ~9.5% for graves). Perhaps worth noting is that relatively more seemingly non-melismatic VV-syllables with a circumflex are excluded in this way (15/32 = ~47% of all the seemingly non-melismatic VV-syllables with circumflex, versus 6/33 = ~18% for acutes, 44/153 = ~29% for unaccented syllables, and 6/19 = ~31.5% for graves). For the overall statistical analysis, however, the choice to exclude these uncertain tokens makes little difference: while the difference in proportion of melismatic setting

TABLE 5      Proportion of melismatic and non-melismatic VV-syllables in DAGM 20 and 21 by accent<sup>a</sup>

|            | Circumflex   | Acute        | Unaccented  | Grave       | SUM         |
|------------|--------------|--------------|-------------|-------------|-------------|
| Melism     | 28 (~62.2 %) | 6 (18.75 %)  | 21 (~16 %)  | 2 (~15 %)   | 57 (~26 %)  |
| Non-melism | 17 (~37.8 %) | 26 (81.25 %) | 109 (~84 %) | 13 (~85 %)  | 165 (~74 %) |
| SUM        | 45 (~20.3 %) | 32 (~14.4 %) | 130 (~58.5) | 15 (~6.8 %) | 222         |

a   Percentages given in the non-marginal cells are proportions of column totals; percentages in marginal cells are proportions across the marginal row and column sums, respectively. This practice applies to all subsequent tables.

with the 57 melismatic tokens, a total of 222 VV-syllables were included in this initial quantitative assessment. The token counts, broken out by accentual type, with approximate percentages, are given in Table 5.

As the values in Table 5 indicate, only about one-fourth (~26 %) of all VV-syllables are set to melisms, with an apparently higher rate of melismatic setting being applied to circumflexes (~62.2 %), but an approximately similar lower rate (less than 20 %) in VV-syllables of any other accent type.<sup>27</sup> This asymmetry is noted in some reports without any descriptive statistics, e.g., Devine & Stephens (1994: 173): “the circumflex is most frequently set to a melism.” Others at least permit a symmetrical interpretation, e.g., Probert (2003: 20): “On long vowels that take two notes the first note is usually higher if the syllable has a circumflex, but the second note if the syllable has an acute.”

The first important conclusion that the data in Table 5 permit is that the categories of accent type and the choice to set a VV-syllable to a melism are robustly *not* independent of one another: Pearson’s  $\chi^2$ -test applied to Table 5 leads to rejection of the null hypothesis of independence ( $\chi^2(3) = 39.674$ ,  $p < 0.001$ ). Examination of the odds ratios for rate of melismatic setting in pairwise comparisons of the different levels of accent type (with a 95 % confidence intervals for the odds ratio), given in Table 6, strongly suggests that circumflex VV-syllables are much more likely to be set to a melism than acute, grave or unaccented syllables, while the latter types are approximately equally unlikely to be set to a melism; because of the relatively small number of graves in the data, the confidence intervals for comparisons involving graves are particularly

between circumflex versus non-circumflex syllables would be somewhat smaller, the differences would nonetheless remain statistically significant.

27   If the values for acute and grave are summed together, the figures amount to 8 melismatic settings (~17 %) versus 39 non-melismatic settings (~83 %).

TABLE 6 Odds ratios derived from Table 5

|            | Circumflex<br>vs. acute | Circumflex<br>vs. grave | Circumflex<br>vs. unaccented | Acute<br>vs. grave | Acute<br>vs. unaccented | Unaccented<br>vs. grave |
|------------|-------------------------|-------------------------|------------------------------|--------------------|-------------------------|-------------------------|
| Odds ratio | 6.94                    | 10.29                   | 8.41                         | 1.49               | 1.2                     | 1.25                    |
| 95% CI     | 2.22–25.04              | 1.97–105.02             | 3.73–19.69                   | 0.22–17.06         | 0.36–3.48               | 0.25–12.22              |

wide.<sup>28</sup> In other words, VV-syllables bearing an acute or grave as well as unaccented VV-syllables do not appear to behave differently in this regard: acute VV-syllables are only slightly more likely (1.2×) than unaccented VV-syllables to be set to a melism.

The values in Tables 5 and 6 thus suggest that acute, grave, and unaccented VV-syllables behave as a natural class vis-à-vis circumflex VV-syllables. (Recall the setting of  $\chi\epsilon\rho\sigma\iota\ \gamma\lambda\alpha\upsilon\kappa\hat{\alpha}\varsigma\ \acute{\epsilon}\lambda\alpha\acute{\iota}\alpha\varsigma$  in Figure 1.) They statistically corroborate Probert’s report that “perusal of the Hellenistic and probably non-strophic compositions in Pöhlmann and West 2001 suggests that a long vowel with acute or (in modern notation) grave accent was not set to a rising sequence of notes any more often than a long unaccented vowel” (Probert 2015: 935, n. 26). We contend that this two-way distinction arises because circumflex VV-syllables host two different tonal targets [H L], whereas the others associate at most a single tone with the entirety of the syllable (~ both morae in a syllable). Previous scholarship also provides reasons to group together two or all three members of that apparent class. First, as noted above, the grave is a tonal allophone of the acute. That underlying phonological identity could motivate grouping acute with grave VV-syllables for further comparison. Turning to the phonetic realization of the grave, we find two competing hypotheses. On the one hypothesis, the grave represents the absence of the underlying acute.<sup>29</sup>

28 An odds ratio may be understood as how many times more likely membership in one category (here, being set to a melism) is for items belonging to some category than for items belonging to another category. For a 2 × 2 table, the sample odds ratio is calculated as the ratio of the values in the first column divided by the ratio of values in the second column. For circumflex vs. acute melismatic vs. non-melismatic syllables, the sample odds ratio is thus (28/17)/(6/26) = 7.14. Here, however, we employ the conditional maximum likelihood estimate for the odds ratio, which is generally superior (especially in cases where sample sizes may be small) to the unconditional maximum likelihood estimate obtained by the simple sample odds ratio (Hauck 1984). Consequently, the odds ratios reported here differ somewhat from what the calculation of the simple sample odds ratio would yield.

29 For the grave as a deleted/delinked acute, see Meillet 1905–1906, Grammont 1948: 388–390, Miller 1976: 16–18, Nagy 2000: 15 n. 19.



TABLE 7 Odds ratios derived from regrouped data from Table 5

|            | Circumflex vs.<br>acute + grave | Circumflex vs.<br>grave + unac-<br>cented | Circumflex vs.<br>acute + grave +<br>unaccented | Acute vs. grave +<br>unaccented | Acute + grave vs.<br>unaccented |
|------------|---------------------------------|-------------------------------------------|-------------------------------------------------|---------------------------------|---------------------------------|
| Odds ratio | 7.82                            | 8.6                                       | 8.29                                            | 1.22                            | 1.06                            |
| 95% CI     | 2.78–24.25                      | 3.87–19.8                                 | 3.84–18.48                                      | 0.37–3.51                       | 0.37–2.76                       |

TABLE 8 Subtable comparing rate of melismatic setting on acute + grave vs. unaccented VV-syllables

|            | Acute + grave | Unaccented  | SUM           |
|------------|---------------|-------------|---------------|
| Melism     | 8 (~17 %)     | 21 (~16 %)  | 29 (~16.4 %)  |
| Non-melism | 39 (~83 %)    | 109 (~84 %) | 148 (~83.6 %) |
| SUM        | 47            | 130         | 177           |

In autosegmental-metrical terms, the high tone has been deleted or delinked, resulting in tonal-phonetic identity between grave and unaccented syllables. That would motivate grouping grave and unaccented VV-syllables. On the other hypothesis,<sup>30</sup> the grave represents a lowered (downstepped) high tone, i.e., a tonal target that is intermediate between acute and unaccented syllables. This would motivate grouping all three members of the apparent class, i.e., acute, grave, and unaccented VV-syllables, as a kind of continuum of single tones (H, M [Mid], and L).

Table 7 shows that regrouping the grave VV-syllables with acute and/or unaccented VV-syllables does not substantially alter the odds ratios vis-à-vis Table 6. Furthermore, a two-sample test for equality of proportions between acute and grave VV-syllables versus unaccented VV-syllables (regrouped 2 × 2 table given in Table 8) offers strong evidence that no true difference in the rate for melismatic setting for acute and VV-syllables on the one hand and unaccented VV-syllables on the other exists (alternative hypothesis: rate of melismatic setting for acute + grave is greater than rate of melismatic setting for unaccented;

30 For the grave as a lowered/downstepped acute, see especially Devine & Stephens 1994: 180–183. Further, see: Ehrlich 1906: 573f.; Ehrlich 1912: 250–265; Debrunner 1929: 54f.; Sturtevant 1940: 100f.; Winnington-Ingram 1955: 62, 66; Erbse 1960: 377f.; Allen 1973: 245–248; Probert 2003: 16f., 20; 2006: 59; 2010: 2; and Gunkel 2014.

$\chi^2 < 0.001$ ,  $p = 0.5$ ). In short, at least as far as rate of melismatic setting is concerned, the acute, grave, and unaccented syllables appear to constitute a natural class.

Given that the data underlying Table 5 contain some instances of settings that are not wholly secure (see discussion at the end of section 2.1 above), the same analytical procedure was carried out, applying, however, a *more conservative criterion* for the inclusion of tokens. Specifically, if all likely melisms in which one of the pitches is missing, as well as all non-melismatic VV-syllables coded in the corpus with “**supplement** = i(nsecure)” are excluded, the figures given in Table 9 result. Under these criteria, only 44 VV-melisms and 160 non-melismatic VV-syllables are available for analysis, resulting in a lower overall proportion of melisms across the board. Nevertheless, the same general trends outlined above remain in evidence: circumflex syllables are set to melisms at noticeably higher rates than VV-syllables bearing other types of accent. Again, the null hypothesis of independence between accent type and melismatic setting can be rejected ( $\chi^2(3) = 35.219$ ,  $p < 0.001$ ). Likewise, the pairwise comparisons between different accent types in terms of odds ratios (Tables 10 and 11) point to the same tendencies seen above, namely, that the odds of a melism occurring on a circumflex VV-syllable are appreciably higher than the odds of a melism occurring on VV-syllables bearing other types of accent, and differences among non-circumflex syllables appear hardly to exist.

The statistical tendencies regarding the rate of melismatic setting in DAGM 20 and 21 could thus hardly be more robust. First, the likelihood of a VV-syllable being set to a melism is demonstrably not independent of the type of accent on that syllable. Furthermore, we observe that circumflexes are set to melisms noticeably more frequently than VV-syllables with an acute, grave, or no accent. In contrast, the differences in rate of melismatic setting among VV-syllables with an acute or grave and VV-syllables without an accent are negligible. Such a distribution speaks against the inference of multiple pitch targets ([L H]) or even a rising contour ([Ø H]) on acute syllables with long vocalic nuclei: if an acute VV-syllable were indeed associated with a [LH] pitch contour, one should rather have expected for the rate of melismatic settings of acutes more closely to resemble the rate for circumflexes than the rate for VV-syllables without an accent.<sup>31</sup> The tentative conclusion that we draw, pend-

31 Given a total of 32 acute VV-syllables as shown in Table 5, if, for instance, the number of melisms on acutes had been instead 11 (34.4%), this rate would be significantly lower than the rate for circumflex VV-syllables ( $\chi^2 = 4.74$ ,  $p = 0.015$ ; one-way two-sample test for equality of proportions) but significantly higher than for unaccented VV-syllables ( $\chi^2 = 4.29$ ,  $p = 0.0192$ ; one-way two-sample test for equality of proportions); applying Šidák's correction

TABLE 9     Proportion of melismatic and non-melismatic VV-syllables in DAGM 20 and 21 by accent (more restrictive criteria for inclusion)

|            | Circumflex   | Acute        | Unaccented    | Grave       | SUM           |
|------------|--------------|--------------|---------------|-------------|---------------|
| Melism     | 21 (~58.3 %) | 4 (13.3 %)   | 17 (~14 %)    | 2 (~15 %)   | 44 (~21.5 %)  |
| Non-melism | 15 (~41.6 %) | 26 (86.6 %)  | 107 (~86 %)   | 13 (~85 %)  | 161 (~78.5 %) |
| SUM        | 36 (~17.6 %) | 30 (~14.6 %) | 124 (~60.5 %) | 15 (~7.3 %) | 205           |

TABLE 10     Odds ratios derived from Table 9

|            | Circumflex vs. acute | Circumflex vs. grave | Circumflex vs. unaccented | Acute vs. grave | Acute vs. unaccented | Unaccented vs. grave |
|------------|----------------------|----------------------|---------------------------|-----------------|----------------------|----------------------|
| Odds ratio | 8.77                 | 8.72                 | 8.65                      | 1               | 0.97                 | 1.03                 |
| 95% CI     | 2.36–41.95           | 1.61–91.01           | 3.51–22.13                | 0.12–12.42      | 0.21–3.33            | 0.2–10.23            |

TABLE 11     Odds ratios derived from regrouped data from Table 9

|            | Circumflex vs. acute + grave | Circumflex vs. grave + unaccented | Circumflex vs. acute + grave + unaccented | Acute vs. grave + unaccented | Acute + grave vs. unaccented |
|------------|------------------------------|-----------------------------------|-------------------------------------------|------------------------------|------------------------------|
| Odds ratio | 8.81                         | 8.69                              | 8.75                                      | 0.97                         | 0.97                         |
| 95% CI     | 2.78–32.25                   | 3.59–21.77                        | 3.72–21.25                                | 0.22–3.28                    | 0.29–2.82                    |

to the *p*-values given two tests on data from the same (hypothetical) sample still allows the corresponding null hypotheses to be rejected. Such (hypothetical) observed data would then fit with proposal 2 (three-way contrast) shown in Tables 3 and 4, and suggest interpreting the acute as [Ø H]. If the true rate of melisms on acute VV-syllables were in fact 34.4% in the performance of Athenaios and Limenios, it is unlikely (i.e., in about 4 in 100 samples of 32 acute VV-syllables) that only six or fewer melisms would be found in a sample of 32 such syllables ( $p = 0.04233$ , one-way binomial test where  $n(\text{successes}) = 6$ , sample size = 32, likelihood of success = 0.344; 95% confidence interval between 0% and 33.7%); one would have expected to find, at minimum, seven melismatic settings. The data on melismatic settings for acute VV-syllables is thus sufficient to conclude that proposal 2 does not fit the data well; there are simply fewer acute melisms found than would plausibly be expected to occur (at the lower end) if melisms on acute syllables were generated at a rate significantly lower than for circumflex VV-syllables but significantly higher than for unaccented VV-syllables.

ing modeling that allows for comparison between the different levels of accent, is thus that acute VV-syllables host only a single tonal target, likely [H], timed to the entire duration of the syllable. If the assumption that acute accents on long syllabic nuclei contain two pitch targets is to be defended, some further layer of explanation would be required to motivate the remarkably low rate of melismatic setting of such syllables in DAGM 20 and 21.

Before continuing to regression modeling of the relation between accent type and rate of melismatic setting, two further potentially confounding effects on the likelihood of a melismatic setting should be addressed. Given that 53.3% (24/45) of the circumflex settings reported in Table 5 are word-final, it might be possible to attribute the higher rate of melismatic settings of circumflex syllables to effects of pitch and duration in final syllables. Here, two factors might conceivably influence melismatic setting in final syllables, so as to make them more likely: 1) final lengthening; 2) a low boundary tone. Given that melisms are much more frequent on VV-syllables than V-syllables (~26% of all VV-syllables [57/222] vs. ~2% of all V-syllables [11/559] in DAGM 20 and 21), duration (at least phonologically if not necessarily phonetically) evidently impacts the likelihood of melismatic setting. Therefore, if other factors exist that might lead to the (phonetic) prolongation of a syllable, a correspondingly higher likelihood of melismatic settings might be expected, and if one type of accent is significantly more frequent on final syllables, its rate of melismatic setting could consequently prove to be higher.

As concerns final lengthening, cross-linguistically, syllable nuclei in the final syllable of a prosodic word often tend to exhibit greater duration relative to other syllable nuclei within the word, especially at the edge of larger prosodic domains (Lunden 2006, Gordon 2007); given that a melismatic setting naturally involves greater duration, syllables realized as phonetically longer might tend to favor melismatic setting. As regards a low boundary tone, several scholars have hypothesized or argued that a low boundary tone ( $L_\omega$ ) occurs at the edge of every prosodic word in Ancient Greek (Devine & Stephens 1994: 180f., 188f. building on Allen 1973: 246; Gunkel 2023), and many recent analyses of (recessive) accentuation explicitly incorporate it (Ito & Mester 2017, Sandell 2020). Although such a low boundary tone (at the right edge of the prosodic word or any other, larger prosodic unit) has not otherwise been incontrovertibly demonstrated to occur in Ancient Greek, under the assumption that it is present, final VV-syllables might conceivably host two tones (H or L, plus  $L_\omega$ ), thereby increasing the likelihood of a melismatic setting.

To test for potential effects of final lengthening and/or low boundary tone, we should compare the frequency of melismatic settings in final vs. non-final VV-syllables. Since accent type may well condition the likelihood of a

TABLE 12     Distribution of circumflexes by melism and syllable position in DAGM 20 and 21 (cf. Table 5)

|           | Melism     | Non-melism  | SUM        |
|-----------|------------|-------------|------------|
| Final     | 14 (50%)   | 10 (~58.8%) | 24 (53.3%) |
| Non-final | 14 (50%)   | 7 (~41.2%)  | 21 (66.6%) |
| SUM       | 28 (62.2%) | 17 (37.8%)  | 45         |

TABLE 13     Distribution of acute + grave + unaccented VV-syllables by melism and syllable position in DAGM 20 and 21 (cf. Table 5)

|           | Melism      | Non-melism  | SUM         |
|-----------|-------------|-------------|-------------|
| Final     | 10 (~35.5%) | 72 (~48.6%) | 82 (~46.3%) |
| Non-final | 19 (~65.5%) | 76 (~51.4%) | 95 (~53.7%) |
| SUM       | 29 (16.4%)  | 148 (83.6%) | 177         |

melismatic setting, the test should be performed separately on circumflex VV-syllables, on the one hand, and other VV-syllables (acute, grave, unaccented), on the other. For circumflex final syllables, since two tonal targets are plausibly present in any case, a boundary tone would in itself be expected to have little impact, but final lengthening could favor a melismatic setting. However, as reflected in Table 12, the percentage of final circumflexes set to a melism (14/24 = 58.3%) is, in fact, lower than the percentage of non-final circumflexes set to a melism (14/21 = 66.6%); the difference is, however, not statistically significant, suggesting that a circumflex is no more or less likely to be set to a melism if it is the final syllable of a lexical word ( $\chi^2 = 0.071, p = 0.7894$ ).

Table 13 shows the same comparison across all non-circumflex VV-syllables. As was the case for circumflexes, non-circumflex VV-syllables are less often set to a melism in a final syllable (10/82 = ~12.2%) than in a non-final syllable (19/95 = 20%). Although the difference between final and non-final syllables is somewhat larger than for circumflexes, the distribution of melisms by final versus non-final position in the word remains statistically insignificant ( $\chi^2 = 1.43, p = 0.232$ ).

The relative impact of accent type (cf. Tables 9–11) and position in the word (cf. Tables 12 and 13) on the likelihood that a given VV-syllable be set to a melism may also be examined by constructing a logistic regression model. Such a logis-

TABLE 14     Coefficient estimates and significance of Bayesian logistic regression model predicting melismatic setting of a VV-syllable

| Coefficient name    | Estimate | Standard error | z value | p (>  z ) |
|---------------------|----------|----------------|---------|-----------|
| (Intercept)         | -1.39633 | 0.27875        | -5.011  | < 0.001   |
| ACCENT = acute      | -0.03996 | 0.50736        | -0.079  | 0.937     |
| ACCENT = circumflex | 2.13308  | 0.38537        | 5.535   | < 0.001   |
| ACCENT = grave      | -0.04153 | 0.74799        | -0.056  | 0.956     |
| IS FINAL SYLLABLE   | -0.50013 | 0.36435        | -1.373  | 0.17      |

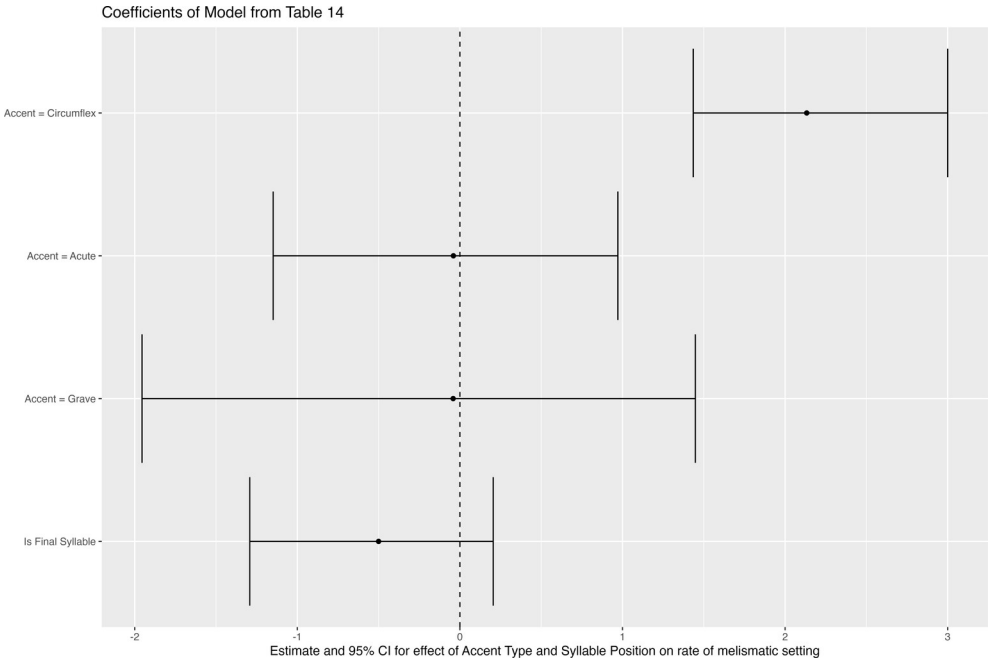


FIGURE 5     95% Confidence intervals for the coefficient estimates from Table 14

tic regression model is a type of classification model that makes predictions as to the likelihood of a given outcome, based on the effects of selected predictor variables, and is thus suitable for testing the effects of those predictor variables (cf. the end of section 2.2 above). To this end, two additional columns were added to a subset of the original database of DAGM 20 and 21 as described in Table 2 above: *is\_melism*, with binary coding (1 = yes, 0 =no) and *is\_final\_syllable*, also with binary coding (1 = yes, 0 = no). The column *is\_melism* then consti-

tutes a binary response variable for the model to be built; as predictor variables, only the binary variable `is_final_syllable` and the factor variable `accent` (with four levels, “unaccented”, “acute”, “circumflex”, and “grave”) were employed; for the latter, the level “unaccented” was set as the reference level (i.e., with treatment contrast comparing all other levels of `ACCENT` to the level “unaccented”). A function call in R to `bayesglm()` from the package *arm* (version 1.31–1, Gelman et al. 2022) was used to generate the model, whose coefficients are shown in Table 13.<sup>32</sup> Note that, since the level “unaccented” constitutes the reference level for the predictor variable `ACCENT`, only coefficients for the levels “acute”, “circumflex”, and “grave” are calculated separately from the intercept. Since the outcomes are assumed to follow a binomial distribution, *z*-tests are used to assess the statistical significance of a coefficient. Figure 5 visualizes 95 % confidence intervals for the estimates of the coefficients (excluding the intercept).

First, there is a statistically significant baseline preference (encompassing unaccented syllables and non-final syllables), for the syllable not to be set to a melism (note the negative estimate for the intercept), which intuitively accords with the fact that melismatic settings are simply proportionally rarer (~21.5 % of all VV-syllables in Table 9). Coefficients with a negative intercept then indicate a decreased likelihood of melismatic setting relative to the baseline, whereas a positive coefficient reflects an increased likelihood. Thus, a syllable with an acute or grave very slightly decreases the likelihood of a melism relative to the base rate, in a way that is statistically not significantly different from the baseline; in other words, VV-syllables with acute or grave accents do not behave differently from unaccented VV-syllables in terms of rate of melismatic setting. Note further that the 95 % confidence intervals for the coefficients shown in Figure 5 span the boundary with 0, and that the confidence intervals for `ACCENT = acute` and `ACCENT = grave` do not or only barely overlap with the 95 % confidence interval for the coefficient estimate of `ACCENT = circumflex`. Likewise, the fact of being a final syllable does not appear to have a significant effect on the rate of melismatic setting; the 95 % confidence interval for `IS_FINAL_SYLLABLE` also crosses 0. Among the individual predictor variables, only `ACCENT = circumflex` achieves a significant effect, and increases the likelihood that a syllable will be set to a melism: its coefficient is positive and does not overlap with 0. A further type II analysis of variance (results reported in Table 15) applied to the logistic regression model confirms that the predictor

32 See further Gelman et al. 2008 and Gelman et al. 2013: §14.8 for background on this Bayesian implementation of generalized linear models through the use of prior distributions imposed on the coefficient estimates (here and below, a Cauchy *t*-distribution with mean = 0).

TABLE 15     ANOVA applied to the model reported in Table 14

| Predictor variable | $\chi^2$ -value | Degrees of freedom | $p$ ( $> \chi$ ) |
|--------------------|-----------------|--------------------|------------------|
| ACCENT             | 36.836          | 3                  | $< 0.001$        |
| IS FINAL SYLLABLE  | 1.973           | 1                  | 0.1601           |

variable ACCENT as a whole has a significant effect on the likelihood of melismatic setting, while the fact of occurring in a final versus non-final syllable does not.

Overall, the data from melisms in VV-syllables suggests that, insofar as occurring word-finally impacts the rate of melismatic setting at all, both accent types are equally affected, and that occurring in a final syllable may somewhat *decrease* the likelihood of being set to a melism, contrary to what the theoretically possible phenomena of final lengthening or boundary tones might predict. However, for present purposes, the major conclusion must be that 1) the accent borne by a VV-syllable significantly affects the likelihood that it will be set to a melism, 2) circumflex VV-syllables are significantly more likely than non-circumflex VV-syllables to be set to a melism, and 3) acute (and grave) VV-syllables are not significantly more (or less) likely than unaccented syllables to condition a melismatic setting. Among the proposals for the representation of the acute on VV-syllables discussed in section 2.2.2, only proposal 3 (acute-unaccented mirror image) predicts that the rate of melismatic setting should pattern in the fashion observed here.

2.3.2     Accent type, melism type, and melism interval

For the 44 instances of melisms on long vocalic nuclei for which the pitch of both notes is attested, measuring the interval of the melism (i.e., the difference between pitch 1 and pitch 2) in Hertz or number of semitones is possible; these tokens correspond to the 44 tokens reported in Table 9 above. The questions with which this section concerns itself are the *direction* (rising or falling) and *size* of interval (in Hertz and semitones) of melisms bearing different types of accent. First, we hypothesize that types of syllables containing multiple phonological pitch targets will exhibit less token variation in the directionality of melism, i.e., a falling tone will consistently map onto a falling melism, while a rising tone (either [L H] or [Ø H]) will consistently map to a rising melism; in other words, opposing tone-tune settings should be rare. In contrast, syllables with only a single or no tonal target will allow for both rising and falling melisms. This overall hypothesis concerns the directionality



of the melism. Second, concerning the size of melisms, we hypothesize that syllables associated with two tonal targets ought to exhibit tendentially larger pitch excursions across a melism than syllables associated with only a single (or no) tonal target(s). In other words, circumflex melisms are predicted to show a larger average distance (measured in Hertz or semitones) between the two pitches constituting the melism in comparison to unaccented VV-syllables; if acute (and grave syllables) contain multiple tonal targets (i.e., acute = [L H]), they ought to behave similarly to the circumflex and differently from unaccented VV-syllables; if acute (and perhaps grave) syllables contain a single tone aligned with the second mora of a VV-syllable (i.e., acute = [Ø H]), the average size of melism should be larger than for unaccented VV-syllables but perhaps not as large as for circumflex VV-syllables.

The data on these 44 melisms in terms of directionality of the melism are summarized in Table 16, with proportions of the column sums reported in the cells. Most immediately striking, in line with what has long been anecdotally observed in the literature on Ancient Greek music, is the fact that circumflex melisms in DAGM 20 and 21 are nearly or perhaps always set to a falling melism (see note a in Table 16 on the single rising melism). On the other hand, acute and unaccented VV-syllables exhibit some variation: both accent types admit of both rising and falling melisms, and, if acute and grave syllables are classed together (given that the latter is an allotone of the former), then acute/grave and unaccented melisms would show approximately equal proportions of rising melisms (about 2/3; 4/6 for acute + grave, 11/17 for unaccented).<sup>33</sup>

Although the small total number of acute ( $N = 4$ ) and grave ( $N = 2$ ) melisms for which both pitches are securely established somewhat limits the confidence with which conclusions can be drawn, the rate at which VV-syllables of

33 Already Winnington-Ingram's (1955) examination of Ancient Greek musical fragments made the observation that acute melisms seem to be set to falling melisms to a surprisingly large extent, under the assumption that an acute VV-syllable bore a rising tone. Specifically, Winnington-Ingram (1955: 64, note 1) writes: "Some philologists tell us that, when a long vowel or diphthong carried an acute accent, there was a rise in pitch from the first to the second *mora*, i.e. that it was the circumflex accent in reverse. We should then expect this rising movement to be reflected in the melody, like the falling movement of the circumflex. A detailed review of the fragments would take up too much space, but it can be briefly stated that, where an acute-accented long syllable or diphthong is set to more than one note, examples of both movements occur, the rising movement being rather more frequent, but not to a degree by which the theory is confirmed. Finally, I would suggest that it might be worth the while of some philologist interested in these matters to look over the musical evidence again." The present study thus makes good on Winnington-Ingram's suggestion.

TABLE 16     Directionality of melism in VV-syllables by accent

|         | Circumflex             | Acute      | Unaccented   | Grave      | SUM          |
|---------|------------------------|------------|--------------|------------|--------------|
| Rising  | 1 <sup>a</sup> (4.8 %) | 2 (50 %)   | 11 (64.7 %)  | 2 (100 %)  | 16 (~36.4 %) |
| Falling | 20 (95.2 %)            | 2 (50 %)   | 6 (35.3 %)   | 0 (0 %)    | 28 (~63.6 %) |
| SUM     | 21 (~47.7 %)           | 4 (~9.1 %) | 17 (~38.6 %) | 2 (~4.5 %) | 44           |

a This single instance of a circumflex syllable set to a rising melism occurs at DAGM 20.18=20 and may be an error. See the apparatus to line 20 in DAGM: 66, as well as West 1992: 291, who considers the conjecture of Reinach (1909–1913: 155), that the *D* is rather an *A*, to be “probable”. This token would thus involve a fall of four semitones, rather than a rise of four semitones (and thus Reinach’s conjecture, if adopted, would not affect the absolute average size of circumflex melisms given below). Compare also DAGM: 72, where the authors more cautiously state “unless the stonemason has made an error, the circumflex is set on a rising interval”. Pöhlmann 1970: 140 adopts a general rule that melisms on circumflex syllables should always be descending.

TABLE 17     Odds ratios for falling melisms on circumflex, acute and unaccented VV-syllables based on Table 16

|            | Circumflex falling<br>vs. acute falling | Circumflex falling vs.<br>unaccented falling | Acute falling vs.<br>unaccented falling |
|------------|-----------------------------------------|----------------------------------------------|-----------------------------------------|
| Odds ratio | 15.87                                   | 32.497                                       | 1.779                                   |
| 95 % CI    | 0.61–1220.46                            | 3.56–1635.66                                 | 0.1–30.66                               |

different accent types are set to rising versus falling melisms is nevertheless significantly affected by the accent of the syllable (see the results of the analysis of variance reported below for the logistic regression model in Table 18). More specifically, a circumflex syllable appears to substantially increase the likelihood that a VV-syllable, if set to a melism, will contain a falling melism; on the other hand, the effect of an acute or grave accent is not statistically distinguishable from the effect of unaccentedness on a melism’s directionality, as shown immediately below. Odds ratios with their 95% confidence intervals given in Table 17 compare the rate of falling melisms between circumflex, acute, and unaccented melisms; perhaps surprising, under the assumption that the acute contains a rising tonal contour, is that the odds of finding a falling melism on an acute VV-syllable are (probably) somewhat higher than for an unaccented VV-syllable.

TABLE 18    Coefficient estimates and significance of Bayesian logistic regression model predicting directionality of a melism on a VV-syllable

| Coefficient name    | Estimate | Standard error | z value | <i>p</i> (>  z ) |
|---------------------|----------|----------------|---------|------------------|
| (Intercept)         | 0.5499   | 0.4784         | 1.150   | 0.2503           |
| ACCENT = acute      | -0.431   | 0.9644         | -0.447  | 0.6549           |
| ACCENT = circumflex | -3.2068  | 0.9516         | -3.37   | < 0.001          |
| ACCENT = grave      | 1.366    | 1.579          | 0.865   | 0.387            |

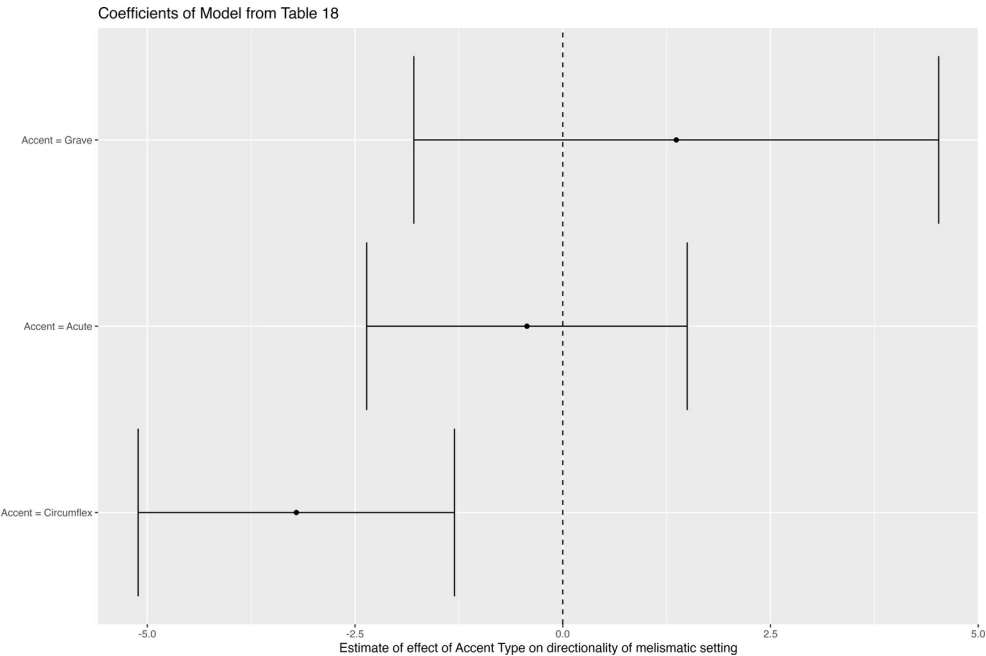


FIGURE 6    95% Confidence intervals for coefficients from Table 18

A logistic regression model constructed in a fashion similar to the model reported in Table 14 above leads to the conclusions just summarized. A response variable **Direction**, with binary coding (0 = falling, 1 = rising), was added to a table containing the size (in Hz and semitones) and accent type of each of the 44 melisms under investigation. As before, the factor **accent** (with four levels, “unaccented”, “acute”, “circumflex”, and “grave” and “unaccented” set as the reference level) was employed as a predictor variable, and the model was generated by a function call to `bayesglm()` in R. The resulting coefficients and

their significance as assessed by *z*-tests are reported in Table 18. A type II analysis of variance indicates that the predictor variable ACCENT as a whole indeed significantly impacts the probability of finding a rising versus a falling melism ( $\chi^2(3) = 21.325, p < 0.001$ ). Figure 6 shows the 95 % confidence intervals for the estimates of the coefficients (excepting the intercept).

In Table 18, the non-significant effect of the intercept indicates lack of a decided baseline preference for either rising or falling melisms; it may be the case that, for melisms without a circumflex accent, the melism's directionality is strongly influenced by the larger musical context (see the discussion in section 2.2.4 below). The relatively large negative coefficient estimate for ACCENT = circumflex, meanwhile, shows that a circumflex accent increases the likelihood of a falling melism to a statistically significant degree; note that the corresponding 95 % confidence lies well below 0. For syllables bearing an acute or grave, on the other hand, no significant effect on the likelihood of generating a rising or falling melism is in evidence, as shown by the relatively smaller coefficient estimates and correspondingly smaller (statistically insignificant) *z* values; the 95 % confidence intervals for both coefficient estimates overlap with the 0 boundary. Moreover, the (small) negative coefficient estimate for the factor ACCENT = acute indicates that a falling melism is slightly more likely to occur on an acute syllable than the baseline (the intercept) alone would predict, and this behavior is precisely contrary to what the hypothesis of a rising tonal contour on acute-accented syllables would predict. Figure 6 shows that the confidence interval for the acute in fact overlaps with the confidence interval for the circumflex: it may be that the true likelihood of finding a falling melism on an acute VV-syllable is more similar to the likelihood of finding a falling melism on a circumflex VV-syllable than the likelihood of finding a falling melism on an unaccented VV-syllable. Such behavior for acute VV-syllables would be comprehensible if the high tone on an acute syllable is timed over the whole syllable, in which case the “pitch-height rule” (especially if the syllable following the acute with a prosodic word is specified as Low) might induce a falling melism in support of a matching tone-tune setting.

Finally, worth considering more closely in light of the small number of acute melisms in this data is the likelihood that the proportionally high number of falling melisms (50 %) might reflect sampling error on a true rate of falling melisms that might be much lower. Keep in mind that an approximately equal number of rising and falling melisms—or better, a proportion of rising and falling melisms statistically indistinguishable from the proportion of rising and falling melisms on unaccented syllables, as shown by the logistic regression model in Table 17—accords with the predictions of the proposal that acute VV-syllables do not contain a rising pitch contour (neither [L H] nor [Ø H]).

A rising tonal contour would be expected to be set to a rising melism at a rate significantly higher than the rate for unaccented VV-syllables, and perhaps not to exhibit such a high level of contrary (in this case, falling) tone-tune mappings. Let us now assume that the true rate for opposing tone-tune mappings in DAGM 20 and 21 is 5 %. This value roughly corresponds to the empirical value of opposing mappings for circumflex VV-syllables in DAGM 20 and 21 ( $N = 21$ ; but cf. note a in Table 16) and is also typologically grounded: in McPherson & Ryan's (2018: 122) survey of rates of opposing settings in tone-tune mappings for nine languages, 5 % opposing settings is the median value.<sup>34</sup> In that case, a one-sided exact binomial test leads to rejection of the null hypothesis that the true rate of opposing (= falling) settings for acute melisms is 5 % or lower ( $p = 0.014$ ; a 95 % confidence interval places the true rate of falling settings between 9.76 % and 100 %).<sup>35</sup> That is, to find even two falling (= opposing) settings of a rising tone, even given a sample size of just four, is a surprisingly large number, if a true rate for opposing settings of 5 % is assumed (and is correspondingly even more surprising, if the true rate of opposing settings in Ancient Greek music is assumed to be even lower, as per note a in Table 16 and note 34).

We now turn to an examination of the size of the intervals of the same 44 melisms. Figure 7 visualizes the direction and size of these melisms by accent type. There, each point represents one of 44 melisms, where the position on the y-axis indicates the size of the melism in Hertz; a greater distance from 0 indicates a larger falling melism (positive values) or larger rising melism (negative values); the size of the points corresponds to the number of melisms having that interval size. Within each subplot corresponding to a different accent type, the black horizontal line shows the unadjusted mean melism interval (i.e., the mean of both rising and falling melisms with their original sign; see Table 19 below), and the vertical colored line extends to within one standard deviation of that mean. This visualization should make evident that circumflex melisms on VV-syllables exhibit a decidedly different distribution in comparison to the

34 The true rate of opposing settings generally in (this genre of) Ancient Greek music may be even lower: considering transitions across syllables within prosodic words, DAGM 20 and 21 contain approximately 300 prosodic words, of which only 2 (cf. footnote 5) set the word's high tone lower than an unaccented syllable (i.e., contain an opposing setting within the word), which would point to a rate of opposing settings of less than 1%. See also Gunkel 2023: 239–244.

35 If the single acute melism on a syllable with a short vocalic nucleus is included (see § 2.2.4 below), which is rising, the sample size rises to 5 and the proportion of opposing settings on acute melisms falls to 40 %. Again, applying a one-way exact binomial test still allows for rejection of the null hypothesis stated above ( $p = 0.0226$ ; 95 % confidence interval between 7.6 % and 100 %), applying Šidák's correction to a significance threshold of 0.05 for two tests (= 0.0253).

other three varieties of accent: with a single exception (discussed in footnote a in Table 16), all circumflex VV-melisms are falling, though only three are especially large (outside one standard deviation from the mean). The melisms set to unaccented, acute, or grave syllables tend, by comparison, to be relatively smaller. Perhaps also remarkable is the relatively wider range of melism direction and size exhibited by unaccented syllables: these cover a span from a rising melism of about 75Hz to a falling melism of about 75Hz. The comparatively smaller range seen among the (few) melisms on acute (largest falling: 25Hz; largest rising: 87Hz) and grave syllables (largest rising: 25Hz) might be interpreted as the effect of a single phonological pitch target that restricts the acceptable range of movement away from the target.

Table 19 reports the mean and median intervals for melisms, both in unadjusted (i.e., with rising melisms as negative values) and absolute values (i.e., having taken the absolute value of all intervals). Overall, the mean absolute interval for a melism on a circumflex syllable (c. 3 semitones) is larger than on acute, unaccented, or grave VV-syllables (1.5–2.25 semitones). Only for the two rising acute melisms does the average interval (3 semitones) compete with the average interval of the circumflex melisms (see footnote 36), but this is due entirely to the relatively large size of a single melism  $e_b-a_b$  of five semitones in the setting of  $\acute{\epsilon}\phi\rho[\acute{o}\acute{\upsilon}\rho\epsilon\iota$  (20.19=22). The noticeably smaller distance from zero for the unadjusted means and medians for acute and unaccented syllables in comparison to circumflex syllables reflects the fact that the latter types contain substantial proportions of both rising and falling melisms.

In order to assess the effect of accent type on the size of intervals in melisms independently of the directionality of melisms, the absolute intervals alone should be compared. Again, a generalized linear model is suitable for this purpose. In this case, the response variable was the absolute value of **Distance\_Hz** (the size of the melism), and **accent** once again served as the only predictor variable, with the level “unaccented” set, as usual, as the reference level. The details of the model generated by a call to `bayesglm()` in R are reported in Table 20; since the response variable in this case is a continuous numeric variable that is assumed to follow a Gaussian distribution, the significance of the coefficients given in Table 20 is determined using a Student’s *t*-test. In this case, only the intercept is significant; none of the other levels of the variable **ACCENT** significantly impact the interval size of a melism, and a two-way ANOVA confirms that **ACCENT** as a whole is not a significant predictor of absolute melism size ( $\chi^2(3) = 2.255$ ,  $p = 0.466$ ). The result in this case is thus effectively null; the relation between accent type and the size of a melism’s interval does not admit of any secure interpretation that supports one phonological interpretation of the accent types over another. One may nonetheless observe that the level

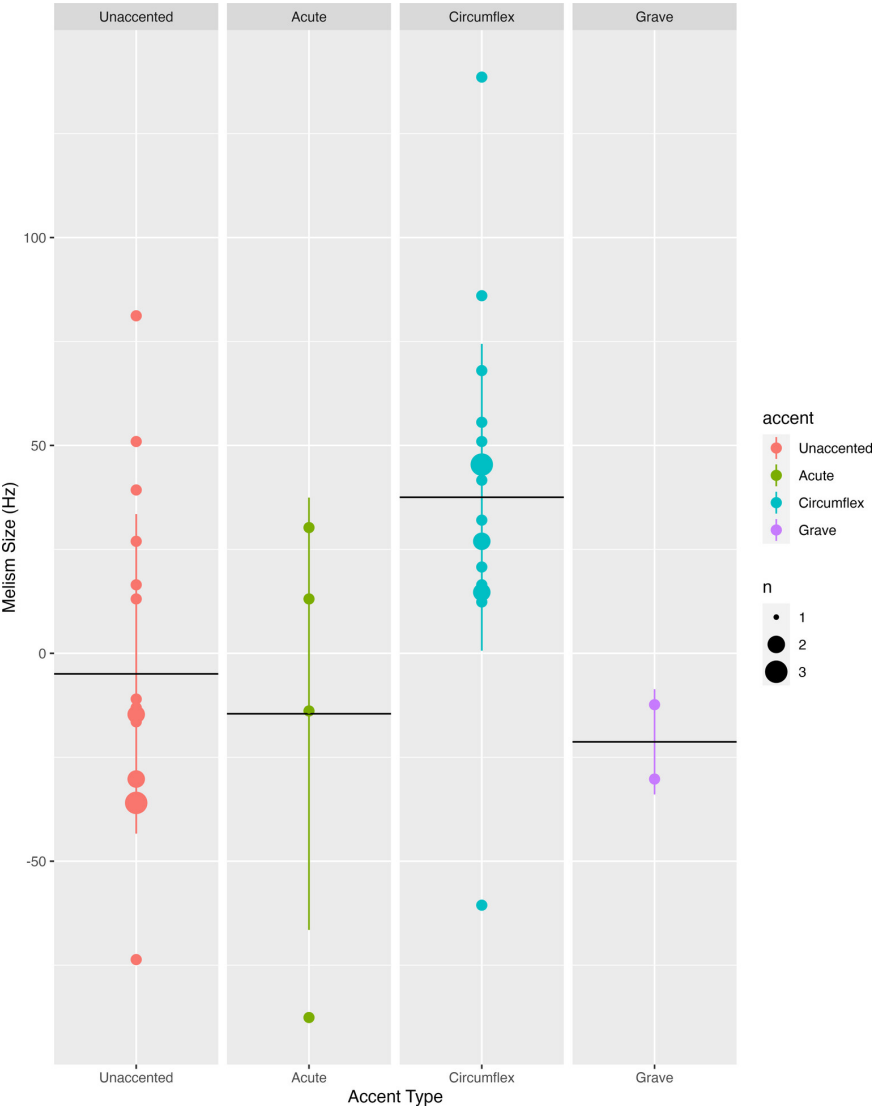


FIGURE 7 Distribution of melism size by accent type (VV-syllables only)

“circumflex” for ACCENT shows the largest coefficient estimate and smallest standard error (besides the intercept); compare the 95% confidence intervals shown in Figure 8. The data thus seem to point in the same direction as the effects found for rate and directionality of melismatic setting: the other three accent types form a natural class against the circumflex. Another study with a larger sample size of melisms might ultimately uncover a significant effect of circumflex accent on interval size in melisms.

TABLE 19     Average interval of melism in VV-Syllables by accent

|                             | Circumflex                  | Acute <sup>36</sup>            | Unaccented <sup>37</sup>         | Grave                         | Acute + grave <sup>38</sup>  |
|-----------------------------|-----------------------------|--------------------------------|----------------------------------|-------------------------------|------------------------------|
| Mean Unad-justed Interval   | 37.53 Hz;<br>2.86 semitones | -14.535 Hz;<br>-0.75 semitones | -4.945882 Hz;<br>-0.17 semitones | -21.295 Hz;<br>-1.5 semitones | -16.78833 Hz;<br>2 semitones |
| Median Unad-justed Interval | 41.63 Hz;<br>3 semitones    | -0.39 Hz;<br>0 semitones       | -14.69 Hz;<br>-1 semitones       | -21.295 Hz;<br>-1.5 semitones | -13.105 Hz;<br>1.5 semitones |
| Mean Absolute Interval      | 43.3 Hz;<br>3.24 semitones  | 36.2 Hz;<br>2.25 semitones     | 31.76 Hz;<br>2.18 semitones      | 21.295 Hz;<br>1.5 semitones   | 28.75 Hz;<br>1.88 Hz         |
| Median Absolute Interval    | 45.39 Hz;<br>3 semitones    | 22.05 Hz;<br>1.5 semitones     | 30.24 Hz;<br>2 semitones         | 21.295 Hz;<br>1.5 semitones   | 21.675 Hz;<br>1.5 Hz         |

Finally, in order to statistically assess the effect of accent type on the size and directionality of a melism in conjunction with one another based on the 44 melisms from Table 16 above, we again generated a generalized linear model (in parallel to the logistic regression and generalized linear models shown in Tables 15, 18, and 20). This model was designed to predict the pitch in Hertz of the second note of a melism (the response variable) based on the pitch in Hertz of the first note and the type of accent of the syllable (the predictor variables,

36     When separated by directionality of melism, the statistics for acute look as follows:

|             | Rising melism ( <i>N</i> = 2) | Falling melism ( <i>N</i> = 2) |
|-------------|-------------------------------|--------------------------------|
| Mean size   | -50.73 Hz; 3 semitones        | 21.66 Hz; 1.5 semitones        |
| Median size | -50.73 Hz; 3 semitones        | 21.66 Hz; 1.5 semitones        |

37     When separated by directionality of melism, the statistics for unaccented syllables look as follows:

|             | Rising melism ( <i>N</i> = 11) | Falling melism ( <i>N</i> = 6) |
|-------------|--------------------------------|--------------------------------|
| Mean size   | -28.36273 Hz; 1.81 semitones   | 37.985 Hz; 2.83 semitones      |
| Median size | -30.24 Hz; 2 semitones         | 33.115 Hz; 2.5 semitones       |

38     When separated by directionality of melism, the statistics for acute and grave taken together look as follows:

|             | Rising melism ( <i>N</i> = 4) | Falling melism ( <i>N</i> = 2) |
|-------------|-------------------------------|--------------------------------|
| Mean size   | -36.0125 Hz; 2.25 semitones   | 21.66 Hz; 1.5 semitones        |
| Median size | -22.05 Hz; 1.5 semitones      | 21.66 Hz; 1.5 semitones        |



TABLE 20    Coefficient estimates and significance of predictors for Bayesian logistic regression model predicting the interval size (in Hz) of melisms on VV-syllables

| Coefficient name    | Estimate | Standard error | <i>t</i> value | <i>p</i> (>   <i>t</i>  ) |
|---------------------|----------|----------------|----------------|---------------------------|
| (Intercept)         | 31.782   | 6.392          | 4.972          | < 0.001                   |
| ACCENT = acute      | 4.323    | 14.556         | 0.292          | 0.768                     |
| ACCENT = circumflex | 11.473   | 8.599          | 1.334          | 0.19                      |
| ACCENT = grave      | -10.095  | 19.409         | -0.52          | 0.606                     |

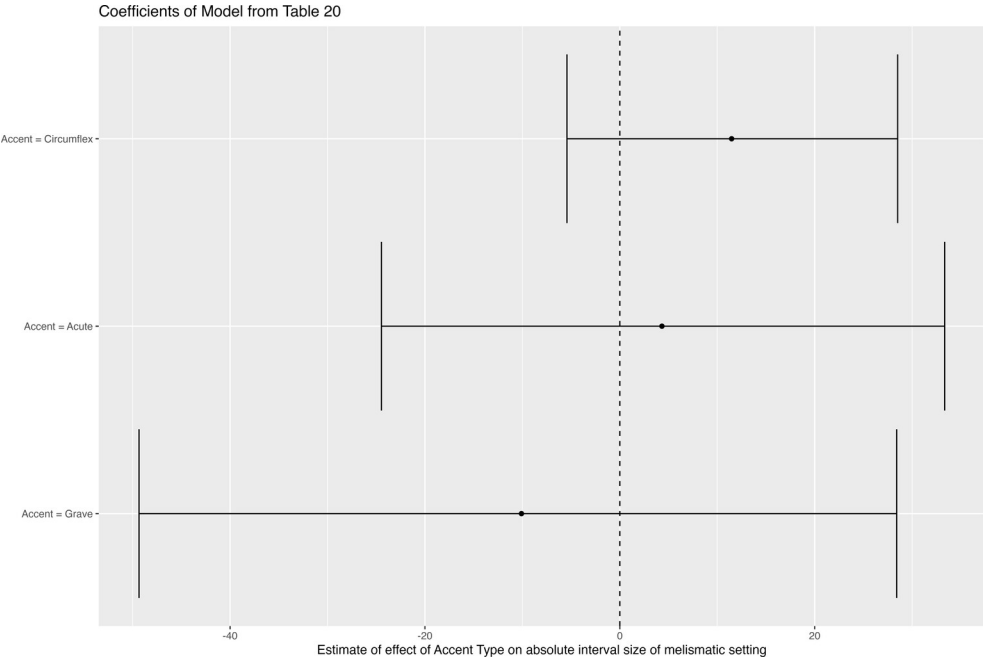


FIGURE 8    95% Confidence intervals for coefficients from Table 20

where the latter, as above, is a factor with four levels). Just as for all other models generated above, “unaccented” was set as the reference level for the variable ACCENT, and the function `bayesglm()` was employed to find the optimal coefficients. Since the response variable in this case is a continuous numeric variable that is assumed to follow a Gaussian distribution, the significance of the coefficients given in Table 21 is determined using a Student’s *t*-test.

As intuitively should be expected, the first pitch in a pair of two pitches constituting a melism plays a decisive role in determining the second pitch; while

TABLE 21    Coefficient estimates and significance of Bayesian generalized linear model predicting pitch (in Hz) of the second pitch

| Coefficient name    | Estimate | Standard error | <i>t</i> value | <i>p</i> (>   <i>t</i>  ) |
|---------------------|----------|----------------|----------------|---------------------------|
| (Intercept)         | 37.5734  | 41.0480        | 0.915          | 0.36563                   |
| FIRST PITCH         | 0.8712   | 0.1575         | 5.532          | < 0.001                   |
| ACCENT = acute      | 9.1977   | 21.2977        | 0.432          | 0.66822                   |
| ACCENT = circumflex | -40.8738 | 12.6751        | -3.225         | < 0.00255                 |
| ACCENT = Grave      | 12.6969  | 28.7894        | 0.441          | 0.66163                   |

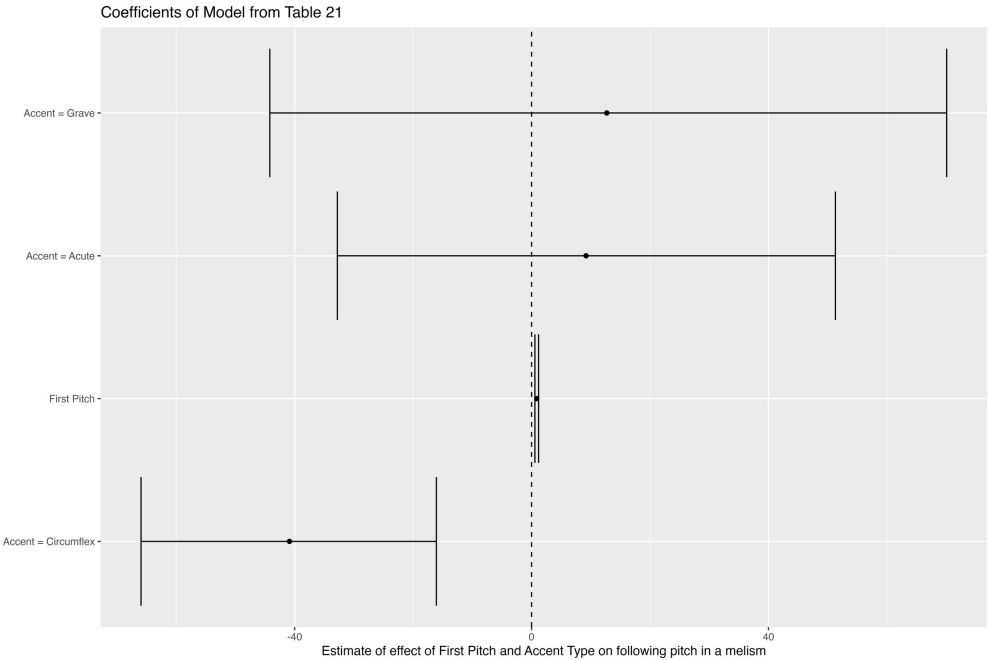


FIGURE 9    95% Confidence intervals for coefficients from Table 21

the estimate for FIRST PITCH is not large, recall here that this coefficient is multiplied by that pitch, and note that its 95% confidence interval does not cross 0. Thus, for example, given a first pitch of 246.94Hz on a syllable with a circumflex, the second pitch is predicted to be 211.8337Hz (= 37.5734 [Intercept] + 246.94×0.8712 + -40.8738 [ACCENT = circumflex]). The estimate for the coefficient of FIRST PITCH is smaller than 1, rather than greater than 1, for the simple reason that falling melisms are more frequent in the data (28/44 =

TABLE 22 ANOVA applied to the model reported in Table 21

| Predictor variable | $\chi^2$ -value | Degrees of freedom | $p (> \chi)$ |
|--------------------|-----------------|--------------------|--------------|
| ACCENT             | 13.858          | 3                  | 0.003        |
| FIRST PITCH        | 30.625          | 1                  | < 0.001      |

~64%). The size of coefficient estimates for the three levels of ACCENT allows for direct comparison and indicates, once again, that the circumflex behaves in a significantly different way from the other three accent varieties. Specifically, a melism set to a circumflex is predicted to fall about 41Hz from the first pitch, just in virtue of being a circumflex. Although a slight rise is associated with acute or grave accents, the small degree makes this effect statistically indistinguishable from the reference level (unaccented), and within the respective 95% confidence intervals, the effect of ACCENT = acute and ACCENT = grave is neither distinguishable from absence of an effect (= indistinguishable from being unaccented) nor clearly distinguishable from the effect of a circumflex (i.e., the confidence intervals overlap). Both predictor variables, FIRST PITCH and ACCENT, have a significant overall effect on determining the second pitch in a melism, as indicated by the type II ANOVA reported in Table 22.

Finally, insofar as acute and grave (if the grave is a downstepped acute, which is uncertain; see footnotes 29 and 30 above) syllables might tend to favor rising melisms, an explanation is available that does not depend upon the assumption of multiple pitch targets: cross-linguistically, H-toned peaks tend to occur late. This phenomenon is known as “peak delay” and has been demonstrated instrumentally for stressed syllables in American English (Silverman & Pierrehumbert 1990), Modern Greek (Arvaniti et al. 1998), and Neapolitan Italian (D’Imperio 2001). Peak delay has also been invoked by Beguš (2016) as an explanation for the fact that the pitch peak of prosodic words in some Vedic Sanskrit dialects was evidently reached after the syllable associated with the phonological accent (i.e., the pitch peak occurred at the beginning of the following syllable, the *svarita*). Given that, as discussed above, the rates for rising and falling melisms do not differ between acute/grave and unaccented VV-syllables, this variation may simply reflect a default preference for rising melisms in absence of a falling pitch contour or result from other factors involved in tone-tune setting (see section 2.3.4).

TABLE 23    Proportion of Melismatic and Non-Melismatic V-Syllables in DAGM 20 and 21 by Accent

|            | Acute         | Unaccented     | Grave        | SUM          |
|------------|---------------|----------------|--------------|--------------|
| Melism     | 1 (~0.67 %)   | 9 (~2.45 %)    | 1 (~2.2 %)   | 11 (~1.9 %)  |
| Non-melism | 148 (~99.3 %) | 357 (~97.55 %) | 43 (~97.8 %) | 548 (98.1 %) |
| SUM        | 149 (~26.7 %) | 366 (~65.5 %)  | 44 (~7.9 %)  | 559          |

2.3.3        Melisms on V-Syllables

In the preceding sections, the examination of melisms was restricted to syllables containing a long vocalic nucleus, for an obvious reason: circumflexes are permitted exclusively on such syllables, and therefore only VV-syllables admit of direct comparison between circumflex melisms and melisms bearing other types of accents. Nonetheless, syllables containing a short vocalic nucleus may be set to a melism, though they are, in the two Delphic hymns considered here, considerably rarer: only 12 are attested. Of those 12, given in Appendix 3, 10 contain an unaccented syllable (of which one is an insecure supplement by the editors of DAGM), one an acute and one a grave; since one of these contains an insecure supplement, we therefore consider only 11 melisms on short vocalic nuclei here. Unsurprisingly, non-melismatic short vocalic nuclei are much more common: excluding cases of insecure supplements, we find, in DAGM 20 and 21, 357 unaccented syllables, 148 acute syllables, and 43 grave syllables (including two cases of a reconstructed grave on a vowel-initial proclitic, following Devine & Stephens 1994: 374–375). The figures for (secure) melismatic and non-melismatic settings of syllables with short vocalic nuclei are provided in Table 23. Between the three categories of accent (acute, unaccented, grave) available for V-syllables, there is no discernible statistically significant effect of accent on rate of melismatic setting, based on Table 23 ( $p = 0.45$ , Fisher’s Exact Test).

In terms of the interval and direction of melisms, the single instances set on an acute or grave syllable, respectively, are rising, by two semitones and one semitone, respectively. In this regard, the two accented examples do not differ whatsoever from the nine unaccented short vowel melisms: all nine of these are rising, on average by 2 semitones (mean = 2, median = 2). These data do not, then, materially alter the picture obtained concerning the interval and direction of melisms based on VV-syllables treated above. If the new data from melisms on short vocalic nuclei are added together with the data on long vocalic nuclei from above, the proportion of rising melisms for both

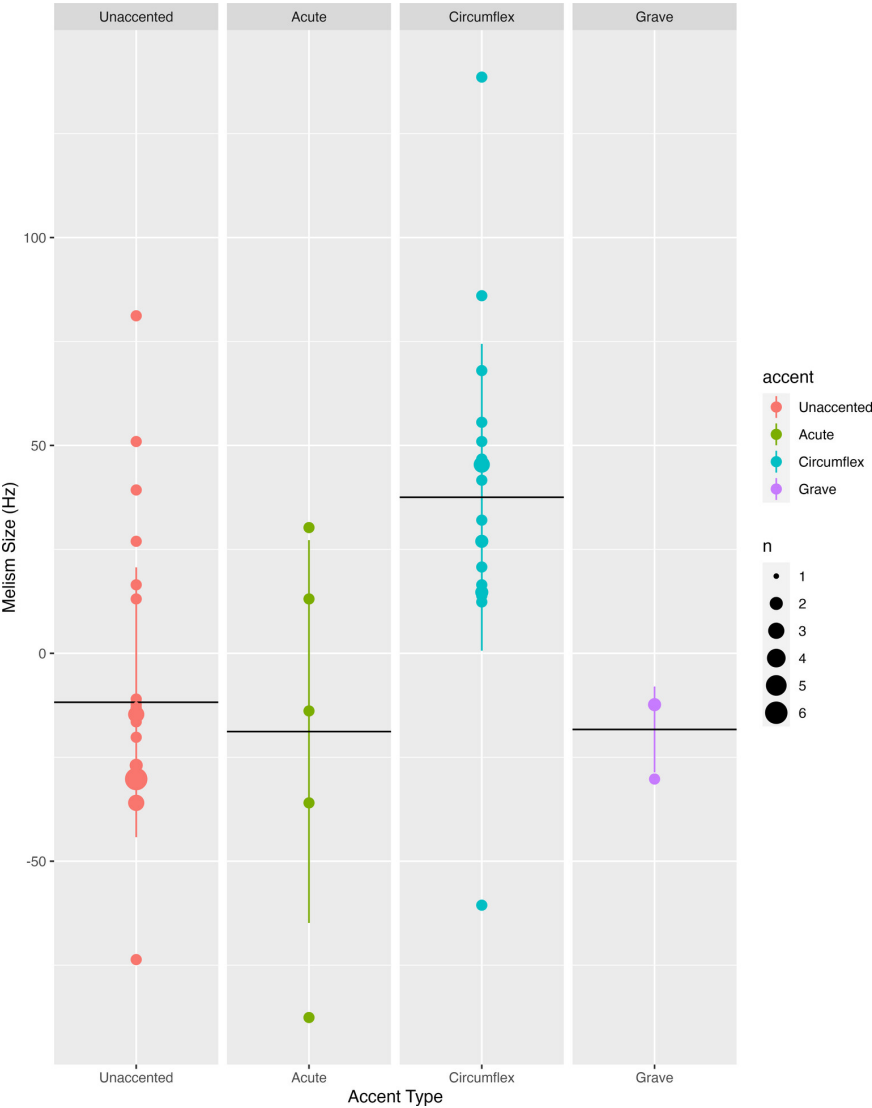


FIGURE 10 Distribution of melism size by accent type (VV- and V-melisms)

acute+grave and unaccented syllables increases, to 75% (6/8) and ~76.9% (20/26) respectively; the difference in proportion of rising and falling melisms on acute/grave versus unaccented syllables, remains, overall, statistically insignificant ( $\chi^2 = \sim 0$ ,  $p = 1$ ). The visualization of the size of melisms in Figure 10, which adds the secure 11 melisms on V-syllables to the 44 melisms on VV-syllables represented in Figure 7 above, shows that the inclusion of the short vowel melisms does not substantially affect the portrait of melisms

on unaccented, acute, and grave syllables. However, through the inclusion of these additional datapoints, the mean melism size for the three non-circumflex melism types becomes more similar (compare the position of the horizontal black lines in Figure 10 with their position in Figure 7).

#### 2.3.4 Why do non-circumflex melisms tend to rise?

The totality of the data inspected and statistically assessed up to this point robustly points to the conclusion that the four accentual categories of Ancient Greek constitute just two natural classes for the purposes of melismatic setting. On the one hand, syllables with a circumflex favor being set to melisms, and typically exhibit a falling melism of about 43Hz (~3 semitones). On the other hand, syllables that are unaccented or bear an acute or grave do not favor melismatic settings; however when set to a melism, a rising melism of about 20Hz in size (~1.5 semitones) would be most typical, though the overall profile exhibits more variation in comparison to the profile of circumflex melisms. That circumflex melisms strongly favor (relatively larger) falling melisms fits hand-in-glove with the universal consensus that the Ancient Greek circumflex exhibited a [HL] pitch contour. However, the fact that non-circumflex melisms are tendentially rising (if to a relatively smaller degree) is perhaps still in need of account.<sup>39</sup>

At the end of section 2.3.2, the phenomenon of peak delay was invoked as a possible grounds for rising melisms on acute (and perhaps grave) syllables. Now we turn to melisms on unaccented VV-syllables and why they tend to rise. As noted in section 2 above, the majority of rising melisms (7/11) on unaccented VV-syllables immediately precede a local peak (acute/circumflex) in the same word, and the majority of falling ones (4/6) immediately follow the local peak in the same word;<sup>40</sup> cf. the setting of the first and last syllable of αἰόλοις in

39 The tendency for all non-circumflex melisms to be set to a rising melism is unlikely to be an accident of sampling. Between melisms on VV- and V-syllables, 34 non-circumflex melisms appear in DAGM 20 and 21, of which 26 are rising (76.5%). Under the assumption that a non-circumflex syllable would have an equal (50%) probability of being set to a rising or falling melism, a two-way exact binomial test indicates that it would then be surprising to find so many rising melisms ( $p = 0.0029$ ; 95% confidence interval between 58.8% and 89.3%).

40 The seven examples of rising melisms on immediately preaccentual unaccented VV-syllables are (Κασταλίδος) εὐ(ύδρου) (i.e., the syllable [seu] of Κασταλίδος εὐύδρου) 20.5=6 f., (Τρι)τω(νίδος) 20.9=10, (βρέμω)ν αἰ(όλοις) 20.12=14, Νη([ρέως]) 21.10=10, (Τρι)τω(νίδος) 21.14 f.=14, (μεγνύμενο)ς αἰ(όλ[οις]) 21.17=16, and Λα(τοῦς) 21.30=26. The four examples of falling melisms on immediately postaccentual unaccented VV-syllables are (ἐριβρό)μου 20.2=2, (αἰό)λοις 20.12=14, (ἄπέσ)τῃ(σας) 20.20=22 f., and (ῥμ)νω 21.2=2.

Figure 3, where preaccentual [a<sub>i</sub>] is set to a rising melism and postaccentual [lo<sub>i</sub>s] is set to a falling one. These eleven peak-adjacent instances of melisms on unaccented VV-syllables find a plausible explanation if we recall the general model of pitch excursion reconstructed for non-oxytone words by Devine & Stephens (1994: 187–189) on the basis of the same musical documents examined here:<sup>41</sup> within the word, pitch rises steadily to the local peak, then falls relatively steeply over the first post-peak mora; the fall usually continues more flatly up to the end of the word, but occasionally ends slightly earlier (see below on “secondary rise”). Given this reconstruction, it is plausible that the eleven peak-adjacent cases just cited reflect the fact that unaccented syllables interpolate with respect to pitch. In other words, if they are located along the preaccentual rising pitch trajectory, they host rising pitch; if they are located along the postaccentual falling pitch trajectory, they host falling pitch. This explanation may be extended to a twelfth case, namely the rising melism *d-e*, on the initial syllable [hi:] of *τεῖς ἀθώπε[υτ]* (20.22=24). Regardless of whether the grave represents a downstepped high tone or a deleted/delinked one (see sections 2 and 2.3.1 above as well as footnotes 35 and 36), [hi:] should be part of the rising pitch trajectory that reaches its peak *a<sub>i</sub>* on the acute-accented syllable: see Figure 11 for the setting *d-e<sub>i</sub>ff a<sub>i</sub> a<sub>i</sub>*.

Three of the remaining five cases involve so-called “secondary rise” (Devine & Stephens 1994: 189–191).<sup>42</sup> As noted in the preceding paragraph, after the peak, pitch typically falls through the end of the word. In the paeans, there are a number of exceptions to this generalization, where pitch falls from the peak to the first post-peak mora/syllable (as usual), but then begins rising again before word-end. In other words, while the turning point between falling and rising tune usually aligns with the end of the acute/circumflex-accented word (plus enclitics, if any), there are a number of exceptions where it occurs one syllable/mora earlier. To cite an example that does *not* involve an unaccented VV-syllable, in the setting of *δεξάμενος ἀμβρόταν d d B c d-e f e* (21.20=18), *δεξάμενος* exhibits secondary rise: the tune falls three semitones (*d B*) from the peak [sá] to post-peak [me], then rises a semitone to [no]—before the end of the word. Descriptively, the rising tune trajectory to the next peak at [bró] begins

41 Note that, in addition to oxytones, Devine & Stephens (1994) exclude “words which have a secondary pitch rise or are likely to be particularly focused and words in initial or final position in a major phrase” (p. 186).

42 The three examples of rising melisms on unaccented VV-syllables that descriptively qualify as secondary rise are (ἐφρ)οῦρεῖ 20.19 f.=21 f., (ἐπέ)ρα(ς ἀσέπτ[ως] 20.23 f.=26, and (ἔ)χεις 21.25=22.

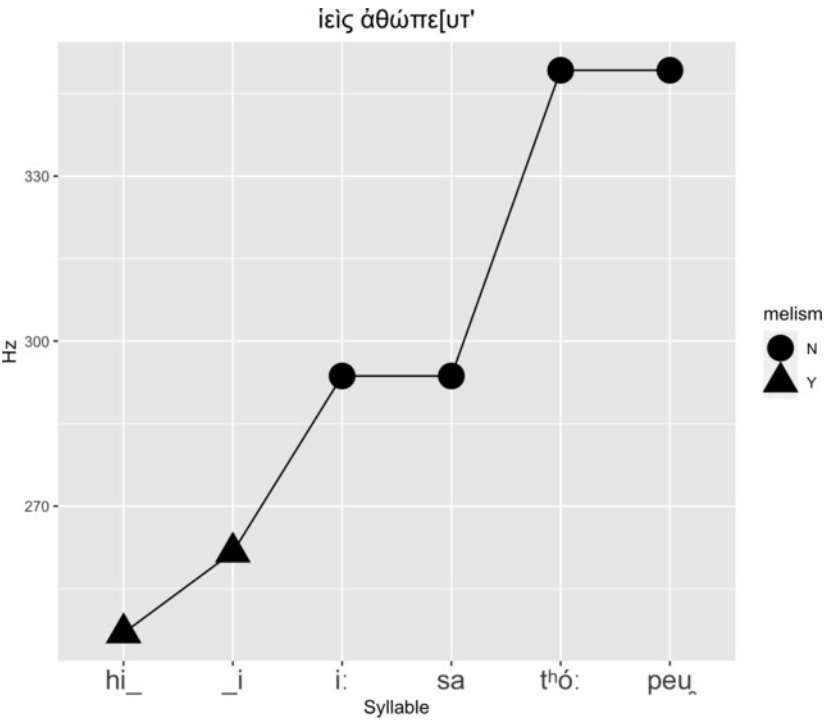


FIGURE 11 Pitch track for ἰεῖς ἄθώπε[υτ] (DAGM 20.22=24)

over the final two morae of the preceding word, specifically with the rising transition over [meno]; see Figure 12.

As pointed out by Devine & Stephens (ibid.), there are at least two good reasons to believe that settings involving secondary rise actually reflect rising pitch trajectories that begin slightly earlier than the usual word-end. First, secondary rise only occurs within the major phrase, never finally in the major phrase, i.e., it appears to be linguistically conditioned. Second, if we set secondary rise aside, relatively secure instances of settings where tone and tune move in opposite directions (i.e., “opposing” settings of rising tone to falling tune, or of falling tone to rising tune; cf. the references in footnote 25) are extremely rare in the paeans; it would thus be surprising if secondary rise involved opposing settings much more frequently. If that is correct, then the setting of those three word-final, unaccented VV-syllables to rising melisms may also be explained as reflecting rising pitch at the beginning of the rising pitch trajectory (cf. the setting of [hi:] in Figure 11). For example, in the setting *fd-efe e* of ἔχεις τρίποδα (21.25=22), the rising melism *d-e* on [k<sup>h</sup>i:s] would reflect the beginning of the trajectory that rises *d-e f* to the peak at [trí]; see Figure 13.



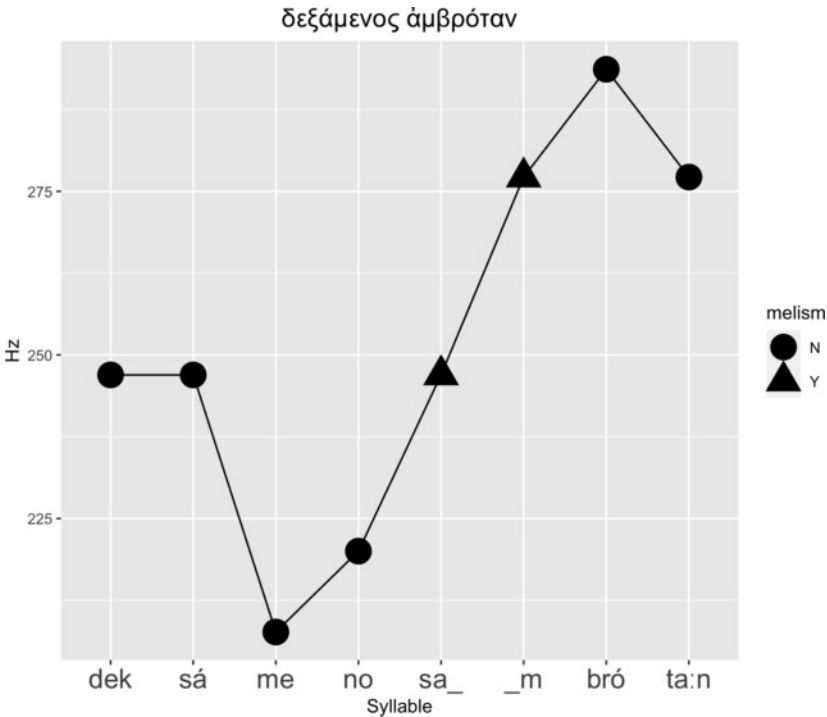


FIGURE 12 Pitch track for δεξάμενος ἀμβρόταν (DAGM 21.20=18)

We are left with two instances of melisms on unaccented VV-syllables that probably should not be explained as reflecting interpolation along the pitch trajectory. Each participates in a setting that is unusual for independent reasons and has received a special explanation in previous scholarship: *δικόρυφον κλειτύν* (21.2=2), which has a falling melism *d-B<sub>6</sub>* on [kli:] that leads into the peak *d* on [tú], may musically express the notion ‘two-peaked’, resulting in the unusual setting;<sup>43</sup> *γαλ[όφωι* (21.14=14), which has a falling melism *f-e* on the syllable [ga:] that leads into the peak *f* on [ló], is said to be set as if it were *γάς λόφωι*, i.e., with “accentual recomposition of compounds” (Devine & Stephens 1994: 167).<sup>44</sup> In sum, the directionality of most (15/17) melisms on unaccented VV-syllables is likely due to interpolation along the pitch trajectory. The fact that more of them rise than fall would simply be because more of them are

43 Thus Reinach (1909–1913: 164, 1926: 18), Pöhlmann (1960: 68), and West (1992: 294), among others.

44 So originally Crusius (1894: 116). See also Devine & Stephens 1994: 350 f. Note that McPherson & Ryan (2018: 133 f.) argue that tone-tune setting in Tommo So is sensitive to both underlying and surface-level tones.

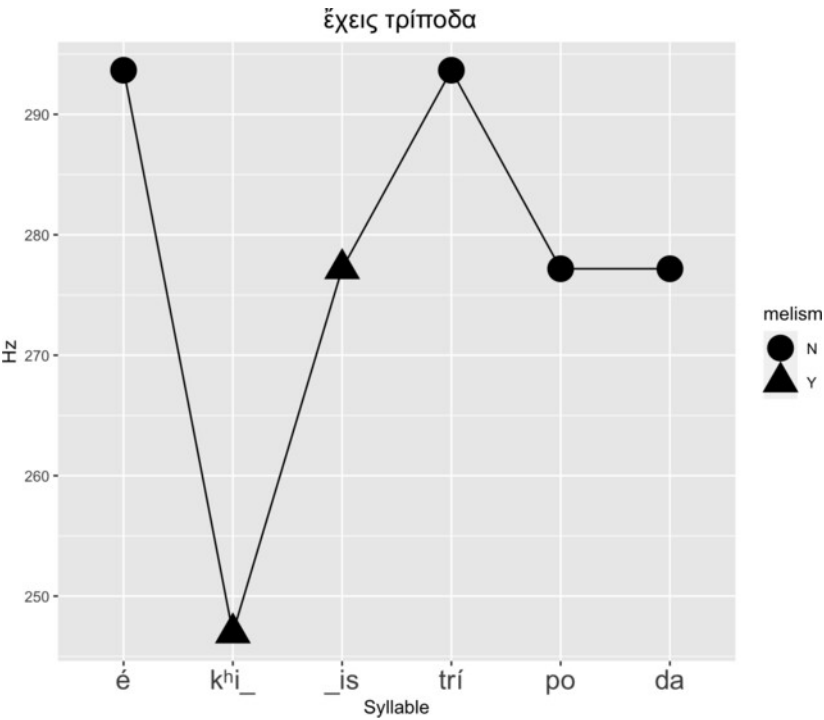


FIGURE 13 Pitch track for ἔχεις τρίποδα (DAGM 21.25=22)

located along the rising trajectory than along the falling one. This is presumably due in part to the fact that the Ancient Greek high tone is restricted to a window consisting of the final two or three syllables of the word, depending on the weight of the final syllable. The statistically significant tendency for non-circumflex melisms to be rising pointed out in footnote 39 may then primarily be attributed to the fact that more unaccented syllables occur before, rather than after, the High tone. In addition to this, as discussed above, grave-accented words form part of the trajectory that rises to the next local peak (acute/circumflex), increasing the length of rising trajectories. Finally, under certain conditions (cf. Devine & Stephens 1994: 191, 355), the turning point between falling and rising pitch may have occurred before the end of the word (secondary rise); this would have further increased the length of rising trajectories and resulted in more rising melisms on unaccented VV-syllables.

In the case of the three secure grave melisms in the corpus (at 20.12 f.=14 f. and 21.9=9), all three are rising and the immediately following pitch is higher, in two instances being an acute syllable. Arguably, the specific graves in all three cases exhibit movement towards the next local peak (i.e., acute or circumflex syllable), as with unaccented melisms that are located along the rising pitch tra-

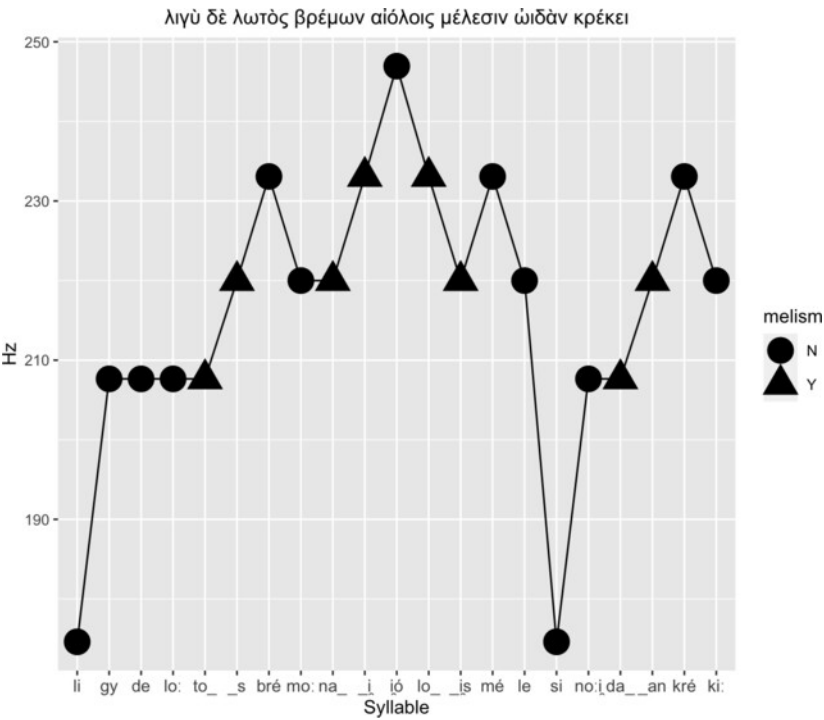


FIGURE 14 Pitch Track for λιγὺ δὲ λωτὸς βρέμων αἰόλοις μέλεσιν ὠιδᾶν κρέκει (DAGM 20.12 f.=14 f.)

jectory. This type of behavior is nicely illustrated in Figure 14, which contains a full sentence in 20.12 f.=14 f., including a total of four melisms (two on graves, two on unaccented syllables). The two rising melisms set on graves here both immediately precede an acute, and of the four acute syllables, three are set at the same height, and one at a yet higher peak, flanked by two melisms on unaccented syllables. Figure 15, containing 21.9=9, shows a setting in which the two acute syllables are set higher than the neighboring unaccented syllables, but the two unaccented syllables [a.el] reach the same height as the earlier acutes (all *f* above middle *c*); the next available pitch is a *d* above middle *c* following four or five missing syllables (including a possible melism on [lô:n], though the note or notes for this syllable are unfortunately not preserved). We venture that the highest pitch peak in 21.9=9 is reached with the circumflex on ἀελλῶν, where the grave melism is used as a turning point from a local trough to rise towards that peak.

Since syllables bearing acute accents usually themselves constitute local pitch peaks, the same account cannot be invoked for explaining the preference for rising melisms among this group. For the five acute melisms, a preference for

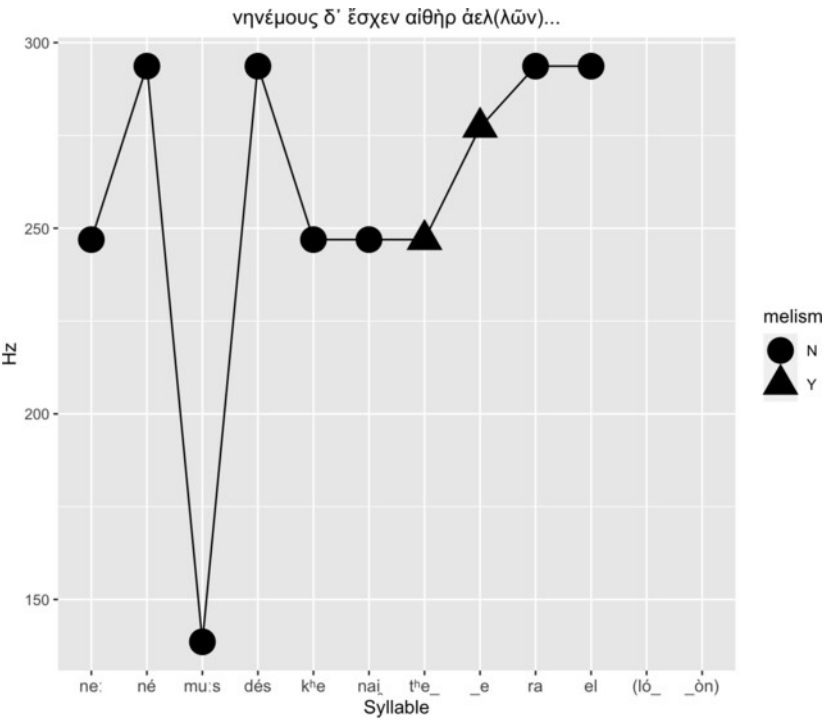


FIGURE 15 Pitch Track for νηνέμους δ' ἔσχεν αἰθήρ ἀελ(λῶν) ... (DAGM 21.9=9)

setting a rise might be justified by reference to peak delay, as discussed above. From that point of view, the two falling acute melisms would be exceptional, for which reason those two examples, at DAGM 20.10=12 and DAGM 21.1=1, merit closer inspection.<sup>45</sup> At 20.10=12, the word τάρων ‘bull:GEN.PL’ constitutes the last word in the sentence beginning with ἁγίοις ‘sacred:DAT.PL’, where the fall in pitch may be conditioned by a fall at the end of the sentence: musically, the final word of the sentence is often equally low or lower than any other pitch in the sentence, and typologically, a low boundary tone might be expected there. In short, the fact that this particular acute occurs so close to the end of a sentence may be the motivation behind a falling melism; see Figure 16. Arguably, a single (rather than double) tonal target on acute syllables would permit greater adjustments in peak timing, e.g., a relatively early realization of the peak in τάρων to avoid “crowding” with a sentence-final Low.

45 Devine & Stephens (1994: 193 f.) consider whether acutes set to falling melisms would have been distinct from circumflexes and speculate that there may have been a difference involving pitch movement over a following sonorant.

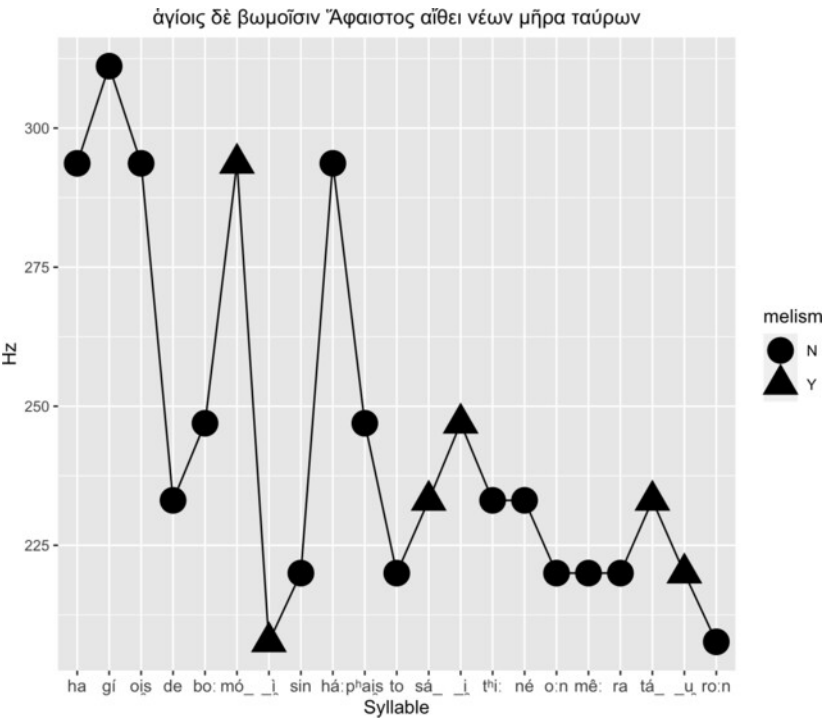


FIGURE 16 Pitch Track for ἁγίοις δὲ βωμοῖσιν Ἄφαιστος αἶθει νέων μῆρα ταύρων (DAGM 20.9–11=10–12)

Meanwhile, the falling acute melism on a VV-syllable found at DAGM 21.1 does not find any further motivation, except that it fits within the downward-sloping trajectory of the paeon's opening, "descending firmly" (West 1992: 294) towards a local trough on the syllable Παρ-; see Figure 17. Conceivably, larger musical considerations may have influenced the particular tone-tune mapping in this instance. The larger point is that acute syllables, when set to melisms, seem more likely to be influenced by considerations of surrounding pitches (preference for fall at the end of a sentence in 20.9–11=10–12, sensitivity to the larger musical context in 21.1). Such flexibility stands in stark contrast to circumflex melisms, which are virtually categorically falling. That is, the falling *tone* of circumflex melisms essentially demands a corresponding fall in *tune*; on the other hand, where *tone* is neutral (certainly in the case of unaccented syllables, *ex hypothesi* for acute and grave syllables), more flexibility in terms of *tune* is admitted.

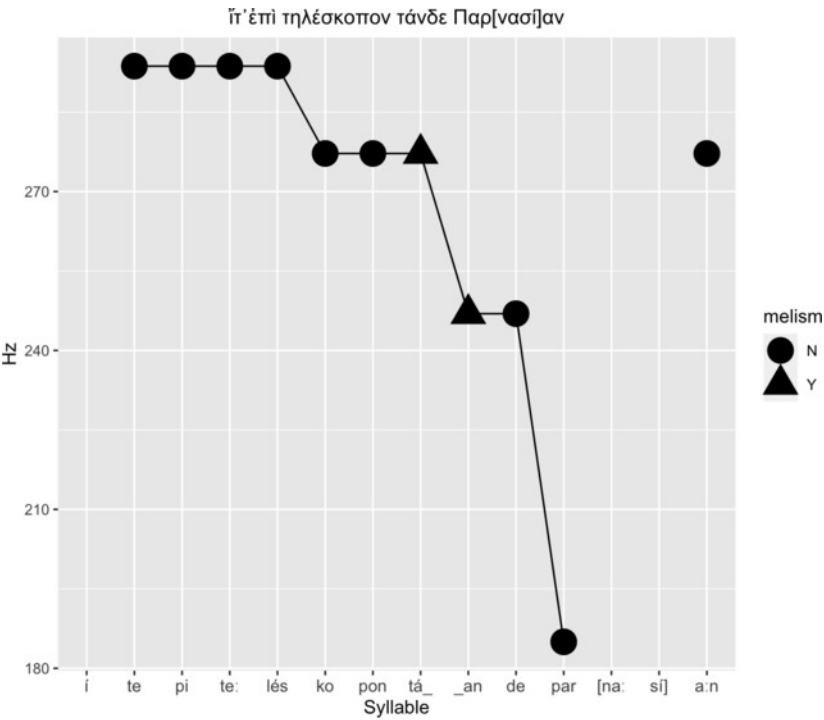


FIGURE 17 Pitch Track for ἵτ' ἐπὶ τηλέσκοπον τάνδε Παρ[νασί]αν (DAGM 21.1)

2.4 Summary of statistical analysis of melisms in DAGM 20 and 21

We may now take stock of how the specific proposals concerning the surface phonological realizations of different accent types on VV-syllables in Ancient Greek outlined in Table 3 above, given the hypotheses about the predicted behavior of melisms ascribed to them in Table 4, have fared in light of the analysis just undertaken. The compatibility of each of the three proposals considered with the three parameters of *rate*, *directionality*, and *size* for melismatic settings in DAGM 20 and 21 assessed above is summarized in Table 24.

First, in terms of melismatic setting, recall that proposal 1 (acute-circumflex mirror image) expects that acute and circumflex VV-syllables should behave similarly or in fully parallel fashions, while proposal 2 expects that acute VV-syllables should behave in a fashion distinct from either circumflex or unaccented VV-syllables, and proposal 3 expects that circumflex VV-syllables should behave distinctly from acute and unaccented syllables, which are expected to exhibit similar patterns. With regard to the parameter of *rate*, the data leave no doubt that acute, grave, and unaccented VV-syllables form a natural class: all three accent types are set melisms at a similarly low rate; in contrast circumflex VV-syllable are set to melisms are a statistically significantly higher rate.

This portrait fits only with proposal three. Importantly, the quantity of the data available for assessing this parameter is sufficiently large and the tendencies sufficiently robust that there is no room for reasoned doubt.

Along the parameter of directionality, we again find that circumflex VV-syllables form one group (~95% falling melisms), while acute and unaccented syllables comprise another group (~50%–66% rising melisms). The proportion of rising acute melisms (50% rising on VV-syllables, 60% overall, if the single melism on a V-syllable is included) is much too low to fit proposal 1 (for which a proportion of melisms closer to 95% might be expected), but also too similar to the rate of rising melisms on unaccented VV-syllables (~66%) to be compatible with proposal 2; only proposal 3 correctly predicts that the patterning of acute VV-syllables should be indistinguishable from the patterning of unaccented VV-syllables. That grave melisms seem exceptionlessly to rise can plausibly be ascribed to interpolation: grave syllables are all located along the pitch trajectory that rises to a local peak. While the amount of data on acute VV-melisms for assessing directionality is small, and is thus compatible with a wider range of outcomes, the specific sample found in DAGM 20 and 21 is significantly different from circumflex VV-syllables, but not from unaccented VV-syllables. We therefore conclude that the data on this parameter are most compatible with proposal 3, and speak against proposal 2 without fully contradicting it.

Finally, the parameter of *size* delivers the least conclusive information overall. Again, in part because the total sample size is relatively small, no statistically significant difference in average size between the types of accent clearly emerges. However, the tendencies in the data are again most compatible with proposal 3 and least compatible with proposal 1: circumflex melisms show the largest mean absolute pitch excursions (43.3 Hz, vs. 36.2 Hz for acute and 31.76 Hz for unaccented melisms; cf. Table 19), and the difference in average pitch excursion between circumflex and acute melisms (7.1 Hz) is larger than the difference in average pitch excursion between acute and unaccented melisms (4.44 Hz). Such a configuration fits best with proposal 3, though proposal 2 (and even proposal 1) cannot be said with great confidence to make fully incorrect predictions. However, when size is considered in combination with directionality, for the purposes of predicting the second pitch of a melism based on its first pitch and accent, the binary distinction between circumflex on the one hand versus acute, grave, and unaccented, on the other, robustly reemerges (cf. Table 21). For this reason, the available data on average interval size lends itself, with the above caveats, to the interpretation that it, too, best fits proposal 3.

In short, proposal 3 evidently wins in a direct comparison: its predictions are strongly compatible—to a high degree of statistical confidence—with the

TABLE 24    Summary of compatibility of proposals from Table 3 with parameters of melismatic setting in DAGM 20 and 21

| Proposal                                            | Rate of melismatic setting | Directionality of melisms | Average interval size of melisms |
|-----------------------------------------------------|----------------------------|---------------------------|----------------------------------|
| 1. Acute-circumflex mirror image (VV-acute = [L H]) | ×                          | ×                         | × (?)                            |
| 2. Three-way contrast (VV-acute = [∅ H])            | ×                          | × (?)                     | ??                               |
| 3. Acute-unaccented mirror image (VV-acute = [H])   | ✓                          | ✓                         | ?                                |

empirical findings regarding rate and directionality, and are best compatible (though with lower confidence) with the parameter of size. In contrast, proposal 1 is clearly incompatible with the findings for rate and directionality, and appears, on balance, to fit the findings for size comparatively poorly. Finally, proposal 2 fails to find any support along the parameter of rate, is clearly less compatible than proposal 3 with the data on directionality, and matches the data on interval size less well than does proposal 3. This overall set of findings and their interpretation thus suggest that the surface representation shown for an acute on a VV-syllable given in Figure 3 above (i.e., a single high tone linked to both morae or the entire duration of a syllable) stands the greatest chance of being correct.

3        Acute = H: typological and analytical considerations

With the above empirical findings now in hand, we turn to further considerations that might offer support for the view of the Ancient Greek acute on a VV-syllable as a level high tone. These considerations concern, on the one hand, the cross-linguistic typology of contour tones (in both canonical tone languages and “pitch-accent” or restricted-tone languages), and on the other, the Greek-internal behavior of accentuation where enclitics are involved. The thrust of the two succeeding sections may be distilled as follows. First, since finding a surface contrast between surface rising and falling tones without a contrastive level high tone on a given syllable type is comparatively uncom-



mon, whereas a contrast between falling and level high tone comparatively common, it is *a priori* more likely that a given language possess a surface [H] vs. [H L] contrast than a [L H] vs. [H L] ( $\sim[\emptyset H]$  vs. [H  $\emptyset$ ]) contrast (section 3.1) on a given syllable type. Second, to assume that an acute on a VV-syllable in Ancient Greek represents a level high tone poses no obstacles to the analysis of enclitic accentuation, and may, depending upon how larger facts of Ancient Greek word prosody are to be analyzed, be preferable to a rising [L H] ( $\sim[\emptyset H]$ ) representation of the acute (section 3.2). Finally, equipped with further reasons to believe, in accord with the empirical results of section 2, that an Ancient Greek acute on a VV-syllable represents a level high tone, we discuss plausible underlying representations able to derive the surface acute–circumflex contrast in section 3.3, in the context of broader facts about Ancient Greek word prosody and the data from typologically similar word prosodic systems in West and North Germanic languages.

### 3.1 *The typology of contour tones and contour tone distribution*<sup>46</sup>

The autosegmental–metrical representations of circumflex and acute shown at the outset in (1) and (2) constitute, on the surface, types of contour tones, insofar as more than one pitch target is contained within a single syllable. If indeed those representations (or parallel underspecified representations [ $\emptyset H$ ] and [H  $\emptyset$ ]) are correct, they imply that accented syllables containing two vocalic nuclei in Ancient Greek knew a two-way contrast between rising tone (acute) and falling tone (circumflex). This section attempts to answer the question, on the basis of the typology of syllable-level pitch contours, of whether, all things being equal, to encounter a language with a surface contrast [L H] ( $\sim[\emptyset H]$ ) vs. [H L] ( $\sim[H \emptyset]$ ) on VV-syllables, as per proposals 1 and 2 above, is more or less likely than encountering a language with a surface contrast [H] vs. [H L], as per proposal 3. The basic question is: do there exist languages that

46 We use the term “contour tone” here to mean either a phonologically unitary tone possessing multiple targets, or a sequence of level tone targets occurring within the same syllable. The definition of contour tone employed here is thus the occurrence of multiple tonal targets within a syllable, regardless of whether those tonal targets may be phonologically decomposed or not, and regardless of whether the syllable or mora is the relevant TBU for a given language. Yip (2002: 28) considers the “true contour tone languages” to be just those languages in which the contours constitute “single phonological objects” or languages in which multiple tones are permitted to attach to a single TBU. If the TBU of Ancient Greek is indeed the mora, since its falling (and possible rising) tone can potentially be analyzed as sequences of level tones (or privative  $\emptyset$  versus H), it would not belong to Yip’s class of “true contour tone languages”; again, however, we speak of surface sequences of tonal targets contained within a single syllable as contour tones.

exclusively allow surface contour tones ([H L] and [L H]) on a given syllable type (especially VV), and if so, how common are they relative to languages that *also* allow level High tones on that syllable type, or only one of the contour tones? To this end, we examine the typology of syllable-level pitch excursions in three different groups of languages: first, some genetically closely related Baltic and Slavic languages, which have been claimed to exhibit only [L H] and [H L] tonal contours on VV- or VR-syllables, and where similar conditions going back to a common ancestor shared with Ancient Greek (Proto-Indo-European) are partly responsible for the occurrence of the tonal patterns (section 3.1.1); second, “pitch accent” or “restricted tone” systems in both non-Indo-European and Germanic languages with historically unrelated word-level tone (section 3.1.2); third, canonical tone languages (section 3.1.3). The evidence across all three of these groups suggests that finding a two-way surface contrast between [H L] (~ [H Ø]) and [L H] (~ [Ø H]) on a given syllable type, as represented in (1) and (2), without also permitting simple level [H], is typologically more marked, though certainly not excluded.

In short, we find evidence for two related typological generalizations: 1) if a language allows contour tones on a given syllable type at all, it more likely allows falling rather than rising contours; 2) if a language permits both rising and falling contours on a given syllable type, it probably permits level high tone on that same syllable type as well. The first of these generalizations is not novel and conforms to well-known markedness hierarchies regarding contour tone (Gordon 2001, Yip 2002: 27–30). The second may be novel—we have not encountered explicit statements to this effect in the literature—but fits with the results of our survey; confirmation of the strength of this generalization must await a fuller investigation.

Before proceeding, to prevent any misunderstanding of the claims and aims of this section, we should first make explicit that we do not wish to argue against the existence of languages that exhibit *only* a surface distinction between contour tones on certain syllable types without *also* permitting a level high tone on those same syllable types. The distribution of [H], [H L], and [L H] in Hup (see Epps 2008: 86–97 and below) may constitute one such case, though the overall situation and distribution of tones is not readily comparable to what is found in Ancient Greek (in contrast to what has been claimed for Lithuanian and varieties of Serbo-Croatian). Perhaps one of the clearest such cases, though, is found in the Nagasaki variety of Japanese (Uwano 1999, Uwano 2007, Uwano 2012, Kubozono 2012, Kubozono 2017), illustrated in (4). As in many varieties of Japanese, High tone is culminative but not obligatory, and in Nagasaki, both an accented class (Type A, with High tone, in (4a)) and unaccented class (no high tone, in (4b)) exists, and within the accented class (in contrast to the

Tōkyō variety), the position of the High is easily predictable: it regularly occurs on the peninitial mora, except in bimoraic words (see (4a.i.)) which may be on a short vowel (as in (4a.ii)), on part of a long vowel (as in (4a.iv)), or part of a syllable closed by a nasal (as in (4a.iii) or (4a.v)).<sup>47</sup> Thus, on VR-syllables (= VV-syllables and VN-syllables) of Nagasaki Japanese, only level low tone, else a rising or falling contour, but no level high tone, may occur; level high tones are found exclusively on syllables with short vowels. Acute on a vowel or nasal indicates that that segment bears High tone, while all other segments are low toned.

- (4) Nagasaki Japanese: [HL] and [LH] but no [H] on VR-Syllables (Kubozono 2017: 2)
- a. **Type A (high tone present, assigned to peninitial or initial mora):**
- i. *á.me* ‘candy’ (level high tone on V-syllable)
  - ii. *o.ná.go* ‘woman’ (level high tone on V-syllable)
  - iii. *koñ.saa.to* ‘concert’ (rising tone on VN-syllable)
  - iv. *paá.tii* ‘party’ (rising tone on VV-syllable)
  - v. *o.rén.zi* ‘orange’ (falling tone on VN-syllable)
- b. **Type B (no high tone):**
- i. *a.me* ‘rain’
  - ii. *o.to.ko* ‘man’
  - iii. *a.sa.ga.o* ‘morning glory’

As the further discussion under section 3.1.2 shows, the distribution of tone on VR-syllables in Nagasaki Japanese seems, however, to be rather exceptional, both among varieties of Japanese (where it may be unique) and among languages with restricted tone systems or persistent word-level pitch accent (where it constitutes a comparatively rare option). The comparative infrequency of the Nagasaki-type distributional patterns plausibly derives from the relative markedness of autosegmental tonal configurations, where level tones are less marked than falling tones, which are in turn less marked than rising tones or tones with both a rise and fall (Yip 2002: 27–30) and a preference for high tone to be associated with a prosodically prominent position, namely, with

47 Nagasaki thus belongs to the group of dialects exhibiting so-called *N*-pattern systems, in which only two different tonal patterns occur in prosodic words (Uwano 1999). The dialect of Kagoshima, for instance, is related to Nagasaki in this regard, though it also differs considerably, by treating the syllable as the TBU and having an obligatory High in every word. Many other dialects, such as those of Tōkyō, Kyōto, or Ōsaka, exhibit multiple different patterns in the position of High tone among accented words (Kubozono 2012: 1397–1398).

stressed syllables (de Lacy 2002) and with the first mora internal to a syllable (Hyman 2007: 10–12).

At this juncture, worth observing is that Nagasaki Japanese's word prosody evidently differs from the Ancient Greek system in at least one crucial respect: in the assignment of default ("recessive") High tone in Ancient Greek, the pattern of heavy and light syllables at the right edge of the word is crucial (hence the analysis of Golston 1990, which relies on moraic trochees, followed, e.g., by Gunkel 2011), not the number of vocalic morae. In the cases in which the Nagasaki-type distributional pattern arises, high tone appears to be predictably assigned by count of licit tone-bearing morae from the left or right edge of the word. As best as we are able to establish at this time, Nagasaki-like syllable level-pitch patterns seem to come about principally in languages that count tone-bearing morae in assigning tone; whether this generalization strictly holds true must await more systematic typological investigation. Finally, while the survey presented here reveals a number of instances from a diverse array of languages (Japanese, Bantu, Cushitic) in which potential syllable-level rising tones are repaired to level high tones, we have uncovered but one case in which the reverse obtains (i.e., a potential falling tone created by H-assignment is repaired to a level high tone, while rising tone remains as such, in Nyamwezi (Bantu); cf. Odden & Marlo 2019: 157).

### 3.1.1 Word prosody in Serbo-Croatian and Lithuanian

The word prosodic systems and varieties of "pitch accent" often claimed to exist in Lithuanian and varieties of Serbo-Croatian have often been adduced as a parallel for the Ancient Greek situation (see already Jakobson 1931 [2002] and 1937 [2002]: 264, further Devine & Stephens 1994: 206–207, and Jasanoff 2017: 33; cf. also Stang 1966: 125–126 for a summary of traditional views on the properties of "acute" and "circumflex" intonations in various Baltic and Slavic languages). Specifically, these languages have been claimed to show a binary contrast between rising and falling tone on VV-syllables, without any level High tone on that syllable type, much like proposals 1 and 2 for the prosody of Ancient Greek. Indeed, traditional descriptions of pitch contours on VV-syllables in these languages (e.g., Kurschat 1876: 58–60; Belić 1909, Stang 1957: 20–22) seem to have informed Jakobson's (1937) analysis of the Ancient Greek acute on VV-syllables as a rising tone. To preview the conclusions of the discussion here: improved phonological description of what entities stand behind the labels "rising" and "falling" (or "circumflex" and "acute", as also traditionally employed for Lithuanian),<sup>48</sup> as well as better instrumental measurements of

48 Note that the terms "circumflex" and "acute" with respect to tonal events in Lithuanian are

syllable prosodies, clearly exclude in some cases and make doubtful in others that these languages possess any syllable-level rising tones whatsoever.

First, in the description of Neo-Štokavian Serbo-Croatian (Standard Serbian), although the terms “falling” and “rising” are in use, they in fact refer to pitch contours that typically span two syllables, not a pitch contour contained within the domain of a single syllable. Zsiga & Zec (2012) have argued, on the basis of production experiments, that these “falling” and “rising” contours describe the alignment of a single lexical High tone relative to stress, and that, in general, the peak of a “rising” High tone is simply phonologically aligned later, namely, on the syllable following the stress (see also Zec & Zsiga 2022).<sup>49</sup> In accord with this analysis, stressed monosyllables permit only the “falling” accent, since here the lexical H and stressed syllable are necessarily aligned. Important for present purposes is simply that Neo-Štokavian evidently has no syllable-level contour tones.

In Čakavian dialects of Serbo-Croatian, meanwhile, the situation is somewhat more complex, and offers a clear potential parallel for Ancient Greek, insofar as the domain for pitch changes is in fact a single syllable. Traditional description of most varieties refers to three accents: the short falling accent (˘), the long falling accent (˘˘) and the long rising accent (ˊ), where the lattermost is sometimes termed “Čakavian acute”. The “short falling” accent clearly reflects High tone aligned with stress on a syllable containing a short vowel, while the properties of the long rising accent (Čakavian acute) have remained unclear. One of the earliest descriptions, Belić 1909 (on the dialect of Novi Vinodolski), holds that the long rising accent is characterized by first a level or falling pitch, followed by a sharp rise, with increasing pitch and loudness in the time course of the vowel. Virtually all attempts at description of the Čakavian acute since the 1970s, whether primarily impressionistic (Houtzagers 1982: 126 on the dialects of Orlec; Budovskaja & Houtzager 1994: 96 on the dialect of Kali; Kalsbeek 1998: 39 on the dialect of Orbanići) or grounded in acoustic measurement (Lehiste & Ivić 1973, dialect of Jelenje; Neweklowsky 1978: 97–102 on the dialect of Nikitsch [Burgenland]), disagree with Belić’s description: the three impressionistic studies characterize the Čakavian acute as lacking

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used in a sense precisely opposite of their (commonly presumed) values in Ancient Greek: the Lithuanian “circumflex” is traditionally described as a rising tone, while the “acute” is in turn described as a falling tone.

49 In terms of phonetic implementation, some peak anticipation tends to occur, such that the F<sub>0</sub> peak in disyllables in fact tends to be found at the end of the stressed (initial) syllable, rather than the peninitial syllable with which the H tone is phonologically (lexically, in Zsiga & Zec’s view) aligned.

in any change of pitch or intensity, while Lehiste and Ivić found an overall rising-falling contour for both the “long falling” and “long rising” accents, distinguished only by the peak of F<sub>0</sub>; Neweklowsky’s speaker exhibited primarily a difference in pitch level between falling and rising accents, with higher average F<sub>0</sub> in the latter. However, most decisive is the detailed acoustic data collected from seven different Čakavian dialects by Langston (2006: 30–70; see especially pp. 40–42), whose evidence unambiguously points to a distinction between a level High tone and a falling (H L) tone on stressed syllables containing long vowels.<sup>50</sup> Čakavian would thus appear to accord with the broader typological generalizations seen below for other “pitch-accent” and canonical tone languages, namely, that rising contour tones occur less commonly and, at least in the canonical tone languages, presuppose the presence of level tones.

As far as Lithuanian is concerned, exactly what acoustic correlates reliably distinguish the “acute” accent (traditionally described as falling tone) and “circumflex” accent (traditionally described as rising tone) remains, unfortunately, unclear. Instrumental measurements reported in Pakerys 1982, Dogil 1999: 278–284, and Kelly 2013 point to considerable interspeaker variation, though they agree on one point: F<sub>0</sub> contours do not reliably distinguish the two accent types. Dogil, for instance, based on his own data and that in Pakerys 1982, finds that vowel spectral quality, particularly in the first half of a VR-syllable, provides the clearest distinction between acute and circumflex; in other words, the acute vowel is “hyperarticulated”. Kushnir (2018: 33–34), in his overview on the phonetics and phonology of Lithuanian prosody, similarly concludes that “greater energy or length” (perhaps a combination of duration, intensity and vowel spectral quality) on either the first or second component of a long vowel or diphthong is the primary distinguishing feature of “acute” and “circumflex” syllables. Whatever the correct phonetic and phonological characterization of the Lithuanian contrast between “acute” and “circumflex” VR-syllables may be, no convincing evidence for treating this distinction as a difference in rising versus falling F<sub>0</sub> contours is available.

50 Langston himself is ultimately somewhat cautious in interpreting his results with respect to the earliest descriptions, allowing that an earlier rising tone may have become a level high tone. However, no positive evidence exists to suggest that the properties of the “long rising” accent underwent a change during the course of the 20th century.

### 3.1.2 Word prosody in other “pitch-accent” and “restricted-tone” languages

In this section, we examine the distribution of syllable-level tone in four language groups (Japanese, Basque, Bantu, and Germanic), and then present results of a further survey based primarily on the examination of the restricted-tone languages mentioned in Hyman 2009. Of real interest are only those languages in which contour tones (rising, falling, or both) occur at all, and especially the languages in which the combination of syllable prosody and the rules for assignment of high tone could at least theoretically generate both rising and falling contours. Throughout, the crucial question is not what underlying tones or pitch accents are available, but rather what sort of contours concretely arise through the association of those tones or pitch accents to licit tone bearing units.

#### 3.1.2.1 *Japanese*

As a specific example of the typologically more marked status of rising tones, consider the survey of Japanese dialects presented in Kubozono 2017. Across these closely related languages, a pitch accent ( $H^*$  or  $H^*L$ ) may be associated with either the first or the second mora of a long vowel; the variety of Kagoshima seems to be unique among varieties of Japanese in treating the syllable as the TBU. Across the vast majority of varieties, if the second mora of a syllable is underlyingly linked to  $H^*$  or assigned  $H^*$  by some rule, the surface result is not a contour tone,  $[LH^*...]$ , but either a falling tone  $[H^*L...]$ , created by shift of the accent or tone, or level  $[H^*...]$  spanning two morae, created by the spreading of accent or tone over both morae in a VR-syllable. In other words, VR-syllables in nearly all Japanese varieties—with the notable exception of Nagasaki Japanese (see (4) above)—tolerate only level tones and falling tones, but not rising tones. Notably, this avoidance of rising tones holds across both *N*-pattern varieties, which show only two distinct tonal patterns (such as Nagasaki), and multiple-pattern varieties (such as Tōkyō).

Example (5) illustrates high-tone spreading within the syllable in an *N*-pattern dialect comparable to Nagasaki Japanese, the endangered Taira dialect of Koshikijima (see Kubozono 2016 for a description). As with the Nagasaki data seen in (4), Taira has just two distinct word-level accent patterns: in Taira, Type A assigns  $H^*$  to the penultimate mora, while Type B assigns  $H^*$  to the final mora. However, in Type A words, if the final is monomoraic and the penult is a VR-syllable, the penult shows a level high tone, not a rising tone; similarly, in Type B words, in case the final syllable is a VR-syllable, the result is never a rising tone, but rather a level high tone.

- (5) No [L H\*], only [H\*] in the Taira Dialect of Koshikijima Japanese (Kubozono 2017: 5–6)
- a. Type A (H\* associates to the penultimate mora, level [H] surfaces)
    - i. /pañ.tu/ → *páñ.tu* ‘pants’
    - ii. /puú.ru/ → *púú.ru* ‘swimming pool’
    - iii. /raí.to/ → *ráí.to* ‘right; light’
  - b. Type B (H\* associates to the final mora, level [H] surfaces)
    - i. /iń/ → *íń* ‘dog’
    - ii. /mi.kań/ → *mí.káń* ‘orange’
    - iii. /sen.seí/ → *sen.séí* ‘teacher’

Examples (6) and (7) illustrate similar sorts of repairs in the Tōkyō dialect. In (6), a word-initial boundary tone /%LH/ regularly deletes its L component in words whose first syllable is bimoraic (a VV- or VR-nucleus), as in examples b. and c.; following the boundary tone, subsequent syllables remain high-toned until a H\* is reached. In (7), loanwords, which often preferentially assign H\* to the antepenultimate mora, will retract the pitch accent by one further mora in order to produce a falling, rather than rising, tone (cf. also Kubozono 1989 on this point). Conversely, falling tones are never repaired via high-tone shift (which would create rising tones) in any dialects, though they may, in rare instances, be repaired by spreading to produce level tones. In short, syllable-level rising tones are almost categorically avoided in varieties of Japanese: only in Nagasaki Japanese (unique in McCawley’s (1978) typology of Japanese dialects for being a “mora-counting, mora language”) are they permitted.

- (6) Boundary Tone /%L H/ → [%H] in Tōkyō Japanese
- a. [gàk.kó:] ‘school’
  - b. [tó:.kjo:] ‘Tōkyō’, not <sup>X</sup>[tǒ:.kjo:]
  - c. [káí.dan] ‘staircase’, not <sup>X</sup>[kǎí.dan]
- (7) Retraction of H\* from Antepenultimate to Preantepenultimate Mora
- a. [róñ.don] ‘London’, not <sup>X</sup>[rǒń.don]
  - b. [sái.da:] ‘cider’, not <sup>X</sup>[sǎi.da:]

### 3.1.2.2 Basque

Word-level prosodies reminiscent of Japanese and Ancient Greek are found in two major dialect areas of Basque: Northern Bizkaian Basque (e.g., Gernika, Lekeitio) and Western Navarrese Basque (e.g., Goizueta, Leitza). The “pitch-accent” systems of the former group share with Tōkyō Japanese a broad lexical



distinction between “accented” and “unaccented” words, and a culminative but not obligatory High tone within a certain prosodic domain (usually labeled the “accentual phrase”); see Elordieta 1997, Hualde 1999, Hualde et al. 2002, or Gussenhoven 2004: Ch. 9 for details. In the latter group, word prosody appears more comparable to many North Germanic systems, with an obligatory and culminative word prominence, which may be associated with different word-level pitch accents; the word-prosodic features of this dialect group are surveyed in Hualde 2007 and Hualde 2012.

In Northern Bizkaian dialects, an “accented” accentual phrase contains a /H\*L/ pitch accent; the TBU is the syllable but where, exactly, H\* associates varies by dialect (e.g., leftmost accented syllable in Gernika, penult in Lekeitio); in “unaccented” accentual phrases, H\* associates to the final syllable. In all cases, L docks to the immediately following syllable. The /%LH/ boundary tone at the left edge of an accentual phrase likewise consistently assigns its two tones to two adjacent syllables. In Western Navarrese dialects, meanwhile, a lexical contrast exists between words with a /H\*L/ pitch accent and words with a /L\*/ pitch accent. As in the Northern Bizkaian dialects, the /H\*L/ pitch accent is usually associated with two adjacent syllables, but may produce a falling contour tone on a monosyllabic word.

For present purposes, the behavior of pitch accents and boundary tones in Northern Bizkaian and Western Navarrese Basque thus yields little in the way of further insights, since contour tones arise only rarely. Most syllables bear but a single pitch target, and since they only contain a single TBU, may be said to constitute level tones, even if phonetic rises or falls within a syllable may occur (see pitch tracks in the references cited above). That the only syllable-level contour tone attested seems to be a falling tone may be regarded as an accident of the fact that the only complex pitch accent in both varieties is /H\*L/, though the fact that this pitch accent itself consists in a falling, rather than rising, contour might be interpreted as consistent with the greater markedness of rising contours generally.

### 3.1.2.3 *Bantu*

The Bantu language family has provided a rich domain for the theoretical investigation of phonological processes involving tone. Many Bantu languages may be correctly characterized as canonical tone languages (see section 3.1.3 below), contrasting at least lexically specific High /H/ and Low /L/ level tones, though underlying contrastive contour tones occur as well. In part because of the rich agglutinative morphology typical of these languages, processes involving the reassociation, spreading, or adjustment of tones are common. For general overviews of the phonology of tone in Bantu languages, see Odden 1995, Yip

2002: Ch. 6, or Odden & Marlo 2019; collections of papers dealing with specific cases are found in Hyman 1976 and Clements & Goldsmith 1984. Data from Bantu languages also substantially informs the generalizations on tone rules drawn in Hyman & Schuh 1974 and Hyman 2007.

Of particular interest for present purposes is the fact that several Bantu languages, including but not limited to Kinga (Schadeberg 1973; cf. Hyman 2006: 241–242), Luganda (Hyman & Katamba 1993), Safwa (Voorhoeve 1973; Hyman 2006: 241), and Tonga (Goldsmith 1984), exhibit obligatory and culminative high tone, much as is the case for an orthotonic word without enclitics in Ancient Greek. In a perhaps even greater number, including Shona, Haya, and Xhosa, the surface distinction between [H] and [L] may be analyzed as an underlying privative distinction between /H/ and /Ø/ (cf. Hyman 2001). Furthermore, in languages such as Kinga, the position of a high tone in the word is, at least within certain lexical classes (e.g., nominals), entirely predictable with respect to word edge and syllable and/or mora-count. These properties may, as illustrated below, interact to create cases in which only contour tones might be able to arise on VV-syllables—at least in theory.

For in fact, although languages like Kinga possess mora-counting rules of H-assignment that would seem to lead to only rising and falling contour tones on VV-syllables, Odden and Marlo (2019: 157) make the following telling observation with respect to word prosody in Bantu languages generally: “Quite frequently, expected rising tones systematically become level H.” Kinga nicely instantiates this phenomenon. Minimal prosodic words of Kinga (Bantu; Schadeberg 1973; cf. Hyman 2006: 241–242, from which the examples in (8) below derive) exhibit obligatory and culminative H tone. High tone (indicated with an acute) in nominals is most frequently assigned to the antepenultimate vocalic mora, as illustrated in (8), though may also occur on the first vowel to the left of the stem or (rarely) on the penultimate mora (Schadeberg 1973: 24–27). Consequently, when the final syllable contains but a single vocalic mora, H is assigned in a way that could yield falling contours [H Ø] on long-voweled penults (as in (8c)), and in case both the final and penult contain a short vowel, given that the antepenult contains a long vowel, H is assigned in a way that the VV-syllable in the antepenult could host a rising contour [Ø H] (as in (8e)). However, Schadeberg (1973: 46) mentions a rule that transforms “any low-high rising sequence to a level high tone”. Although not explicitly illustrated in his presentation of the data, Schadeberg’s rule makes the situation in Kinga quite like the situation illustrated for Tōkyō Japanese above: when high-tone assignment would result in a rising contour, the rising contour is repaired to a level high tone.

- (8) Kinga: H assigned to the antepenultimate vocalic mora, but syllable-level [Ø H] becomes [H]
- |                   |                   |                     |
|-------------------|-------------------|---------------------|
| a./ókóheka/       | → [ókóheka]       | ‘to laugh’          |
| b./ókóvala/       | → [ókóvala]       | ‘to count’          |
| c./ókugéenda/     | → [ókugéenda]     | ‘to go’             |
| d./ókugeendélela/ | → [ókugeendélela] | ‘to walk around’    |
| e./ókohwáánana/   | → [ókohwáánana]   | ‘to become similar’ |

The situation in Safwa appears similar to that of Kinga, with the further caveat that Voorhoeve (1973: fn. 2) is not entirely certain as to what the main phonetic correlates of (syllable-level) prosodic prominence are, although he provides an analysis in terms of tone. As with Schadeberg’s treatment of Kinga, Voorhoeve writes VV̇ sequences (e.g., /uhajeéndile/ ‘you walked’, with H assigned to the antepenultimate vocalic mora), but neither he nor Odden (1988) explicitly characterize such /VV̇/ sequences as having a rising contour on the surface. To assume that Safwa possesses only a surface [H L] vs. [L H] contrast on VV-syllables is not warranted without more precise description.

Tonga, meanwhile, under the analysis of Goldsmith 1984, is characterized as possessing a consistent /HL\*/ prominence tone that appears in the neighborhood of every “accented” vowel. In nominals, four major distinct patterns are observed, according to the position and number of “accented” vowels in the stem (prefixes appear to generally be unaccented). When the stem is disyllabic, accent may occur on either the first or the second stem syllable. In the former case, as in (9a), H spreads leftwards and L\* spreads rightwards; in the latter case, L\* associates to the final syllable, and H spreads leftwards, as in (9b). A stem may also be unaccented altogether, in which case all syllables receive L tone, as in (9c). Most complex is the case in (9d), where both the first and last syllable of the stem are accented: without any prefixes, the expected falling tone on the first syllable is simplified to L\*; with a prefix, there are enough hosts, in principle for all tones, but because H tones spread leftward, the antepenultimate syllable would potentially become rising, but the creation of the rising tone is likewise avoided, and one instead finds a downstepped level H on the antepenult and penult as a consequence of the now floating L tone.

- (9) Disyllabic and Trisyllabic Nominal Stems in Tonga (\* following a vowel indicates an accented vowel)
- |                          |              |                 |
|--------------------------|--------------|-----------------|
| a. Class A: /i-ma-ka*ni/ | → [ímákàni]  | ‘news, affairs’ |
| b. Class B: /i-mu-sune*/ | → [ímúsúnè]  | ‘ox’            |
| c. Class C: /i-ma-tongo/ | → [ìmàtòngò] | ‘ruins’         |

## d. Class D

- i. /a\*cisya\*/ → [àcísyà] ‘uncle’
- ii. /ba-a\*cisya\*/ → [bá'ácísyà]

Other Bantu languages that do allow for surface contour tones, in accord with Odden & Marlo's (2019) generalization, typically present only falling, never rising contours. This is the case, for instance, in Haya (Hyman & Byarushengo 1984) and Luganda (Hyman & Katamba 1993). Zulu comes close to this portrait as well, inasmuch as a V-syllable associated with a /LH/ sequence “has a rising pitch level which levels out to a level high pitch in some instances depending on the rate of speech” (Laughren 1984: 213); V-syllables in Zulu do, in any case, admit of both surface [H] and [HL] as well. One clear example of rising tone avoidance in a Bantu language with fewer restrictions on the occurrence of High tone comes from Ekegusii: “There are no Rising tones in Ekegusii. Thus, when a process such as gliding and compensatory lengthening occur [*sic*], the resulting syllable surfaces with a level High tone and not a Rising tone” (Bickmore 1999: 112).

Thus, Bantu languages—both of the canonical tone and restricted tone varieties—frequently avoid the creation of rising tones, and in languages in which rising tones do occur, both level High and falling tones are admitted on the same types of syllables that allow for rising tones. One partial exception, mentioned in Odden & Marlo (2019: 157), is found in Nyamwezi, where potential falling tones are repaired to level H, but expected rising /LH/ faithfully surfaces as such. We have not, though, in our survey of the literature, found an unambiguous case of a Bantu language in which exclusively rising and falling tones occur on some given syllable type. Given the prosodic diversity within the language family, such a language, like Nagasaki Japanese, may well exist, but would seem to be in the minority.

3.1.2.4 *Germanic*

Many varieties of North Germanic languages (Swedish, Norwegian) exhibit consistent word-level pitch accents and also exhibit a contrast between two different word-level pitch patterns (in the literature commonly labeled “accent 1” and “accent 2”). A similar situation obtains in some West Germanic varieties, specifically in some Riparian, Moselle Franconian, and Limburgian dialects spoken in western Germany, Belgium, and the southern Netherlands. The historical development of the contrasts in tonal accent in North and West Germanic must be independent, and the exact patterns of pitch excursion and their phonological analysis vary from variety to variety. In both the North and West Germanic languages, the particular lexical prosodic properties may be fur-

ther impacted by information structure (e.g., whether a word is under focus or not), clause type (e.g., declarative vs. interrogative), and sentence position (utterance-final position or not). For overviews of these word-prosodic systems, see Köhnlein 2020 or Riad & Peters 2021. As generally representative of the syllable-level  $F_0$  contours that may or may not occur in these varieties, we consider primarily the situation in Central Swedish, as analyzed by Riad (2012, 2014), and Cologne German, as analyzed by Gussenhoven & Peters 2004 and Peters 2006.

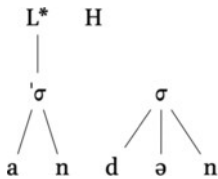
In the many varieties of Swedish and Norwegian with distinctive word-level pitch patterns, the stressed syllable,  $\tau$ , is the tone bearing unit (cf. Gussenhoven 2004: 211–212): that is, the single stressed syllable of the word strictly associates with the starred tone, while any other non-boundary tones float before or after the starred tone, thereby impacting the word's overall  $F_0$  contour, without being strictly aligned to any other prosodic unit (mora or syllable). Under Riad's privative tonal analysis of Central Swedish, lexical items belonging to the “accent 1” group (e.g., /'and-/ ‘duck’) are toneless, while lexical items belonging to the “accent 2” group have a lexical / $H^*$ / tone, perhaps underlyingly linked to the stressed syllable (e.g., /'andə-/ ‘spirit’). In all prosodic words without a focal accent, the intonation / $H+L(^*)$ / is present, with the result that words with accent 1 have Low tone on their stressed syllable, with preceding High (when the stress is not on the initial syllable), while words with accent 2 have High tone on their stressed syllable, trailed by Low pitch (i.e., the / $H$ / of / $H L^*$ / is deleted on account of the OBLIGATORY CONTOUR PRINCIPLE, and / $L^*$ / is blocked from associating to the stressed syllable by the lexical / $H^*$ /, though it is not deleted). When a word receives a focal accent, then a / $L^*H$ / intonation appears instead; this focal intonation is a rising contour, but in words with accent 1, only the  $L^*$  portion actually associates to the stressed syllable, as shown by variability in the timing of the realization of the  $F_0$  peak from the  $H$  in utterances of different lengths (Riad 2014: 187, citing Bruce 1987).<sup>51</sup> In words with accent 2, the lexical  $H^*$  remains associated with the stressed syllable, while / $L(^*)+H$ / trails afterwards, producing a “two-peaked”  $F_0$  contour when accent 2 words have a focal accent. In short, insofar as the situation in Central Swedish is representative of North Germanic word prosody generally—which appears to be the case, since dialectal differences appear to consist mainly in their intonational lexicon, not in the TBU or the way in which tones associate to it—there are no proper syllable-level contour tones at all, only level  $L^*$  and  $H^*$ .

51 Certain varieties, rather than exhibiting a categorical contrast in type of pitch accent depending upon the prominence of the word, instead show differences in scaling of the same pitch accent (cf. Riad 2021: 273).

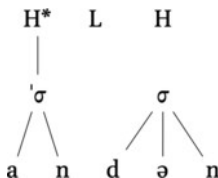
Examples (10) and (11) provide autosegmental diagrams for the tone associations in the definite forms of ‘duck’ [¹and-ən] and ‘spirit’ [²andə-n] in their focal accent (10) and word accent (11) forms. Again, while up to three tones are present within the domain of the prosodic word, only a single tone is properly associated to a TBU, 'σ, at all. When realized in isolation, the word receives the focal accent and also a L% boundary tone corresponding to the end of an intonational phrase.

(10) Focal Accent in Central Swedish

a. Accent 1 [¹and-ən] ‘duck-the’

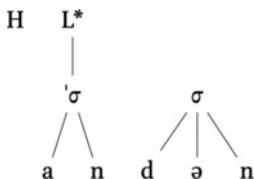


b. Accent 2 [²andə-n] ‘spirit-the’

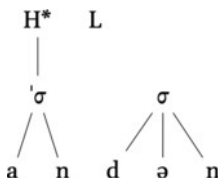


(11) Word Accent in Central Swedish

a. Accent 1 [¹and-ən] ‘duck-the’



b. Accent 2 [²andə-n] ‘spirit-the’



As far as word prosody in Central and South Low Franconian varieties is concerned, there exists a similar two-way distinction between lexical items with

distinct pitch patterns. For the present purpose of investigating the extent to which only contour tones occur over syllables, most relevant are dialects in which the TBU appears to be the mora (or alternatively, permits contour tones on syllables whatsoever). The Cologne dialect of Central Franconian fulfills this criterion: VR-syllables show Fo contours compatible with analysis as contour tones. More specifically, the Cologne dialect has been treated in Gussenhoven & Peters 2004 and Peters 2006 as fundamentally distinguishing between lexical items with a lexical tone (“accent 2”) and lexically toneless forms that receive intonational tones only (“accent 1”). Under their analysis, the toneless accent 1 forms, when in nuclear position, may host either a falling /H\*L/ (declarative) or rising /L\*H/ (interrogative) pitch accent; /<sup>1</sup>luus/ ‘smart, clever’ thus may surface as either [lùs] or [lúus]. In contrast, lexical items possessing accent 2 have a lexical tone, which, however, is subject to assimilation to the starred tone of the relevant nuclear intonation. Hence, /<sup>2</sup>luus/ ‘louse’ obtains a level high tone, [lúus], in a declarative context, after which a fall in Fo occurs (i.e., the /L/ of /H\*L/ floats), but a level low tone, [lùs], in an interrogative context, after which a rise in Fo occurs. In short, VR-syllables in Cologne Franconian permit a range of contour and level tones; compare Figure 18.6 in Riad & Peters 2021: 286, reproduced in Figure 18 here.

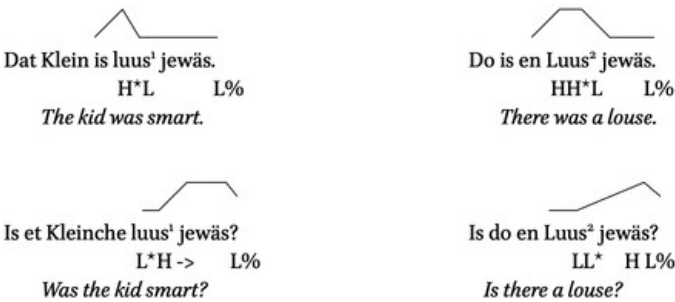


FIGURE 18 Accent 1 (left) and Accent 2 (right) nuclear intonational contours for declaratives (upper) and interrogatives (lower) in Cologne  
AFTER RIAD & PETERS 2021: 286

Thus, insofar as Central Swedish and Cologne Franconian are representative of the types of syllable-level Fo patterns encountered in Scandinavian and Franconian West Germanic generally, two general behaviors are observable: either no syllable-level contour tones occur (Scandinavian) or level high tones occur alongside contour tones (Franconian). In neither case are level high tones absent from syllable types that permit contour tones.

### 3.1.2.5 *Further languages*

We have just surveyed four language groups (Japanese, Basque, Bantu, and Germanic) in which “pitch-accent” or restricted tone systems at least partially comparable to Ancient Greek word prosody are found. In our examination of the relevant literature thus far, the only unambiguous case of a language with [H L] and [L H] contours, but no level [H], on some given syllable type, remains Nagasaki Japanese. The relative rarity of the type of situation encountered in Nagasaki Japanese is further confirmed by a more superficial survey conducted on a wider array of languages.

Of particular note is the situation in two Cushitic languages that exhibit restricted tone, Somali and Oromo. Both of these languages share the feature of limiting high tone to a window of two syllables at the right edge of the word; in Somali, H is further obligatory and culminative. Owens 1985 is the primary source for Oromo word prosody; many descriptions and phonological treatments of tone assignments in Somali are available.

In the description of Oromo word prosody provided by Owens (1985: 26–59), three general types of surface word-level tone patterns occur, and the same lexeme may potentially surface with all of these patterns, depending upon (morpho)syntactic context. First, a word may surface entirely with low tone; second, a word may surface with its “basic (High/Low) tone”, which is marked on the penultimate syllable, after which the final syllable is always high-toned, regardless of the tone on the penultimate syllable; third, in nominals used predicatively, the “basic (High/Low) tone” surfaces on the penultimate syllable, but the final syllable remains low-toned. In short, (morpho)syntactic contexts predict whether the final syllable will be high- or low-toned, and in some contexts a distinction between H and L on the penult may surface, while in other contexts, a penultimate H is suppressed. Most relevant for present ends is simply the observation that high tone on long vowels is always level, never rising or falling (Owens 1985: 56).

While some amount of variation in descriptions of Somali word prosody exists, most examinations thereof suggest that potential rising tones are either systematically transformed into level high tones, or at least tend to “flatten” the contour, whereas falling contours persist faithfully as such. Under the analysis of Hyman (1981), H may be assigned to either the final or penultimate vocalic mora; perhaps tellingly, Hyman’s (1981: 173) discussion, as well the earlier work of Armstrong (1934) and Andrzejewski (1956, 1964), explicitly refers to a falling contour when H is assigned to the penultimate vowel making up the first part of a long vowel or diphthong, but nowhere is a rising contour clearly indicated when H is assigned to the final mora of a long-voweled final syllable. Banti (1988) goes yet further, formulating a rule of “rising tone simplification”, which



results in underlying /V́V́/ sequences surfacing as [V́V́]. Orwin (1994: 164–165) likewise indicates that the realization of an underlying /LH/ contour on a VV-syllable will either be a level high, or, if there is an F<sub>0</sub> rise within the syllable, that this rise is not as steep as the fall on syllables with underlying /HL/, which are exceptionlessly realized as [H L]. Yip (2002: 140) suggests that the leftward spreading of H within a syllable to avoid rising contours is merely “optional”.

Beyond these two Cushitic languages, we have attempted to determine what surface tones occur on syllables in a variety of restricted tone languages, departing mainly from the languages cited in Hyman 2009: 220–224; we refer the reader to the references there. Specifically, we checked descriptions for the following languages: Puinave, Prinmi, Lamba, Manding, Hup, Crow, Acoma, Kaure, Kham, Arigibi Kiwai, Yucatec Mayan, Ocaina, and Obokuitai. With the exception of Hup (discussed immediately below) and partially for Yucatec Mayan, all of these languages either lack contour tones on syllables altogether (Manding, Kaure(?), Arigibi Kiwai, Yucatec Mayan, Ocaina), show only either a rising or a falling tone alongside level tones on all syllable types on which the contour tone may occur (Lamba, Crow, Acoma, Kham, Kaure(?), Obokuitai), or show a level high tone alongside any syllable type on which both contour tones occur (Puinave, Prinmi). Yucatec Mayan in general has only level tones on VV-syllables, but when a high-toned VV-syllable occurs phrase-initially or phrase-finally, a rising or falling contour is found, likely due to the effect of boundary tones.

Hup (Naduhup family, spoken in Brazil), meanwhile, presents an interesting case. Epps (2008: 89) writes: “Hup has two phonemic tones, realized as rising and high, which occur exclusively on stressed syllables; the Hup system is accordingly defined as a word accent system, as discussed above (section 2.3.2). Phonetically, Hup also has a falling contour tone, which is an allophone of the high tone (note, however, that it is not entirely clear which should be considered underlying; see below). Unstressed syllables take a default low tone.” Specifically, the surface distribution is as follows: rising [LH] is possible on any syllable type, but is very rare on open (CV) syllables; falling [HL] occurs on open (CV) syllables and syllables closed by a [+voice] consonant (CVC<sub>[+voice]</sub>); level [H] occurs only on syllables closed by a voiceless consonant, but never on other syllable types. Epps concludes from the fact that rising [LH] may occur on syllables closed with a voiceless consonant that the best way to interpret the complementary distribution of [H] and [HL] is as underlying /H/, where the falling contour follows from the absence of another specified tone where one could be permitted. Regardless, at least syllables of the type CVC<sub>[+voice]</sub> apparently show only contour tones, [L H] and [H L]. This case is not, however, neatly comparable with the possible [L H] vs. [H L] distinction on VV-syllables

of Ancient Greek, because the Hup situation involves allophony exclusively, whereas the Ancient Greek situation involves a contrast on final VV-syllables.

### 3.1.3 The typology of contour tones in canonical tone languages

We look now to the availability, frequency, and language-internal distribution of contour tones among languages known to possess lexically specified tonal contrasts that are not subject to considerations of culminativity, obligatoriness, or metricality. In terms of availability, the presence of contour tones in such canonical tone languages usually presupposes the presence of level tones, and languages with just two (surface) tones typically contrast the level tones, [H] and [L], on all syllables of all structures, not only a rising tone and a falling tone (Yip 2002: 24–25).<sup>52</sup> Considering the typological frequency of different types of contour tones, Zhang (2000, 2001, 2002) and Yip (2002: 27) explicitly state that falling tones are encountered much more commonly than rising tones. Thus, insofar as a given canonical tone language exhibits contour tones at all, it is much more likely to be falling tone: Zhang's survey of 198 canonical tone languages found 39 without any rising tones; in contrast, only 3 of the languages included lacked a falling tone, with Bamileke-Dschang (Hyman 1985, Manfredi 1993) constituting a rare case in which surface rising tones, but no surface falling tones, are to be found. Further data pointing towards the predominance of falling tones appears in Cheng (1973), in which the number of type of tones (level, falling, rising, concave, convex) was counted across varieties of Chinese, where a given variety might possess more than one sort of tone (e.g., Cantonese has two rising tones); there, the number of falling tones (1125) significantly outnumbers the number of rising tones (790).<sup>53</sup>

### 3.1.4 Summary

The brief typological survey just conducted has attempted to establish how common it is for a language—especially of the “pitch-accent” or “restricted-tone” variety, with word prosody comparable in certain respects to that of Ancient Greek—possesses syllable-level rising [H L] and falling [L H] tones, and, at least on certain syllable types (VV, VR, etc.) does not likewise admit of

52 Amahuaca may constitute one exception to this generalization, if it is correctly described as exhibiting a consistent contrast between a rising tone and a falling tone, rather than a level high and level low tone (Maddieson 1978).

53 Yip (2002: 30), cautions, however, that the number of falling tones in Cheng's survey might be an overcount, in some cases incorrectly identifying a level tone followed by a low boundary tone as a falling tone.

level high tone on that same syllable type. Precisely such a constellation seems to arise comparatively rarely, for two reasons: 1) level tones are less marked than falling tones, which are, in turn, less marked than rising tones; 2) where rising tones have the potential to arise through processes of tone assignment, they are often repaired to level (high) tones or falling tones. The first point is uncontroversial; the second is substantiated (at least) by evidence from varieties of Japanese and the Bantu languages. Thus, languages of the Nagasaki Japanese type, in which only [H L], [L H], or [L] occur on VR-syllables, while far from impossible, appear to constitute typological rarities; in the present survey, only Somali and Hup (and Amahuaca, among the canonical tone languages) make for further possible examples (and relatively weak ones at that). Nonetheless, a more comprehensive investigation of contour tone contrasts outside of canonical tone languages remains a desideratum.

The larger conclusion that applies to the interpretation of Ancient Greek word prosody is then the following: insofar as the language-internal phonological facts do not unambiguously demand the assumption of a rising [L H] versus falling [H L] surface contrast on VV-syllables, it is more probable, from the typological point of view, that a level [H] versus falling [H L] surface contrast on VV-syllables obtained. Indeed, as the examples from Kinga and Tōkyō Japanese, for instance, show, it is entirely possible that processes of high tone association in Ancient Greek might predict the creation of rising contours on VV-syllables, but such rising contours might be blocked and instead realized as a level high tone. Indeed, that VV-syllables of Ancient Greek, one way or another, could only bear [L] ( $\sim \emptyset$ ), [H], or [H L] tones is precisely the conclusion to which the empirical investigation in section 2 above arrived. In this sense, the few relatively unambiguous empirical datapoints that we now possess on Ancient Greek pitch contours appear to felicitously accord with the larger typology of surface tonal contrasts.

### 3.2 *Obligatory contour principle effects in Ancient Greek enclitic accentuation*

Having now established that a contrast of level high tone (= acute) versus falling tone (= circumflex) on VV-syllables is not only better compatible than a contrast of rising (= acute) versus falling (= circumflex) with the empirical evidence from tone-tune setting, but also with typological considerations, we return to language-internal phonological evidence, specifically to the behavior of accent in enclitic sequences, to test whether the assumption “Acute = H” causes undesirable issues. Ancient Greek possesses a number of enclitic lexemes incapable of constructing independent minimal prosodic words ( $\omega_{\min}$ ) without the support of an orthotonic word. Following Goldstein (2015: 47–60), the enclitics of

Ancient Greek constitute affixal clitics (in the classificatory scheme of Anderson 2005), building recursive prosodic words with their host (under the MIN-MAX model of Ito & Mester 2009, 2020, 2021), in a fashion essentially similar to the behavior of clitics in Neapolitan Italian (for this data, see Peperkamp 1997: 178–191); cf. also Revithiadou 2018 for an analysis of aspects of enclitic accentuation in Ancient Greek that employs a recursive prosodic word.

The word-level prosodic structure of ἄνθρωπός τις ‘some man’ is given in (12) as a simple example: ἄνθρωπος ‘man’ forms a minimal prosodic word ( $\omega_{[+min, -max]}$ ), to which (an) additional syllable(s) may be attached, with recursion at the level of the prosodic word in the process, terminating in the maximal prosodic word ( $\omega_{[-min, +max]}$ ).<sup>54,55</sup> Perhaps the most striking prosodic effect conditioned by the enclitic is that, under certain circumstances, an additional acute appears on the final syllable of the host; in some cases in which a “secondary acute” is not permitted on the final syllable of a host, it may surface on the final syllable of a disyllabic enclitic (e.g., φίλος τις ‘some friend’ vs. φίλος ἐστί ‘he is a friend’). Such “secondary acutes” thus do not occur in all cases, but depend upon the accentuation of the host and composition of its final syllable (see Probert 2003: 147–148 and Roussou & Probert 2023: § 2.7–9, 4 for an overview of the basic rules); the eight different patterns that occur are summarized in Table 25.

54 For the reader who may prefer to operate with the Clitic Group (see Nespor & Vogel 1986 [2007]: 145–162) as the prosodic constituent dominating the orthotonic word and enclitic (see also arguments against recursive prosodic words in Vogel 2009 and Vogel 2020), they may feel free to replace the  $\omega_{[-min, -max]}$  and  $\omega_{max}$  with a clitic group; nothing in the present argument crucially hinges on whether clitic groups or recursive prosodic words are employed. We do, however, insist that a right constituent boundary is assumed to occur between the orthotonic word and an enclitic, as well as between an enclitic and further successive enclitics (i.e., the structure is not “flat”), though multiple enclitic sequences of Ancient Greek are not discussed in detail here (see most recently the philological investigation in Roussou & Probert 2023).

55 One potential argument in favor of employing a recursive prosodic word for Ancient Greek concerns the phenomenon of grave formation. Grave formation applies to an acute on the final syllable of both an orthotonic word alone or on the final syllable of an enclitic. If a recursive prosodic word is adopted, then the process of grave formation can be considered to apply at the right edge of a  $\omega_{[+max]}$  (regardless of whether the  $\omega$  is  $[+min]$  or  $[-min]$ ) when not also at the right edge of an utterance. If the clitic group is instead adopted, then either an orthotonic word without any enclitics must assumed to form a clitic group on its own, or grave formation must be assumed to apply both at the right edge of the prosodic word and the right edge of a clitic group. Either of these alternatives is analytically less efficient, requiring either the construction of otherwise unmotivated prosodic structure (i.e., a clitic group without any clitics) or the doubling of a phonological process at the edge of two different prosodic domains.

(12) Word-level Prosodic Structure of ἄνθρωπός τις

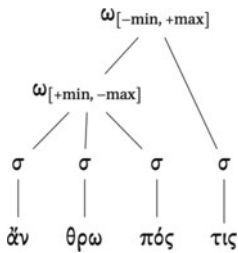


TABLE 25    Accentual behavior of host + single enclitic sequences in Ancient Attic-Ionic Greek

| Host     | Host+enclitic | Host accent       | Host final σ shape | # Clitic σ |
|----------|---------------|-------------------|--------------------|------------|
| πατήρ    | πατήρ τις     | Final Acute       | N/A                | 1/2        |
| καλῶς    | καλῶς ἐστί    | Final Circumflex  | VV                 | 1/2        |
| φίλος    | φίλος τις     | Penult Acute      | N/A                | 1          |
| φίλος    | φίλος ἐστί    | Penult Acute      | N/A                | 2          |
| οἶκος    | οἶκος τις     | Penult Circumflex | VV, VC             | 1/2        |
| κῆρυξ    | κῆρυξ τις     | Penult Circumflex | ṼCC                | 1          |
| κῆρυξ    | κῆρυξ ἐστί    | Penult Circumflex | ṼCC                | 2          |
| ἄνθρωπος | ἄνθρωπός τις  | Antepen. Acute    | N/A                | 1/2        |

Although a consensus on the rules of metrical structure construction and tone assignment involved in enclitic accentuation has not been reached, especially where multiple enclitics are involved (see Steriade 1988: 283–296, Sauzet 1989: 97–101, Golston 1990: 74–76, Steriade 2014: 13–19, or Revithiadou 2018: 11–17 for attempts at phonological analyses; see Probert 2016 or Roussou & Probert 2023 on longer enclitic sequences), under some approaches, where the secondary acute is or is not permitted may potentially shed light on the nature of the acute in VV-syllables. To review the facts, a secondary acute occurs on the final syllable of a host if the acute of the host falls on the antepenult (like ἄνθρωπος) or the penult bears a circumflex (and the final syllable is not of the shape [-Ṽ(η)ks] or [-Ṽps]), e.g., οἶκος τις ‘some house’ (contrast κῆρυξ τις ‘some herald’, without secondary acute). In all other cases, (circumflex final syllable, acute final syllable, or acute penult, regardless of the final syllable’s shape) no secondary acute occurs. In what follows, we presume that any H tone assigned within the host as an independent prosodic word ( $\omega_{[+min, -max]}$ ) cannot be affected by the addition of an enclitic; we remain agnostic here about the exact source of the

additional H tone, e.g., whether the enclitic brings with it a floating /H/ that attempts to associate recessively (as assumed by Golston 1990; but see the critique of Revithiadou 2018: 18), or each further constituent created by addition of an enclitic must receive stress and associates an /HL\*/ pitch accent with the enclitic (as proposed by Sauzet 1989).

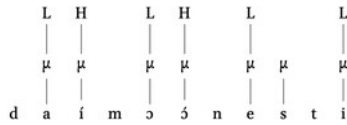
Under any set of assumptions, the absence of a secondary acute on hosts with an acute final syllable is unsurprising: since the final TBU already hosts a H tone, it cannot host another. In the case of a final circumflex, e.g.,  $\pi\acute{\upsilon}\rho$  ‘fire’, the addition of an enclitic with its H tone might create a level-toned (acute) host, if the circumflex is represented as  $H\emptyset$  and each mora can act as a TBU:  $/\pi\acute{\upsilon}\rho\ \tau\iota/ \rightarrow {}^X\pi\acute{\upsilon}\rho\ \tau\iota$ . This possibility is readily excluded, however, in one of two ways. Under the assumption that the circumflex consists in a fully specified HL sequence (as the evidence in § 2 suggests), another H tone cannot be associated without crowding three tones onto a VV-syllable. Alternatively,  ${}^X\pi\acute{\upsilon}\rho\ \tau\iota$  may be ruled out by invoking the OBLIGATORY CONTOUR PRINCIPLE (OCP) with respect to the tonal tier: no two identical elements may occur adjacent to one another, in this case, two high tones. The relevant phonological constraint would thus be OCP(H): two adjacent H tones are banned. That  ${}^X\pi\acute{\upsilon}\rho\ \tau\iota$  does not occur is thus unsurprising.

Meanwhile, in cases like  $\acute{\alpha}\nu\theta\rho\omega\pi\acute{o}\varsigma\ \tau\iota\varsigma$  or  $\omicron\acute{\iota}\chi\acute{o}\varsigma\ \tau\iota\varsigma$ , a secondary acute is permitted on the host because its final syllable does not appear to be already occupied by any tone, and the placement of a further H tone on the final syllable in such cases would not violate OCP(H): an L tone (or  $\emptyset$ ) intervenes, either on the unaccented syllable (in  $\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$ ) or the second portion of the circumflex (in  $\omicron\acute{\iota}\chi\omicron\varsigma$ ), yielding the sequence [H L/ $\emptyset$  H] on the host. The two remaining configurations are not immediately captured by reference to the availability of the host’s final syllable and OCP(H), and behave in parallel: neither the type  $\delta\acute{\alpha}\iota\mu\omega\upsilon\iota\ \acute{\epsilon}\sigma\tau\acute{\iota}$  ‘it is a *daimōn*’ (with acute penult) nor the type  $\chi\acute{\eta}\rho\upsilon\varsigma\ \acute{\epsilon}\sigma\tau\acute{\iota}$  ‘it is a herald’ (with circumflex penult, like  $\omicron\acute{\iota}\chi\acute{o}\varsigma\ \acute{\epsilon}\sigma\tau\iota$  ‘it is a house’) permit a secondary acute on the final syllable of the host, but both types (in contrast to hosts with acute or circumflex on the final syllable) admit of a secondary acute on the second syllable of disyllabic enclitics. We do not venture an account here as to why secondary acutes in the host in  ${}^X\chi\acute{\eta}\rho\acute{\upsilon}\xi\ \tau\iota\varsigma$  or  ${}^X\chi\acute{\eta}\rho\acute{\upsilon}\xi\ \acute{\epsilon}\sigma\tau\iota$  are not permitted; OCP(H) is clearly not violated in such configurations, and given the presence of an HL contour on the penult, the final syllable would seem to be available for tone.<sup>56</sup>

56 Bally (1945: 26) suggests that the short vowel in the sequence VCC is too short to host high tone (“accent de hauteur”), and interprets the intraparadigmatic alternation between [i:]/[u:] and [i]/[u] in cases like NOM  $\chi\acute{\eta}\rho\acute{\upsilon}\xi$ ,  $\phi\omicron\acute{\iota}\nu\acute{\iota}\xi$ ; GEN  $\chi\acute{\eta}\rho\acute{\upsilon}\chi\omicron\varsigma$ ,  $\phi\omicron\acute{\iota}\nu\acute{\iota}\chi\omicron\varsigma$  as evidence for vowel shortening in (final) syllables closed by [ks] or [ps]. Sandell (2023: 577–581) pro-

In cases like δαίμων ἐστί, meanwhile, the absence of a secondary acute is not readily explained if an acute on VV-syllables is assumed to be realized as LH (~ØH): the unexpected <sup>x</sup>δαίμων ἐστί would contain the sequence LH LH (~ØH ØH), as illustrated in (13a), which would not be penalized by OCP(H). It thus follows that a secondary acute from enclitics would be expected on words like δαίμων, were the acute in such cases indeed LH. See, for this argument, Steriade 2014: 13–14, and already similar suggestions in Steriade 1991 and Devine & Stephens 1994: 369–370 with respect to words having an acute VC-penult.<sup>57</sup> In contrast, if the acute is assumed to be a level H, as in (13b). (whether associated to a syllable, or, as shown here, to both morae of a VV-syllable), then OCP(H) will correctly predict blocking of a secondary acute on the final syllable of the orthotonic word (and instead a H tone on the final syllable of a disyllable enclitic), thus yielding expected δαίμων ἐστί.

(13) a. <sup>x</sup>δαίμων ἐστί [daímɔ́ɔn esti]<sup>58</sup>



b. δαίμων ἐστί [daímɔɔn estí]



poses that the final syllable in such forms is occupied by a L\* tone spread across the final and penultimate syllables as a consequence of the need to associate tone to all syllables with stress (given a foot structure and stress assignment [(‘kê:).(rùk)<s>])).

57 Nevertheless, the uncertainties involved in the proper treatment of enclitic accentuation in relation to a general analysis of accentuation in orthotonic words mean that this line of argumentation is not watertight. In particular, under treatments of recessive accentuation that posit a /HL\*/ word-level pitch accent, the final syllable of words like δαίμων will already host a L\* tone when an enclitic enters the derivation. The fact that the syllable μων already hosts a tone might then prevent its association with a H tone induced by the enclitic, regardless of the tonal profile of the first syllable, δαί (cf. Golston 1990: 76 and similarly Golston & Paulsen 2022).

58 The [n] at the right edge of δαίμων is represented as weightless (i.e., not assigned a mora), for the reason that word-final consonants likely behave as extrametrical in Ancient Greek (see again Golston 1990; cf. Hayes 1995: 56–61 on prosodic extrametricality generally) and, in the example at hand, this segment would presumably be parsed into the onset of a syllable headed by [e], and therefore not be subject to receiving weight-by-position.

Since there are at least two further conceivable outputs, namely  $^{\text{X}}\delta\alpha\acute{\iota}\mu\omega\nu \acute{\epsilon}\sigma\tau\iota$  (with acute on the first syllable of the two-syllable enclitic; note, though, that disyllabic enclitics never bear a H tone on their first syllable in any configuration) or  $^{\text{X}}\delta\alpha\iota\mu\omega\nu \acute{\epsilon}\sigma\tau\iota$  (with no secondary acute whatsoever), briefly sketching a descriptive constraint-based analysis of enclitic accentuation is in order. Again, without developing a larger-scale phonological analysis of Ancient Greek word prosody, providing a fully motivated analysis of the enclitic facts is not possible, for which reason we rely here primarily on descriptive phonotactic markedness constraints. The relevant constraints are defined in (14), with a ranking provided in (15).

(14) Constraints Employed for Deriving Enclitic Accentuation<sup>59</sup>

- a.  $^*LLL]_{\omega}$ : no sequences of three Low- and/or  $\emptyset$ -toned syllables at the right edge of a prosodic word of any level of recursion.
- b.  $^*]_{\omega} \acute{\sigma} (\dots)]_{\omega\text{-non-min}}$ : no High tone on the first syllable immediately to the right of a prosodic word constituent where that syllable is contained within a non-minimal prosodic word.
- c.  $OCP(H)-\omega_{[+min]}$ : no adjacent High tones on the tonal tier within a minimal prosodic word.
- d.  $^*\acute{\sigma}]_{\omega\text{-max}}$ : no High tone on a syllable immediately preceding the syllable at the right edge of a maximal prosodic word.

(15) Ranking of Constraint in (14) for Deriving Enclitic Accentuation:

$\{^*LLL]_{\omega}, OCP(H)-\omega_{[+min]}\} \gg ^*]_{\omega} \acute{\sigma} (\dots)]_{\omega\text{-non-min}} \gg ^*\acute{\sigma}]_{\omega\text{-max}}$

Among these constraints, (14a) and (14c) may be seen as independently motivated. The avoidance of a long sequence of syllables without any High tone at the right edge of the prosodic word fits the facts of the Law of Limitation in independent prosodic words; it may constitute a form of (tonal) ANTI-LAPSE (cf. Steriade 2014, Ito & Mester 2017). As for (14c), the invocation of OBLIGATORY CONTOUR PRINCIPLE effects on the tonal tier needs no extensive justification, being cross-linguistically common; the specific restriction of this OCP effect to the minimal prosodic word is motivated by the observation that two High tones can, in fact, occur on adjacent syllables when sequences involving more than one enclitic arise (e.g.,  $\kappa\alpha\acute{\iota} \tau\acute{\epsilon} \mu\epsilon \varphi\eta\sigma\acute{\iota}$ ), but in such cases, the right edge of a prosodic constituent always intervenes. Constraints (14b) and (14d), meanwhile, constitute mere phonotactic observations that correspond to the

59 Again, we exclude  $\kappa\eta\rho\nu\acute{\xi}$ -type forms from present consideration.



relative preferences in the placement of secondary acutes: in single enclitic sequences, to avoid violations of  $*LLL]_{\omega}$ , the secondary High ideally falls on the last syllable of the orthotonic word (leading to no violations), otherwise on the last syllable of a disyllabic enclitic (violating  $*\acute{\sigma}]_{\omega-max}$ ); in multiple enclitic sequences, if a violation of  $*LLL]_{\omega}$  would result, a further High tone occurs preferentially on the second syllable of a disyllabic enclitic (violating  $*\acute{\sigma}]_{\omega-max}$ ), otherwise on the only syllable of an enclitic (thus violating  $*]_{\omega} \acute{\sigma} (...)]_{\omega-non-min}$ ) that can resolve a violation of  $*LLL]_{\omega}$  without also violating  $\acute{\sigma}]_{\omega-max}$ .<sup>60</sup>

The tableau shown in (16) illustrates the outcome  $\delta\acute{\alpha}\iota\mu\omega\nu\ \acute{\epsilon}\sigma\tau\acute{\iota}$ . Presuming that the new H on a VV-syllable would produce a level high tone, in accord with the empirical investigation from section 2, candidate c. is excluded (cf. (13a) above). The fully faithful candidate, a., is ruled out by the resulting sequence of three unaccented syllables, and candidate d. by the placement of a High tone immediately after the right edge of a prosodic word. Thus, only candidate b., with High tone on the final syllable of the maximal prosodic word, remains, despite its violation of the lowest-ranking constraint. The combined effect of  $*]_{\omega} \acute{\sigma} (...)]_{\omega-non-min}$  and  $*\acute{\sigma}]_{\omega-max}$  is similar to Steriade’s (2014)  $*CLITIC$  (“avoid High tone on enclitics”).

(16) / $\delta\acute{\alpha}\iota\mu\omega:n\ esti/ \rightarrow [[\delta\acute{\alpha}\iota\mu\omega:n]_{\omega-min}\ esti]_{\omega-max}$

| / $\delta\acute{\alpha}\iota\mu\omega:n\ esti/$ |                                                                                                                                                                | $*LLL]_{\omega}$ | OCP(H)-<br>$\omega_{[+min]}$ | $*]_{\omega} \acute{\sigma} (...)]_{\omega-non-min}$ | $*\acute{\sigma}]_{\omega-max}$ |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|------------------------------------------------------|---------------------------------|
| a.                                              | $[[\delta\acute{\alpha}\iota\mu\omega:n]_{\omega-min}\ esti]_{\omega-max}$                                                                                     | $*!$             |                              |                                                      |                                 |
| b.                                              |  $[[\delta\acute{\alpha}\iota\mu\omega:n]_{\omega-min}\ esti]_{\omega-max}$ |                  |                              |                                                      | *                               |
| c.                                              | $[[\delta\acute{\alpha}\iota\mu\acute{\omega}:n]_{\omega-min}\ esti]_{\omega-max}$                                                                             |                  | $*!$                         |                                                      |                                 |
| d.                                              | $[[\delta\acute{\alpha}\iota\mu\omega:n]_{\omega-min}\ \acute{\epsilon}\sigma\tau\acute{\iota}]_{\omega-max}$                                                  |                  |                              | $*!$                                                 |                                 |

60 Conceivable potential motivations underlying (14b). and (14c). could be a preference for High tone to align as close to the right edge of a prosodic word (at any level of recursion) as possible (i.e., ALIGNRIGHT-H/ $\omega$ ; contrast Ito & Mester 2017: 8, who explicitly assume that High tone in Ancient Greek prefers to associate as far to the *left* in a prosodic word as possible) and avoidance of High tone on the final syllable of a (recursive) prosodic word on account of a boundary tone (compare Revithiadou 2018: 15–17, who instead suggests that “secondary acute” on the final syllable of a disyllable enclitic is itself due to a H% boundary tone, and Gunkel 2023, where possible musical evidence for low boundary tones in the minimal prosodic word is treated; but contrast Sandell 2023: 520–523 on contradictions that the assumption of boundary tones in the prosodic word might create for properly generating both grave accent and explaining the application of the  $\sigma\omega\tau\eta\rho\alpha$ -Rule).

This simple, *ad hoc* analysis does, though, extend well to multiple enclitic sequences. Data here is drawn from the patterns of accentuation attested in multiple enclitic sequences of the Venetus A manuscript of the *Iliad*, summarized in Probert 2016 and Roussou & Probert 2023. We illustrate here in (17) and (18) that the same constraints correctly predict the attested outcome in the sequence καί τέ με φησί (*Iliad* 1.521); note that the accentuation given here is different than the accentuation printed in many textual editions, but we follow Roussou & Probert in holding that καί τέ με φησί represents the likely original, correct accentuation. One further constraint that must be introduced (cf. also frequent cases like σύ πέρ μιν, ὅτε κέν τοι, or οὐ ἐθέλιν ἐστι, in which a secondary acute is added to the penultimate or antepenultimate syllable) is that a sequence of two adjacent unaccented enclitics is not permitted ( $^{*}\acute{(\acute{\sigma})}]_{\omega\text{-non-min}} \acute{(\acute{\sigma})}]_{\omega\text{-non-min}}$ ), which is also inviolable. In the cyclic calculation of prosodic structure and High assignment, first in (17), when a second monosyllabic enclitic appears, another High is necessary to avoid a violation of  $^{*}\acute{(\acute{\sigma})}]_{\omega\text{-non-min}} \acute{(\acute{\sigma})}]_{\omega\text{-non-min}}$  (see candidate a.), but placing the additional High on the final syllable results in violations of both  $^{*}]_{\omega} \acute{\sigma} (...)]_{\omega\text{-non-min}}$  and  $^{*}\acute{\sigma}]_{\omega\text{-max}}$ . In (18), the addition of a further disyllabic enclitic conditions a secondary acute on the final syllable of the new enclitic, φησί. Leaving the rightmost two enclitics without a High tone would result in violations of both the ban on adjacent unaccented enclitics and the ban on a long stretch of unaccented syllables (see candidate a.). Adding a further High tone anywhere other than at the very end of the sequence then results in comparatively more violations of  $^{*}]_{\omega} \acute{\sigma} (...)]_{\omega\text{-non-min}}$  (see candidates c.–e.), while the winner, candidate b., again suffers instead a violation of  $^{*}\acute{\sigma}]_{\omega\text{-max}}$ .

(17)  $/[[[\text{kaí}]_{\omega\text{-min}} \text{té}]_{\omega} \text{me}] / \rightarrow [[[[[\text{kaí}]_{\omega\text{-min}} \text{té}]_{\omega} \text{me}]_{\omega}$

|    | $/[[[\text{kaí}]_{\omega\text{-min}} \text{té}]_{\omega} \text{me} /$                   | $^{*}\acute{(\acute{\sigma})}]_{\omega\text{-non-min}}$<br>$\acute{(\acute{\sigma})}]_{\omega\text{-non-min}}$ | $^{*}\text{LLL}]_{\omega}$ | OCP(H)-<br>$\omega_{[+min]}$ | $^{*}]_{\omega} \acute{\sigma} (...)]_{\omega\text{-non-min}}$ | $^{*}\acute{\sigma}]_{\omega\text{-max}}$ |
|----|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|----------------------------------------------------------------|-------------------------------------------|
| a. | $[[[[[\text{kaí}]_{\omega\text{-min}} \text{te}]_{\omega} \text{me}]_{\omega}$          | $^{*}!$                                                                                                        |                            |                              |                                                                |                                           |
| b. | $\text{☞} [[[[[\text{kaí}]_{\omega\text{-min}} \text{té}]_{\omega} \text{me}]_{\omega}$ |                                                                                                                |                            |                              | $^{*}$                                                         |                                           |
| c. | $[[[[[\text{kaí}]_{\omega\text{-min}} \text{té}]_{\omega} \text{mé}]_{\omega}$          |                                                                                                                |                            |                              | $^{*}!^{*}$                                                    | $^{*}$                                    |
| d. | $[[[[[\text{kaí}]_{\omega\text{-min}} \text{te}]_{\omega} \text{mé}]_{\omega}$          |                                                                                                                |                            |                              | $^{*}$                                                         | $^{*}!$                                   |

(18)  $/[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ me}]_{\omega} \text{ p}^h\epsilon:\text{si}/ \rightarrow [[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ me}]_{\omega} \text{ p}^h\epsilon:\text{sí}]_{\omega-\max}$

|    | $/[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ me}]_{\omega} \text{ p}^h\epsilon:\text{si}/$               | $*\acute{\sigma}(\acute{\sigma})_{\omega-\text{non-min}}$<br>$\acute{\sigma}(\acute{\sigma})_{\omega-\text{non-min}}$ | $*\text{LLL}]_{\omega}$ | $\text{OCP(H)-}$<br>$\omega_{[+\min]}$ | $*]_{\omega} \acute{\sigma}(\dots)_{\omega-}$<br>$\text{non-min}$ | $*\acute{\sigma}]_{\omega-\max}$ |
|----|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------|-------------------------------------------------------------------|----------------------------------|
| a. | $[[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ me}]_{\omega} \text{ p}^h\epsilon:\text{si}]_{\omega-\max}$ | $*!$                                                                                                                  | $*!$                    |                                        |                                                                   |                                  |
| b. | $[[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ me}]_{\omega} \text{ p}^h\epsilon:\text{sí}]_{\omega-\max}$ |                                                                                                                       |                         |                                        | $*$                                                               | $*$                              |
| c. | $[[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ mé}]_{\omega} \text{ p}^h\epsilon:\text{si}]_{\omega-\max}$ |                                                                                                                       |                         |                                        | $**!$                                                             |                                  |
| d. | $[[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ me}]_{\omega} \text{ p}^h\epsilon:\text{si}]_{\omega-\max}$ |                                                                                                                       |                         |                                        | $**!$                                                             |                                  |
| e. | $[[[[kaí]_{\omega-\min} \text{ té}]_{\omega} \text{ mé}]_{\omega} \text{ p}^h\epsilon:\text{sí}]_{\omega-\max}$ |                                                                                                                       |                         |                                        | $**!$                                                             | $*$                              |

Let us not, however, lose sight of the larger point. However precisely the accentuation of enclitic sequences in Ancient Greek is best to be motivated and analyzed, the discussion here should make clear that positing a level High, rather than rising LH ( $\sim\emptyset$ H) tone for the acute on a VV-syllable, creates no new obstacles. In other words, interpreting the acute as a level High tone is, at minimum, compatible with the facts of enclitic accentuation. Just in case the absence of a secondary acute on the final syllable of the host in sequences like  $\delta\acute{\alpha}\iota\mu\omega\nu \tau\iota\varsigma$  or  $\delta\acute{\alpha}\iota\mu\omega\nu \acute{\epsilon}\sigma\tau\iota$  cannot, ultimately, be motivated by claiming that the final syllable of the host is already occupied by tone, a version of the OCP can be invoked (cf. again Sauzet 1989, Golston 1990, Steriade 1991, and Steriade 2014) to keep a secondary acute off of the host—but this maneuver, crucially, is only possible in the event that the acute on VV-syllables is regarded as a level High tone (cf. the autosegmental representations in (13) above).

3.3 *On the relation of underlying representations to the acute–circumflex contrast*

Up to this point, the evidence from tone-tune setting strongly supports the view that an acute accent, on the surface, always represents a level high tone, and further, that there was a contrast between level high tone and falling tone on VV-syllables. The same view is suggested by the typology of tone and is at least compatible with the analysis of enclitic accentuation sketched above. That view raises a further question: what is or are the source(s) for this surface contrast? In this section, we reexamine the phonologically conditioned distribution of acute and circumflex accents in light of the evidence that the acute always represents a level high tone, with an eye to the relevant underlying representations.

Beginning with the proposal of Allen (1973: 231–234; 246) that (minimal) prosodic words of Ancient Greek were generally characterized by the presence of a High to Low falling “contonation”, the presence of a High to Low pitch

movement has been widely adopted and reaffirmed in many studies of Ancient Greek prosody and diverse phonological analyses of accent assignment (e.g., Sauzet 1989, Golston 1990, Devine & Stephens 1994: 154; 180, Steriade 2014, Ito & Mester 2017). It is unproblematic to assume such a high–low movement in all words having “recessive” accentuation, which uncontroversially constitutes the default pattern of Ancient Greek word prosody (see Probert 2006 for extensive argumentation). Recessively accented words bear an acute on the antepenult if the final is light (ἄνθρωπος with an extrametrical final consonant [see fn. 64 above]), and an acute on the penult if the final contains a long vowel or heavy diphthong (ἄνθρώπου, οἴκοι ‘at home’); in both of these cases, whether the antepenult or penult, respectively, contains one or more vocalic morae is irrelevant. When, however, the final is heavy in virtue of containing a consonant cluster (i.e., sequence [-Ṽ(N)p/ks]), a short-voweled penult shows an acute (πολύανθραξ ‘having much coal:NOM.SG’), while a VV-penult shows a circumflex (πολυπίδαξ ‘having many wells:NOM.SG’). In all of these cases, at least one vocalic mora follows a High tone, and, with the exception of cases like πολυάνθραξ, at least two follow; there is thus ample space for a sequence of two tones /H+L/, and cases like πολυπίδαξ might be interpreted as realizing the entire /H+L/ sequence on a single syllable. Furthermore, the accentuation of finite verbs, which is almost exceptionlessly recessive, likewise exceptionlessly exhibits a circumflex when monosyllabic forms are involved (e.g. root aorists like βῆ or στῆ), which indicates that a falling intonation constitutes the default pitch pattern (see Steriade 2014 or Sandell 2023: 503, following Vendryes 1904: 111, for this argument), comparable to the status of “accent 1” in Scandinavian or Franconian “pitch-accent” systems. Thus, the default word-prosodic pattern of Ancient Greek may be defined as assignment of H to the syllable immediately to the left of the rightmost heavy syllable (within a two-syllable window at the right edge), with L assigned to its right—either on the succeeding syllable or within the same syllable, in the event that no other syllable or only some dispreferred host for L is available (e.g., βῆ, πολυπίδαξ). With Golston (1990), H associates to the antepenult and L to the penult when the final two syllables are light.

The parallel conclusion to be drawn is that VV-monosyllables with acute (e.g., πούς) or words with a high tone closer to the right edge of the word than would be generated by recessive accentuation (e.g., ἀστερίσκος, σωτήρ, ποδῶν) must be exceptionally marked in some fashion. One approach might be to mark a given vowel or mora as lexically associated with a high tone, /H\*/; under this analysis, a stem like /pód-/ ‘foot’ or a derivational suffix like /-ísko-/ ‘DIM’ would bear /H\*/; while a stem like /antʰrɔ:po-/ ‘man’ would be unmarked. In this regard, the tonally marked items would have a status similar to words with

“accent 2” in Scandinavian and Franconian word prosody (under the analyses of Riad and Peters, respectively, discussed under § 3.1.2 above). As long as faithful preservation of a /H\*/ tone is highly ranked, it will yield non-recessive accentuation. Forms like σωτήρ, with level high on a VV-final syllable (in isolation) show that no repairs are necessarily undertaken to ensure that a fall in pitch (H–L movement) occurs in the prosodic word; however, when just such forms are extended by one syllable with one vocalic mora—as in ACC.SG σωτήρα—the H–L contour reappears. If /sɔ:tér-/ is assumed to lexically bear /H\*/, then this tone can either be transformed into a [H\*L] falling tone on the penult in order to create a H–L falling “contonation”, or it is possible for H\* to be displaced by default /H+L/, without altering the syllabic seat of the High pitch. Under the Sauzet–Golston interpretation of the default pitch accent as /H+L\*/ (followed i.a. by Gunkel 2011 and Sandell 2023: Ch. 6), marking the second stem vowel of /sɔ:tér-/ as lexically *stressed* would potentially lead to incorrect predictions, if /H+L\*/ can displace /H\*/: the result would be <sup>X</sup>σωτήρα (like ἄνθρωπος).

At the same time, the existence of falling tone within the final syllable itself, as in ποδῶν, poses a conundrum. In this latter case, either the entirety of the default /H+L/ pitch accent must be compelled to associate to the final syllable—contrary to the apparent avoidance of contour tones in most recessively accented forms (e.g., ἄνθρωπος, not <sup>X</sup>άνθρωπος). Under a tonal analysis, the contrast σωτήρ versus ποδῶν has two potential solutions: 1) not only lexical /H\*/ exists, but also lexical /H\*L/ (cf. Golston & Paulsen 2022); 2) lexical /H\*/ may be underlying associated to either the first or second mora of a VV-sequences, but in the latter case, a ban on surface rising tones induces leftward spread of /H\*/ within the syllable (as in Kinga or Tōkyō Japanese; see under § 3.1.2 above). An unsatisfying feature of both approaches is that the additional lexical tonal contrast or the need to adopt a distinction in association at the moraic level would exist almost exclusively to account for the GEN.PL suffix -ῶν (and the much less frequently encountered GEN/DAT.DU ending -οῖν), which sometimes surfaces as accented even in paradigms that are otherwise recessive or seem to have a lexical /H\*/ on a stem-final syllable (e.g., μυριάς, μυριάδος, but μυριαδῶν).<sup>61</sup> While both GEN.PL -ῶν and GEN/DAT.DU -οῖν contain

61 Pace Golston & Paulsen (2022: 2), Attic adjectival paradigms that exhibit a persistent surface circumflex on the final syllable, such as χρῦσοῦς ‘golden:M.NOM.SG’, plausibly do contain a synchronically recoverable sequence of two underlying vowels that contract. The argument is in part phonological, in part morphological, in part morphosyntactic. On the phonological side, the vowels in the masculine and neuter forms (usually [o:] in the final syllable) largely arise in (conservative) Attic through vowel contraction or lengthening to satisfy a minimal word requirement. Furthermore, in the specific case of χρῦσοῦς,

historical contraction products, in the case of the GEN.PL especially the contraction /-â:n/ < \*/é/óh<sub>1/3</sub>om/ (Kümmel 2013) is so old that positing underlying vowel sequences (e.g., /-óâ:n/ rather than /-â:n/) would constitute an extreme abstraction as well.

To attempt to capture the acute (level H) versus circumflex (falling HL) contrast in final syllables through the construction of different types of feet, as has been proposed for the Scandinavian and Franconian (see Morén-Duolljá 2013 on Götaland Swedish, Köhnlein 2016 on the German dialects of Cologne and Arzbach) unfortunately does not appear to make for a promising resolution in the Ancient Greek case. The analysis of Köhnlein (2016), for instance, essentially relies on a lexical distinction between syllabic trochees and moraic trochees, where only the head of a syllabic trochee is able to support a contour tone in virtue of containing two “strong” morae. When a lexical item instead builds a moraic trochee, a foot composed of a single heavy syllable is then only able to support a level tone. With Golston 1990, a default moraic trochee is probably the easiest means of accounting for (most of) the facts of recessive accentuation, and non-recessive forms with level High on VV-syllables (like σῶτήρ) could be interpreted as forming a default moraic trochee on a syllable where lexical /H\*/ is present. Note here that a lexical contrast in tone cannot be abandoned because simply leaving /sɔ:tɛ:r-/ unmarked altogether would result in recessively accented <sup>x</sup>σῶτήρ.

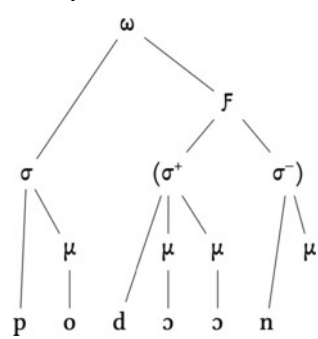
Meanwhile, the primary obstacle towards treating a form like ποδῶν as containing a syllabic trochee, however, with a headless syllable built around the consonant [n] (see the diagram in (19), modeled on Köhnlein 2016: 97) is the fact that Ancient Greek is a language with final consonant extrametricality. That is, the language seems to categorically avoid aligning the right edge of a foot with the right edge of a (minimal) prosodic word, and thus does not parse consonants at the right edge of the word into syllables in order to achieve this end. Of course, the GEN.PL ending could be lexically marked as containing an

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when taken together with the F.NOM.SG χρῦσῆ, the final vowel of the stem preceding the inflectional endings /-os/ or /-ε:/ must be an /e/: no other vowel in contraction with the vowels of the endings would yield the alternation [o:]~[ε:]. On the morphological side, to adjectival stems ending in a consonant, the same endings (M.NOM.SG /-os/, F.NOM.SG /-ε:/, etc.) are found, and thus should be expected to be underlyingly present in adjectival stems ending in vowels. Finally, adjectival paradigms contain the realization of gender features independent of the stem itself. Altogether, these considerations lead to ‘golden-M-NOM’ as /k<sup>h</sup>ru:se-o(-)s/. In terms of the accent, through some combination of lexical properties of the stem and adjectival inflectional endings that need not concern us here, /k<sup>h</sup>ru:séðs/, with H on the penultimate vowel results; after post-lexical vowel contraction, χρῦσοῦς surfaces.

underlying syllabic trochee ( $/\sigma^+ \sigma^-/$ , in Köhnlein’s notation), but this would achieve no more analytical efficiency than introducing an underlying  $/H^*L/$  specification or permitting the underlying association of  $/H^*/$  with different morae.

(19) ποδῶν ‘foot:GEN.PL’ with right-aligned syllabic trochee containing a headless syllable



The conclusion to which we ultimately arrive is that Ancient Greek, as in the “traditional” tonal analyses of Scandinavian and Franconian accentuation, knew at least a contrast between a default word intonation ( $/H+L/$ , likely  $/H+L^*/$ ) and a lexical tone,  $/H^*/$ . Some further element of lexical contrast in prosody—be it in terms of tone, underlying moraic association, or underlying foot structure—seems necessary to account for the relatively marginal contrast between falling tone and level high tone in final syllables that cannot be attributed to effects of post-lexical vowel contraction alone (i.e., the σωτήρ vs. ποδῶν distinction). None of these analytical alternatives is obviously better than the other, but the additional complications involved in the assumption of underlying differences in foot structure make it perhaps less promising. In its relative simplicity of representation, a further contrastive tone,  $/H^*L/$  (as also posited by Golston & Paulsen 2022) is attractive, and, depending upon the extent to which products of vowel contraction were restructured during the 1st millennium BCE prior to the reinterpretation of high tone as stress (see Sandell 2023: 624–628 for discussion of this development, with references), the further lexical falling tone  $/H^*L/$  would be rescued from its marginal status.

#### 4 Summary and conclusions

We have sought here to address a fundamental question concerning the prosody of Ancient Greek: what (surface) phonological properties should be ascribed to the “acute” accent? While the circumflex accent has universally and uncontroversially—in accord with the descriptions of Alexandrian grammarians—been interpreted as defined by falling pitch (a [H L] tonal contour), an unambiguous description of the acute that fits with its ability to occur on both short and long vocalic nuclei, has been wanting. Perhaps the predominant view in the literature on Greek prosody, going back at least to Vendryes 1904 and Jakobson (1931, 1937), derives the properties of the acute by way of contrast with the circumflex. On that view, the acute on long vocalic nuclei is a rising [L H], the mirror image of the [H L] circumflex. On short vocalic nuclei, the acute is simply [H], in accordance with traditional grammatical descriptions; see examples (1)–(3).

We have explicitly argued against the view that the Ancient Greek acute in any context was a rising contour tone with two phonological pitch targets, [L H], and we have adduced crucial empirical evidence, building upon some earlier observations to that effect: tone-tune setting practices in Ancient Greek music. That testimony, on the assumption that Ancient Greek musical *tunes* were sensitive to the surface realizations of phonological word-level *tones* present in the language, accords most readily with a surface contrast between level high and falling tone on VV-syllables. The investigation focused on effects of accentuation on the setting of syllables to two-note melisms in the Delphic hymns (DAGM 20 and 21), the two oldest and most informative compositions that reflect tone-tune matching. In several respects, accentuation interacts with melismatic settings in such a way that two neat natural classes may be discerned: on the one hand, syllables with a circumflex; on the other, syllables with an acute, its allotone the grave, and no accent. Statistical analysis of their distributions presents evidence, often very robust, for the following differences: 1) syllables with a circumflex are significantly more likely to be set to a melism than non-circumflex syllables; 2) the directionality of circumflex melisms is more predictable than non-circumflex melisms (circumflex melisms consistently fall; non-circumflex melisms may either rise or fall); and 3) the average size of the interval between two pitches in a melism is largest for circumflex melisms, and only the circumflex is a useful predictor of the second note in a melism given the first note. In light of the consensus that the circumflex was characterized by a [H L] sequence, we interpret the persistent distinction between circumflex and non-circumflex syllables (and equally importantly, the absence of any significant distinctions with the latter



group) as evidence that circumflex-accented syllables had two tonal targets, whereas unaccented, acute, and grave syllables did not; they had at most a single phonological tonal target. In the case of the acute, the phonological target is best understood as a level High tone, [H].

One question not fully resolved here concerns a tendency exhibited by non-circumflex melisms, which may occur on syllables with both long and short vocalic nuclei (though the latter are comparatively rare): rising melisms are somewhat more common than falling melisms. Two hypotheses were entertained here: first, in the case of acute syllables, the phenomenon of peak delay might result in a slight phonetic rise, to which tune settings might be sensitive; second, in the case of unaccented and grave syllables, the tendency for tunes to rise towards local tune peaks (defined by acute or circumflex syllables) could favor a rising melism when a given unaccented or grave syllable precedes an acute or circumflex. It is important to note that the two explanations are not mutually exclusive: peak delay and interpolation may both have been at play.

Beyond the empirical evidence from tone-setting, we further considered the typology of surface syllable-level tonal distinctions in a variety of languages, focusing especially on “pitch-accent” or “restricted-tone” systems with superficial similarities to the word prosody encountered in Ancient Greek. Since the “traditional” interpretation of the acute-circumflex distinction on VV-syllables would entail that VV-syllables show only rising [L H] or falling [H L] tonal patterns, we attempted to establish how typologically common such a configuration, in which a given syllable type admits only of contour tones but no level high tone, actually is. Our survey, which considered the Balto-Slavic, Japanese, Basque, Bantu, and Germanic language families, as well as other restricted-tone systems cited in Hyman 2009, found just such a situation to be decidedly rare: it unambiguously obtains in one language (Nagasaki Japanese), and may hold with some qualifications in two others (Somali, Hup). In contrast, systems of the following sort are better represented: level tones only, level vs. falling tones, and level vs. falling vs. rising tones. Importantly, we observed that within the synchronic phonology of a language, processes that repair potential rising tones are also well-attested. In typological terms, then, the “traditional” interpretation of the acute-circumflex distinction as a rising-falling one would be unusual, but possible.

Finally, we considered what impact the assumption of a consistent level high tone might have on aspects of the phonological analysis of accentual phenomena in Ancient Greek. For one, the behavior of accentuation in host + enclitic sequences does not encounter any additional complications, and regarding the acute as a level high tone may even prove necessary in order to account for certain facts. In terms of underlying representations, a lexical level /H\*/

is largely adequate for accounting for most cases of non-recessive accentuation, while a default /H+L\*/ intonation largely suffices for lexically unmarked forms.<sup>62</sup> In this regard, the global word-prosodic distinctions found in Ancient Greek appear comparable to the tone-accent systems encountered in Scandinavian languages and Franconian dialects of West Germanic.

Ultimately, while the data and arguments put forth here strongly suggest that an acute on a VV-syllable is to be defined, at the surface level of representation, as an H linked either to a syllable node or to both vocalic morae of the syllable (see the representations in (3)), other representations at underlying or intermediate levels of representation are not necessarily excluded. For instance, if the mora is assumed to constitute the TBU of Ancient Greek, long vowels or diphthongs with lexically specified acutes versus lexically specified circumflexes might be distinguished by the mora (first or second) to which the H tone is underlyingly associated. If that is the case, the formation of a rising tone through insertion of default L tones must be blocked, with the H instead spreading over both morae (*pace* Olander 2009: 62 and Jasanoff 2017: 11 *i. a.*). Taking the syllable to be the TBU more elegantly aligns with the fact that the acute apparently consists of a single phonological pitch target, and thus potentially simplifies the more general analysis of Ancient Greek word prosody. Regardless of the details to be worked out in future phonological treatments, we strongly recommend the following to all analysts: the defining surface feature of acute accentuation on all syllable types was a level high tone.

### Note on corpus and statistical analysis

The electronic corpus constructed by the second author as well as all code in R used for conducting statistical analyses and generating figures is available at: [https://osf.io/wke79/?view\\_only=132b6e6530cc4e8986bba9fc3979af97](https://osf.io/wke79/?view_only=132b6e6530cc4e8986bba9fc3979af97).

The corpus is also available at: [github.com/dcgunkel/agmusic](https://github.com/dcgunkel/agmusic).

All code in R, written primarily by the first author, which was used for conducting statistical analyses and generating figures, is also available at: [github.com/rpsandell/melisms](https://github.com/rpsandell/melisms).

62 We caution again, however, that the aims of this study are not to provide a comprehensive phonological analysis of all facts of Ancient Greek prosody. In particular, the mobility of high tone in monosyllabic nominal stems is completely ignored here.

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A.1      **Appendix 1: list of certain and relatively secure melisms in DAGM 20 and 21**

The melisms listed here constitute the 44 certain and 13 relatively secure cases of melisms on VV-nuclei discussed in section 2.2. Each individual melism is shown here as two adjacent lines in the table, in accord with the structure of the corpus as described in section 2.1. Here, we provide the document, line, note, note value, syllable (in the romanization employed in the corpus), and accent of each part of the melism; in the less secure cases, the note or note and note value is missing for half of the apparent melism. These are further marked with an asterisk following the transcription in the syllable column. Compare Appendix 3 for melisms on V-nuclei.

| Document | Line | Note | Note value | Syllable | Accent |
|----------|------|------|------------|----------|--------|
| 20       | 1    |      | NA         | kO*      | c      |
| 20       | 1    |      | NA         | kO*      | c      |
| 20       | 2    | d    | 1          | mou      | u      |
| 20       | 2    | c    | 1          | mou      | u      |
| 20       | 3    | d    | 1          | Poi      | c      |
| 20       | 3    | c    | 1          | Poi      | c      |
| 20       | 3    | c    | 1          | dai      | c      |
| 20       | 3    | AF   | 1          | dai      | c      |
| 20       | 4    | eF   | 1          | tAs      | c      |
| 20       | 4    | c    | 1          | tAs      | c      |
| 20       | 5    | c    | 1          | tais     | c      |
| 20       | 5    | AF   | 1          | tais     | c      |
| 20       | 5    | f    | 1          | s eu     | u      |
| 20       | 5    | g    | 1          | s eu     | u      |
| 20       | 6    | c    | 1          | prO      | c      |
| 20       | 6    | AF   | 1          | prO      | c      |
| 20       | 7    | eF   | 1          | tei      | c      |
| 20       | 7    | d    | 1          | tei      | c      |

(*cont.*)

| Document | Line | Note | Note value | Syllable | Accent |
|----------|------|------|------------|----------|--------|
| 20       | 8    | g    | 1          | Kai*     | c      |
| 20       | 8    |      | 1          | Kai*     | c      |
| 20       | 9    | f    | 1          | tO       | u      |
| 20       | 9    | g    | 1          | tO       | u      |
| 20       | 10   | f    | 1          | moi      | c      |
| 20       | 10   | B    | 1          | moi      | c      |
| 20       | 10   | dF   | 1          | s ai     | a      |
| 20       | 10   | d    | 1          | s ai     | a      |
| 20       | 10   | dF   | 1          | tau      | a      |
| 20       | 10   | c    | 1          | tau      | a      |
| 20       | 11   | d    | 1          | mou      | c      |
| 20       | 11   | dF   | 1          | mou      | c      |
| 20       | 12   | c    | 1          | n ai     | u      |
| 20       | 12   | dF   | 1          | n ai     | u      |
| 20       | 12   | dF   | 1          | lois     | u      |
| 20       | 12   | c    | 1          | lois     | u      |
| 20       | 13   | B    | 1          | dAn      | g      |
| 20       | 13   | c    | 1          | dAn      | g      |
| 20       | 15   | aF   | 1          | tOn      | c      |
| 20       | 15   | f    | 1          | tOn      | c      |
| 20       | 18   | dF   | 1          | tois     | c      |
| 20       | 18   | f    | 1          | tois     | c      |
| 20       | 18   | f    | 1          | neis*    | u      |
| 20       | 18   |      | 1          | neis*    | u      |
| 20       | 19   | d    | 1          | tei      | c      |
| 20       | 19   | c    | 1          | tei      | c      |
| 20       | 19   | f    | 1          | s hei    | c      |
| 20       | 19   | eF   | 1          | s hei    | c      |
| 20       | 19   | eF   | 1          | Prou     | a      |
| 20       | 19   | aF   | 1          | Prou     | a      |
| 20       | 20   | f    | 1          | rei      | u      |
| 20       | 20   | g    | 1          | rei      | u      |
| 20       | 20   | d    | 1          | tE       | u      |
| 20       | 20   | B    | 1          | tE       | u      |
| 20       | 21   |      | 1          | sU*      | u      |
| 20       | 21   | d    | 1          | sU*      | u      |

(cont.)

| Document | Line | Note | Note value | Syllable | Accent |
|----------|------|------|------------|----------|--------|
| 20       | 22   | d    | 1          | T' hI    | u      |
| 20       | 22   | eF   | 1          | T' hI    | u      |
| 20       | 23   | a    | 1          | tA       | c      |
| 20       | 23   | aF   | 1          | tA       | c      |
| 20       | 24   | d    | 1          | rA       | u      |
| 20       | 24   | eF   | 1          | rA       | u      |
| 20       | 27   |      | 1          | dA*      | a      |
| 20       | 27   | aF   | 1          | dA*      | a      |
| 21       | 1    | e    | 1          | tAn      | a      |
| 21       | 1    | d    | 1          | tAn      | a      |
| 21       | 2    | d    | 1          | klei     | u      |
| 21       | 2    | BF   | 1          | klei     | u      |
| 21       | 2    | e    | 1          | nOn      | u      |
| 21       | 2    | BF   | 1          | nOn      | u      |
| 21       | 6    | g    | 1          | kA       | c      |
| 21       | 6    | eF   | 1          | kA       | c      |
| 21       | 6    | d    | 1          | gou      | c      |
| 21       | 6    | BF   | 1          | gou      | c      |
| 21       | 8    | A    | 1          | pAs      | c      |
| 21       | 8    | E    | 1          | pAs      | c      |
| 21       | 9    | d    | 1          | TE       | g      |
| 21       | 9    | e    | 1          | TE       | g      |
| 21       | 10   | c    | 1          | nE       | u      |
| 21       | 10   | f    | 1          | nE       | u      |
| 21       | 11   |      | 1          | s oid*   | c      |
| 21       | 11   | A    | 1          | s oid*   | c      |
| 21       | 13   | e    | 1          | nA       | c      |
| 21       | 13   | E    | 1          | nA       | c      |
| 21       | 14   | f    | 1          | gA       | u      |
| 21       | 14   | e    | 1          | gA       | u      |
| 21       | 15   | d    | 1          | tO       | u      |
| 21       | 15   | e    | 1          | tO       | u      |
| 21       | 17   |      | 1          | dei*     | c      |
| 21       | 17   | BF   | 1          | dei*     | c      |
| 21       | 17   | A    | 1          | s ai     | u      |
| 21       | 17   | BF   | 1          | s ai     | u      |

(*cont.*)

| Document | Line | Note | Note value | Syllable | Accent |
|----------|------|------|------------|----------|--------|
| 21       | 21   | c    | 1          | T' hO    | c      |
| 21       | 21   | B    | 1          | T' hO    | c      |
| 21       | 23   | f    | 1          | tO       | c      |
| 21       | 23   | e    | 1          | tO       | c      |
| 21       | 25   | d    | 1          | Keis     | u      |
| 21       | 25   | e    | 1          | Keis     | u      |
| 21       | 26   |      | 1          | nA*      | u      |
| 21       | 26   | E    | 1          | nA*      | u      |
| 21       | 27   | eF   | 1          | nO       | c      |
| 21       | 27   | d    | 1          | nO       | c      |
| 21       | 30   | e    | 1          | lA       | u      |
| 21       | 30   | f    | 1          | lA       | u      |
| 21       | 31   | f    | 1          | gAs*     | c      |
| 21       | 31   |      | 1          | gAs*     | c      |
| 21       | 32   | f    | 1          | TE       | c      |
| 21       | 32   | d    | 1          | TE       | c      |
| 21       | 33   |      | 1          | sU*      | a      |
| 21       | 33   | f    | 1          | sU*      | a      |
| 21       | 34   |      | 1          | nOn*     | c      |
| 21       | 34   | e    | 1          | nOn*     | c      |
| 21       | 35   | E    | 1          | reis*    | u      |
| 21       | 35   |      | 1          | reis*    | u      |
| 21       | 35   | a    | 1          | gA*      | c      |
| 21       | 35   |      | 1          | gA*      | c      |

*Legend:* a = acute; c = circumflex; g = grave; u = unaccented

A.2      Appendix 2: intervals of VV-melisms in DAGM 20 and 21

| Document | Line | Syllable | Accent | Distance in Hz | Distance in semitones |
|----------|------|----------|--------|----------------|-----------------------|
| 20       | 2    | mou      | u      | 26.94          | 2                     |
| 20       | 3    | Poi      | c      | 26.94          | 2                     |
| 20       | 3    | dai      | c      | 45.39          | 4                     |
| 20       | 4    | tAs      | c      | 41.63          | 3                     |
| 20       | 5    | tais     | c      | 45.39          | 4                     |
| 20       | 5    | s eu     | u      | -35.97         | 2                     |
| 20       | 6    | prO      | c      | 45.39          | 4                     |
| 20       | 7    | tei      | c      | 14.69          | 1                     |
| 20       | 9    | tO       | u      | -35.97         | 2                     |
| 20       | 10   | moi      | c      | 86.01          | 6                     |
| 20       | 10   | s ai     | a      | -13.86         | 1                     |
| 20       | 10   | tau      | a      | 13.08          | 1                     |
| 20       | 11   | mou      | c      | 13.86          | 1                     |
| 20       | 12   | n ai     | u      | -13.08         | 1                     |
| 20       | 12   | lois     | u      | 13.08          | 1                     |
| 20       | 13   | dAn      | g      | -12.35         | 1                     |
| 20       | 15   | tOn      | c      | 55.57          | 3                     |
| 20*      | 18*  | tois*    | c*     | -60.58*        | 4*                    |
| 20       | 19   | tei      | c      | 26.94          | 2                     |
| 20       | 19   | s hei    | c      | 32.03          | 2                     |
| 20       | 19   | Prou     | a      | -87.6          | 5                     |
| 20       | 20   | rei      | u      | -35.97         | 2                     |
| 20       | 20   | tE       | u      | 39.29          | 3                     |
| 20       | 22   | T' hI    | u      | -14.69         | 1                     |
| 20       | 23   | tA       | c      | 20.76          | 1                     |
| 20       | 24   | rA       | u      | -14.69         | 1                     |
| 21       | 1    | tAn      | a      | 30.24          | 2                     |
| 21       | 2    | klei     | u      | 50.94          | 4                     |
| 21       | 2    | nOn      | u      | 81.18          | 6                     |
| 21       | 6    | kA       | c      | 68             | 4                     |
| 21       | 6    | gou      | c      | 50.94          | 4                     |
| 21       | 8    | pAs      | c      | 46.41          | 5                     |
| 21       | 9    | TE       | g      | -30.24         | 2                     |
| 21       | 10   | nE       | u      | -73.66         | 5                     |



(cont.)

| Document | Line | Syllable | Accent | Distance in Hz | Distance in semitones |
|----------|------|----------|--------|----------------|-----------------------|
| 21       | 13   | nA       | c      | 138.59         | 12                    |
| 21       | 14   | gA       | u      | 16.48          | 1                     |
| 21       | 15   | tO       | u      | −30.24         | 2                     |
| 21       | 17   | s ai     | u      | −11            | 1                     |
| 21       | 21   | T' hO    | c      | 12.35          | 1                     |
| 21       | 23   | tO       | c      | 16.48          | 1                     |
| 21       | 25   | Keis     | u      | −30.24         | 2                     |
| 21       | 27   | nO       | c      | 14.69          | 1                     |
| 21       | 30   | lA       | u      | −16.48         | 1                     |
| 21       | 32   | TE       | c      | 46.72          | 3                     |

*Nota bene:* Distance in Hz was obtained by subtracting the second pitch from the first pitch; thus, a positive value means a falling pitch [i.e., the first pitch was higher than the second], while negative value means a rising pitch [i.e., the first pitch was lower than the second].

A.3      **Appendix 3: list of melisms on short vocalic nuclei in DAGM 20 and 21**

The melisms listed here constitute the 11 relatively secure cases of melisms on V-nuclei discussed in section 3.2.3. Each individual melism is shown here as two adjacent lines in the table, in accord with the structure of the corpus as described in section 3.1. Compare Appendix 1 for melisms on VV-nuclei.

| Document | Line | Note | Note value | Syllable | Accent |
|----------|------|------|------------|----------|--------|
| 20       | 5    | c    | 1          | del      | u      |
| 20       | 5    | d    | 1          | del      | u      |
| 20       | 5    | d    | 1          | sin      | u      |
| 20       | 5    | eF   | 1          | sin      | u      |
| 20       | 6    | c    | 1          | man      | u      |
| 20       | 6    | d    | 1          | man      | u      |
| 20       | 12   | B    | 1          | tos      | g      |
| 20       | 12   | c    | 1          | tos      | g      |

(cont.)

| Document | Line | Note | Note value | Syllable | Accent |
|----------|------|------|------------|----------|--------|
| 20       | 17   | f    | 1          | n am     | a      |
| 20       | 17   | g    | 1          | n am     | a      |
| 21       | 12   | d    | 1          | s am     | u      |
| 21       | 12   | e    | 1          | s am     | u      |
| 21       | 13   | d    | 1          | kun      | u      |
| 21       | 13   | e    | 1          | kun      | u      |
| 21       | 13   | B    | 1          | kar      | u      |
| 21       | 13   | c    | 1          | kar      | u      |
| 21       | 20   | d    | 1          | s am     | u      |
| 21       | 20   | e    | 1          | s am     | u      |
| 21       | 24   | d    | 1          | kos      | u      |
| 21       | 24   | e    | 1          | kos      | u      |
| 21       | 28   | A    | 1          | s ap     | u      |
| 21       | 28   | BF   | 1          | s ap     | u      |
| 21       | 28   | G    | 1          | t' am    | u      |
| 21       | 28   | A    | 1          | t' am    | u      |